# 第四篇 常微分方程

## 第十章 常微分方程

### §1. 常微分方程的概念

习题. P.231

1. 指出下列各题中的函数(或隐函数)是否为所给微分方程的解:

(1) $xy' = 2y$. $y = 5x$.

解: $xy' - 2y = x(5) - 2(5x) = -5x \neq 0$. ∴ 不是解.

(2) $y'' + 4y = 0$. $y = 6\sin 2x - 2\cos 2x$.

解: $y'' + 4y = (-24\sin 2x + 8\cos 2x) + 4(6\sin 2x - 2\cos 2x) = 0$. ∴ 是解.

(3) $y'' - 2y' + y = 0$. $y = 3xe^x + x$.

解: $y'' - 2y' + y = (3e^x + 3xe^x + 1)' - 2(3e^x + 3xe^x + 1) + (3xe^x + x)$

$= 3e^x + 3e^x + 3xe^x - 6e^x - 6xe^x - 2 + 3xe^x + x$

$= x - 2 \neq 0$. ∴ 不是解.

(4) $(2xy^2 - y)dx + (y^2 + x + y)dy = 0$. $x^2 + y - \frac{x}{y} + \ln y = 0$.

解: 对 $x^2 + y - \frac{x}{y} + \ln y = 0$ 两边求微分:

$2x\,dx + dy - \frac{1}{y^2}(y\,dx - x\,dy) + \frac{1}{y}dy = 0 \Rightarrow 2xy^2\,dx + y^2\,dy - y\,dx + x\,dy + y\,dy = 0$

即 $(2xy^2 - y)dx + (y^2 + x + y)dy = 0$. ∴ 是解.

2. 验证下列函数是所给微分方程的特解:

(1) $\begin{cases} y' - 2xy = x \\ y(0) = 1 \end{cases}$ $y = \frac{3}{2}e^{x^2} - \frac{1}{2}$.

解: $y' = \frac{3}{2}e^{x^2} \cdot 2x = 3xe^{x^2}$. $y(0) = \frac{3}{2} - \frac{1}{2} = 1$.

$y' - 2xy - x = 3xe^{x^2} - 2x\left(\frac{3}{2}e^{x^2} - \frac{1}{2}\right) - x = 0$. ∴ 是方程的特解.

(2) $\begin{cases} y'' - 3y' + 2y = 5 \\ y|_{x=0} = 1, \quad y'|_{x=0} = 2. \end{cases}$ $y = -5e^x + \frac{7}{2}e^{2x} + \frac{5}{2}$.

解: $y' = -5e^x + 7e^{2x}$. $y|_{x=0} = -5 + \frac{7}{2} + \frac{5}{2} = 1$. $y'|_{x=0} = -5 + 7 = 2$.

$y'' = -5e^x + 14e^{2x}$.

$y'' - 3y' + 2y = -5e^x + 14e^{2x} + 15e^x - 21e^{2x} - 10e^x + 7e^{2x} + 5 = 5.$

∴ 是方程的解.

## §2. 一阶常微分方程

习题 P. 246

1、曲线 $f(x) = y$ 经过点 $(e, -1)$，且在任一点处的切线斜率为该点横坐标的倒数，求该曲线的方程.

解：设曲线 $y = f(x)$，在 $(x, y)$ 处切线斜率 $k = f'(x) = \frac{1}{x}$.

即：$\begin{cases} \frac{dy}{dx} = \frac{1}{x} \\ y(e) = -1 \end{cases} \Rightarrow \begin{cases} dy = \frac{1}{x}dx \\ y(e) = -1 \end{cases} \Rightarrow \begin{cases} y = \ln x + C \\ y(e) = -1 \end{cases} \Rightarrow C = -2$

∴ 曲线方程：$y = \ln x - 2$

2. 已知曲线 $y = f(x)$ 在任意一点 $x$ 处的切线斜率都比该点横坐标的立方根多 1，

(1) 求出该曲线方程所有满足的形式，并在直角坐标系中画出示意图；

(2) 若已知该曲线经过 $(1, 1)$ 点，求该曲线的方程.

解：设曲线 $y = f(x)$，在 $(x, y)$ 处切线斜率 $k = f'(x) = \sqrt[3]{x} - 1$.

即满足：$y' = \sqrt[3]{x} - 1 = \frac{dy}{dx}$

(1) $\frac{dy}{dx} = \sqrt[3]{x} - 1 \Rightarrow dy = (\sqrt[3]{x} - 1)dx \Rightarrow y = \int (x^{\frac{1}{3}} - 1)\,dx$

$\therefore y = \frac{3}{4}x^{\frac{4}{3}} - x + C.$

(2) $y(1) = 1 \Rightarrow 1 = \frac{3}{4} - 1 + C \Rightarrow C = \frac{5}{4}$. $\therefore y = \frac{3}{4}x^{\frac{4}{3}} - x + \frac{5}{4}$.

3、求下列微分方程的通解.

(1) $x\frac{dy}{dx} = y\ln^2 y$.

解：$\frac{dy}{y\ln^2 y} = \frac{dx}{x} \Rightarrow \frac{d\ln y}{\ln^2 y} = \frac{dx}{x} \Rightarrow -\frac{1}{\ln y} = \ln|x| + C$.

$\therefore \ln y = -\frac{1}{\ln|x| + C}$.

(2) $(y+1)^2\frac{dy}{dx} + x^3 = 0$.

解：$(y+1)^2 dy = -x^3 dx \Rightarrow \frac{1}{3}(y+1)^3 = -\frac{1}{4}x^4 + C$.

$\therefore \frac{(y+1)^3}{3} + \frac{x^4}{4} = C.$

(3) $\frac{dy}{dx} = x\frac{dy}{dx} + a\left(y^2 + \frac{dy}{dx}\right)$

解: $(1-x-a)\frac{dy}{dx} = ay^2 \Rightarrow \frac{dy}{y^2} = \frac{a}{1-a-x}dx.$

$\Rightarrow +\frac{1}{y} = +a\ln|1-a-x| + C.$

$\therefore y\left[C + a\ln|1-a-x|\right] = 1.$

(4) $\sin x\cos y\,dx + \sin y\cos x\,dy = 0.$

解: $\frac{\sin x}{\cos x}dx + \frac{\sin y}{\cos y}dy = 0. \Rightarrow \frac{d\cos x}{\cos x} + \frac{d(\cos y)}{\cos y} = 0$

$\Rightarrow \ln|\cos x| + \ln|\cos y| = \widetilde{C}.$

$\therefore |\cos x|\cdot|\cos y| = C.$

(5) $\sec^2 x\tan y\,dx + \sec^2 y\tan x\,dy = 0.$

解: $\frac{\sec^2 x\,dx}{\tan x} + \frac{\sec^2 y\,dy}{\tan y} = 0. \Rightarrow \frac{d\tan x}{\tan x} + \frac{d\tan y}{\tan y} = 0$

$\Rightarrow \ln|\tan x| + \ln|\tan y| = \widetilde{C}.$

$\therefore |\tan x|\cdot|\tan y| = C.$

(6) $\frac{dy}{dx} = 2^{x+y}$

解: $\frac{dy}{dx} = 2^x\cdot 2^y \Rightarrow \left(\frac{1}{2}\right)^y dy = 2^x dx. \Rightarrow \frac{\left(\frac{1}{2}\right)^y}{\ln\frac{1}{2}} = \frac{2^x}{\ln 2} + \widetilde{C}.$

$\therefore 2^x + 2^{-y} = C.$

(7) $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}.$

解: $\frac{dy}{\sqrt{1-y^2}} = \frac{dx}{\sqrt{1-x^2}} \Rightarrow \arcsin y = \arcsin x + C$ 以及 $y = \pm 1.$

(8) $(e^{x+y} - e^x)dx + (e^{x+y} + e^y)dy = 0.$

解: $e^x(e^y-1)dx + e^y(e^x+1)dy = 0. \Rightarrow \frac{e^x dx}{e^x+1} + \frac{e^y dy}{e^y-1} = 0.$

$\ln|e^x+1| + \ln|e^y-1| = \widetilde{C}$

$\therefore (e^x+1)\cdot|e^y-1| = C$ 为 $y=0$.

(9) $\cos y\,dx + (1+e^{-x})\sin y\,dy = 0$

解: $\frac{dx}{1+e^{-x}} + \frac{\sin y\,dy}{\cos y} = 0 \Rightarrow \frac{e^x dx}{1+e^x} - \frac{d\cos y}{\cos y} = 0$

$\Rightarrow \ln(1+e^x) - \ln|\cos y| = \tilde{C}$

$\therefore (1+e^x)/|\cos y| = C$ 即: $1+e^x = C\cdot|\cos y|$

$1+e^x = C\cos y \Rightarrow e^x dx = C(-\sin y)dy \Rightarrow e^x dx = -\frac{1+e^x}{\cos y}\sin y dy$

$1+e^x = -C\cos y \Rightarrow e^x dx = C\sin y dy \Rightarrow e^x dx = -\frac{1+e^x}{\cos y}\sin y dy$

(10) $(x+1)\frac{dy}{dx} + 1 = e^{-y}\sin x$.

解: $e^y(x+1)dy + e^y dx = \sin x\,dx \Rightarrow d[(x+1)e^y] = d(-\cos x)$

$\Rightarrow e^y(1+x) = -\cos x + \tilde{C}$. $\therefore e^y = -\frac{\cos x}{1+x} + \frac{C}{1+x}$

或: $e^y(1+x) + \cos x = C$.

4. 求下列微分方程的解:

(1) $\frac{dy}{dx} = e^{2x+y}$. $y(0)=0$.

解: $\frac{dy}{dx} = e^{2x}\cdot e^y \Rightarrow \frac{dy}{e^y} = e^{2x}dx$

$\Rightarrow -e^{-y} = \frac{1}{2}e^{2x} + C \Rightarrow e^{-y} + \frac{1}{2}e^{2x} + C = 0$

$y(0)=0 \Rightarrow C = -\frac{3}{2} \Rightarrow e^{-y} + \frac{1}{2}e^{2x} - \frac{3}{2} = 0$.

$\therefore e^{2x} - 3 + 2e^{-y} = 0$.

(2) $x\,dy + 2y\,dx = 0$. $y(1)=1$.

解: $x\,dy = -2y\,dx \Rightarrow \frac{dy}{2y} + \frac{dx}{x} = 0 \Rightarrow \frac{1}{2}\ln|y| + \ln|x| = C$.

$y(1)=1 \Rightarrow C=0 \Rightarrow \frac{1}{2}\ln|y| + \ln|x| = 0$.

$\therefore x^2 y = 1$.

(3) $\cos y\,dx + (1+e^{-x})\sin y\,dy = 0$; $y(0) = \frac{\pi}{4}$.

解: $\frac{dx}{1+e^{-x}} + \frac{\sin y\,dy}{\cos y} = 0 \Rightarrow \frac{e^x dx}{1+e^x} + \frac{-d(\cos y)}{\cos y} = 0$

$\Rightarrow \ln(1+e^x) - \ln|\cos y| = C$. $y(0)=\frac{\pi}{4} \Rightarrow C = \ln 2 + \frac{1}{2}\ln 2 = \frac{3}{2}\ln 2$.

$\therefore (1+e^x)/|\cos y| = 2\sqrt{2}$. 即: $1+e^x = 2\sqrt{2}\cdot|\cos y|$.

(4) $\sin x\cos y\,dx + \sin y\cos x\,dy = 0$. $y(0)=\frac{\pi}{4}$.

解: $\frac{\sin x}{\cos x}dx + \frac{\sin y}{\cos y}dy = 0 \Rightarrow -\frac{d\cos x}{\cos x} - \frac{d\cos y}{\cos y} = 0$.

$\Rightarrow \ln|\cos x| + \ln|\cos y| = \tilde{C}$. $\Rightarrow |\cos x|\cdot|\cos y| = C$.

$y(0)=\frac{\pi}{4} \Rightarrow C = \frac{\sqrt{2}}{2}$. $\therefore |\cos x\cdot\cos y| = \frac{\sqrt{2}}{2}$.

5. 镭的衰变速度与它的现存量成正比. 设 $t_0$ 时有镭 $Q_0$ 克, 经1600年它的量减为一半, 求镭的衰变规律.

解: 设镭的现存量 $Q(t)$. 衰变速度 $\frac{dQ}{dt} = -kQ$. $Q(t_0)=Q_0$.

$\frac{dQ}{dt} = -kQ \Rightarrow \ln Q = -kt + C$. $\Rightarrow Q = e^{-kt+C}$.

$Q(t_0)=Q_0 \Rightarrow Q_0 = e^{-kt_0+C} \Rightarrow C = \ln Q_0 + kt_0$.

$\therefore Q = Q_0 e^{-k(t-t_0)}$

由 $\frac{1}{2}Q = Q_0 e^{-k(1600-t_0)} \Rightarrow$

当 $t-t_0=1600$ 时, $Q=\frac{1}{2}Q_0 \Rightarrow k = \frac{\ln 2}{1600}$.

$\therefore$ 衰变规律: $Q(t) = Q_0 e^{-\frac{\ln 2}{1600}(t-t_0)}$.

6. 将A物质转化为B物质的化学反应速度与B物质的浓度成反比, 设反应开始时有B物质20%, 半小时后有B物质25%, 求B物质的浓度的变化规律.

解: 设B物质的浓度 $u(t)$. 化学反应速度 $\frac{du}{dt} = \frac{k}{u}$. $u(0)=0.2$, $u(\frac{1}{2})=0.25$

$u\,du = k\,dt \Rightarrow u^2 = 2kt + C$.

$u(0)=0.2 \Rightarrow 0.04 = C \Rightarrow u^2 = 2kt + 0.04$

$u(\frac{1}{2})=0.25 \Rightarrow 0.0625 = k + 0.04 \Rightarrow k = 0.0225$.

$\therefore$ 变化规律: $u^2(t) = 0.045t + 0.04$

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7. 按复利计算，$t$ 时刻某种物的增加速度与当时的数量 $N(t)$ 成正比，设 $N(0)=N_0$，证明：

$$\left[\frac{N(t_2)}{N_0}\right]^{t_1}=\left[\frac{N(t_1)}{N_0}\right]^{t_2}.$$

证明：某种的数量 $N(t)$，$t$ 增加速度 $\frac{dN}{dt}=kN$，$N(0)=N_0$.

$$\frac{dN}{N}=k\,dt \Rightarrow \ln N=kt+C \Rightarrow N=e^{kt+C}.$$

$$N(0)=N_0 \Rightarrow N_0=e^{C} \Rightarrow C=\ln N_0 \Rightarrow N=N_0e^{kt}$$

$$\begin{cases} N(t_1)=N_0e^{kt_1} \\ N(t_2)=N_0e^{kt_2} \end{cases} \Rightarrow \frac{N(t_1)}{N_0}=e^{kt_1},\quad \frac{N(t_2)}{N_0}=e^{kt_2}.$$

$$\frac{\ln\left(\frac{N(t_1)}{N_0}\right)}{\ln\left(\frac{N(t_2)}{N_0}\right)}=\frac{kt_1}{kt_2}=\frac{t_1}{t_2} \Rightarrow t_2\ln\left[\frac{N(t_1)}{N_0}\right]=t_1\ln\left[\frac{N(t_2)}{N_0}\right].$$

$$\therefore \left[\frac{N(t_2)}{N_0}\right]^{t_1}=\left[\frac{N(t_1)}{N_0}\right]^{t_2}.$$

8. 一个 $1000\,m^3$ 的大厅中的空气内含有 $a\%$ 的废气，现以 $1\,m^3/$分钟 的输入新鲜空气，混合后的空气以同样的速率排出，求 $t$ 时刻空气内含有的废气浓度，并求使废气浓度减少一半所需的时间。

解：设 $t$ 时刻空气中含有的废气浓度为 $u(t)$，则大厅中的废气量为 $V\cdot u(t)=x(t)$.

在 $dt$ 时间间隔内，排出的废气量为：$1\times dt\times u(t)=u(t)\,dt$.

减少的废气量为：$-dx=-V\,du$.

$$\therefore u\,dt=-V\,du \Rightarrow \frac{du}{u}=\frac{-1}{V}dt \Rightarrow \ln u=\frac{-1}{V}\cdot t+C.$$

$$u(0)=a \Rightarrow C=\ln a \Rightarrow \ln u=\frac{-t}{V}+\ln a$$

$$\therefore u(t)=a\,e^{\frac{-t}{V}}=a\,e^{-\frac{t}{1000}}.$$

若 $u(t_0)=\frac{1}{2}a$，则：$\frac{1}{2}a=ae^{-\frac{t_0}{1000}} \Rightarrow t_0=1000\ln 2$（分）.

9. 设 $[t,t+dt]$ 中的人口增长量与 $P_{max}-P(t)$ 成正比，试导出相应的人口模型，画出人口变化情况的草图并与 Malthus 和 Verhulst 人口模型加以比较。

解：设人口模型 $P(t)$，则人口增长量 $\frac{dP}{dt}=k[P_{max}-P(t)]$.

$\Rightarrow \frac{dp}{p_{max}-p} = k\,dt. \quad \Rightarrow \ln(p_{max}-p(t)) = -kt+C.$

$\Rightarrow p(t) = p_{max} - e^{-kt+C}$ 若 $p(t_0)=p_0 \Rightarrow C = kt_0 + \ln(p_{max}-p_0)$

$\therefore p(t) = p_{max} - (p_{max}-p_0)e^{-k(t-t_0)}$

当 $t\to+\infty$ 时, $p(t)\to p_{max}$.

Malthus 人口模型: $p(t) = p_0 e^{\lambda(t-t_0)}$. 当 $t\to+\infty$ 时, $p(t)\to+\infty$.

Verhulst 人口模型: $p(t) = \dfrac{p_{max}}{1+\left(\frac{p_{max}}{p_0}-1\right)^{-\lambda(t-t_0)}}$ 当 $t\to+\infty$ 时, $p(t)\to+\infty$.

(Verhulst: $\frac{dp}{dt} = \lambda\left[1-\frac{p(t)}{p_{max}}\right]p(t) = \frac{\lambda p(t)}{p_{max}}[p_{max}-p(t)]$

$= \lambda\cdot\frac{p(t)}{p_{max}}[p_{max}-p(t)]$

本例中: $\frac{dp}{dt} = k[p_{max}-p(t)]$.)

10. 半径为 1 m, 高为 2 m 的直立的圆柱形容器中充满水, 按去底部的一个半径为 1 cm 的塞子开始流出, 试导出水面高度 h 随时间变化的规律, 并求水完全流空所需的时间 (水面比水口高 h 时, 出水速度 $v=\sqrt{2gh}$)

解: 水面的流量为: $v_1\cdot S_1 = -\frac{dh}{dt}\cdot\pi R^2$ (R 为圆柱半径)

出水口的流量为: $v_2\cdot S_2 = \sqrt{2gh}\cdot\pi r^2$. (r 为塞子半径)

两者应相等 (不可压缩液体): $-\frac{dh}{dt} = \frac{r^2}{R^2}\sqrt{2gh}$.

$\Rightarrow -\frac{dh}{\sqrt{h}} = \sqrt{2g}\cdot\frac{r^2}{R^2}dt \quad \Rightarrow -2\sqrt{h} = \sqrt{2g}\cdot\frac{r^2}{R^2}t + C$ $(C=-2\sqrt{H})$

$h(0)=200 \Rightarrow C=-2\sqrt{200}=-20\sqrt{2} \Rightarrow -2\sqrt{h} = \sqrt{2g}\left(\frac{r}{R}\right)^2 t - 20\sqrt{2}$.

$\therefore h(t) = \left[\frac{\sqrt{2g}}{20000}t - 10\sqrt{2}\right]^2$. $h(t) = \left[\sqrt{H} - \frac{\sqrt{g}}{\sqrt{2}}\left(\frac{r}{R}\right)^2 t\right]^2$

当 $h(t)=0$ 时, $T = \left(\frac{R}{r}\right)^2\sqrt{\frac{2H}{g}} = \left(\frac{100}{1}\right)^2\sqrt{\frac{2\times200}{980}} \doteq 6389\,(\text{sec}) \approx 1\text{小时}46\text{分}29\text{秒}$

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11. 求下列齐次方程的通解：

(1) $x\frac{dy}{dx} = y - \sqrt{x^2+y^2}$

解：令 $y = ux$，则 $\frac{dy}{dx} = x\frac{du}{dx} + u$，代入方程得：

$$x\left(x\frac{du}{dx}+u\right) = ux - \sqrt{x^2+u^2x^2} \Rightarrow x^2\frac{du}{dx} = -x\sqrt{1+u^2}$$

$$\Rightarrow \frac{du}{\sqrt{1+u^2}} = -\frac{dx}{x}$$

令 $u = \tan\theta$，则 $\frac{\sec^2\theta\, d\theta}{\sec\theta} = -\frac{dx}{x}$

$$\Rightarrow \sec\theta\, d\theta = -\frac{dx}{x} \Rightarrow \frac{d\sin\theta}{1-\sin^2\theta} = -\frac{dx}{x}$$

$$\Rightarrow \frac{d\sin\theta}{1+\sin\theta} + \frac{d\sin\theta}{1-\sin\theta} = -\frac{2dx}{x} \Rightarrow \ln\frac{1+\sin\theta}{1-\sin\theta} = -2\ln x + C$$

$$\Rightarrow x^2\left[\frac{1+\sin\theta}{1-\sin\theta}\right] = C \Rightarrow x^2\frac{1+\frac{u}{\sqrt{1+u^2}}}{1-\frac{u}{\sqrt{1+u^2}}} = C \Rightarrow x\left[\sqrt{1+u^2}+u\right] = C$$

$\therefore \sqrt{x^2+y^2} + y = C.$

(2) $(x^2+y^2)dx + (xy-2x^2)dy = 0.$

解：令 $y = ux$，则 $\frac{dy}{dx} = x\frac{du}{dx} + u$，代入方程得：

$$x^2(1+u^2)dx + (ux^2-2x^2)\cdot(x\,du + u\,dx) = 0$$

$$\Rightarrow (1+u^2)dx + (u-2)\cdot x\,du + u(u-2)dx = 0$$

$$(2u^2-2u+1)dx + x(u-2)du = 0$$

$$\Rightarrow \frac{(u-2)du}{2u^2-2u+1} + \frac{dx}{x} = 0 \Rightarrow \frac{(2u-1)du - 3du}{u^2-u+\frac{1}{2}} + \frac{4dx}{x} = 0$$

$$\Rightarrow \frac{d(u^2-u+\frac{1}{2})}{u^2-u+\frac{1}{2}} - 3\cdot\frac{d(u-\frac{1}{2})}{(u-\frac{1}{2})^2+\frac{1}{4}} + 4\cdot\frac{dx}{x} = 0$$

$$\Rightarrow \ln\left(u^2-u+\frac{1}{2}\right) - 6\arctan(2u-1) + 4\ln x = C$$

即：$\ln(x^2y^2 - x^3y + \frac{1}{2}x^4) - 6\arctan\frac{2y-x}{x} = C$

(3) $(x^3+y^3)\,dx - 2xy^2\,dy = 0$

解: $\frac{dy}{dx} = \frac{x^3+y^3}{2xy^2} = \frac{1+\left(\frac{y}{x}\right)^3}{2\cdot\left(\frac{y}{x}\right)^2}$ 令 $\frac{y}{x} = u$, 则有:

$u + x\frac{du}{dx} = \frac{1+u^3}{2u^2} \Rightarrow x\frac{du}{dx} = \frac{1+u^3}{2u^2} - u = \frac{1-u^3}{2u^2}$.

$\Rightarrow \frac{2u^2\,du}{1-u^3} = \frac{dx}{x} \Rightarrow -\frac{2}{3}\ln(1-u^3) = \ln x + C$.

$\Rightarrow \left(1-\frac{y^3}{x^3}\right)\cdot x^{\frac{3}{2}} = C \quad \therefore \left(x^3-y^3\right)^2 = Cx^3$.

(4) $x\frac{dy}{dx} = y\ln\frac{y}{x}$.

解: 设 $y = ux$, 则: $x\left(\frac{du}{dx} + u\right) = ux\cdot\ln u \Rightarrow x\frac{du}{dx} = u\ln u - u$

$\frac{du}{u[\ln u - 1]} = \frac{dx}{x} \Rightarrow \frac{d(\ln u)}{\ln u - 1} = \frac{dx}{x}$

$\Rightarrow \ln[\ln u - 1] = \ln x + C \Rightarrow \frac{\ln u - 1}{x} = C$.

$\therefore \ln\frac{y}{x} = 1 + Cx$.

(5) $\frac{dy}{dx} = \frac{y^2 + x\sqrt{x^2+y^2}}{xy}$

解: 设 $y = ux$, 则: $u + x\frac{du}{dx} = \frac{x^2u^2 + x\sqrt{x^2(1+u^2)}}{ux^2} = \frac{u^2+\sqrt{1+u^2}}{u} = u + \frac{\sqrt{1+u^2}}{u}$

$\Rightarrow x\frac{du}{dx} = \frac{\sqrt{1+u^2}}{u} \Rightarrow \frac{u\,du}{\sqrt{1+u^2}} = \frac{dx}{x}$

$\Rightarrow \frac{d(1+u^2)}{\sqrt{1+u^2}} = 2\frac{dx}{x} \Rightarrow 2\sqrt{1+u^2} = 2\ln x + C$

$\Rightarrow \sqrt{1+u^2} = \ln x + C \quad \therefore 1 + \frac{y^2}{x^2} = (\ln x + C)^2$.

12. 求下列齐次方程的解:

(1) $\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}$. $y(1) = 2$.

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解：设 $y=ux$，则 $u+x\frac{du}{dx}=\frac{1}{u}+u \Rightarrow x\frac{du}{dx}=\frac{1}{u}$

$u\,du=\frac{dx}{x} \Rightarrow \frac{1}{2}u^2=\ln x+C \Rightarrow u^2=2\ln x+C$

$\frac{y^2}{x^2}=2\ln x+C$，$y(1)=2 \Rightarrow C=4$

$\therefore y^2=2x^2\ln x+4x^2 \Rightarrow y=x\sqrt{2\ln x+4}$

(3) $(x^2+2xy-y^2)dx+(y^2+2xy-x^2)dy=0$，$y(1)=1$

解：令 $y=ux$，则 $dy=x\,du+u\,dx$，代入方程得：

$(x^2+2ux^2-u^2x^2)dx+(u^2x^2+2x^2u-x^2)(x\,du+u\,dx)=0$

$(1+2u-u^2)dx+(u^2+2u-1)\cdot x\,du+u(u^2+2u-1)dx=0$

$(1+2u-u^2+u^3+2u^2-u)dx+x(u^2+2u-1)du=0$

$(u^3+u^2+u+1)dx+x(u^2+2u-1)du=0$

$\Rightarrow \frac{(u^2+2u-1)du}{u^3+u^2+u+1}+\frac{dx}{x}=0 \Rightarrow \left(\frac{2u}{u^2+1}-\frac{1}{u+1}\right)du+\frac{dx}{x}=0$

$\Rightarrow \frac{d(u^2+1)}{u^2+1}-\frac{d(u+1)}{u+1}+\frac{dx}{x}=0 \Rightarrow \ln(u^2+1)-\ln(u+1)+\ln x=C$

$\Rightarrow \frac{u^2+1}{u+1}\cdot x=C \Rightarrow \frac{(\frac{y}{x})^2+1}{\frac{y}{x}+1}\cdot x=C \Rightarrow \frac{x^2+y^2}{x+y}=C$

$y(1)=1 \Rightarrow C=1$，$\therefore x^2+y^2=x+y$

(4) $(x^2+y^2)dx+(xy-2x^2)dy=0$，$y(1)=0$

解：令 $y=ux$，则 $\frac{dy}{dx}=u+x\frac{du}{dx}=\frac{x^2+y^2}{2x^2-xy}=\frac{1+u^2}{2-u}$

$\Rightarrow x\frac{du}{dx}=\frac{1+u^2}{2-u}-u=\frac{1+u^2-2u+u^2}{2-u}=\frac{2u^2-2u+1}{2-u}$

$\Rightarrow \frac{(u-2)du}{2u^2-2u+1}+\frac{dx}{x}=0 \Rightarrow \frac{d(u^2-u+\frac{1}{2})}{u^2-u+\frac{1}{2}}-3\frac{d(u-\frac{1}{2})}{(u-\frac{1}{2})^2+\frac{1}{4}}+\frac{4dx}{x}=0$

$\Rightarrow \ln(u^2-u+\frac{1}{2})-6\arctan(2u-1)+4\ln x=C$

$\Rightarrow \ln(x^2y^2-x^3y+\frac{1}{2}x^4)-6\arctan\frac{2y-x}{x}=C$

$y(1)=0 \Rightarrow \ln\frac{1}{2} - 6\arctan(-1) = C \Rightarrow C = -\ln 2 + \frac{3\pi}{2}$.

$\therefore \ln(x^2y^2 - x^3y + \frac{1}{2}x^4) - 6\arctan\frac{2y-x}{x} = -\ln 2 + \frac{3\pi}{2}$

即: $\ln(2x^2y^2 - 2x^3y + x^4) - 6\arctan\frac{2y-x}{x} - \frac{3\pi}{2} = 0$.

13. 将下列方程化为齐次方程并求出通解:

(1) $(x+2y+1)dx + (2x-y-3)dy = 0$.

解: $\frac{dy}{dx} = -\frac{x+2y+1}{2x-y-3}$. 令 $\begin{cases} x = x_1 + \xi \\ y = y_1 + \eta \end{cases}$

$\begin{cases} x+2y+1 = x_1 + 2y_1 + \xi + 2\eta + 1 \\ 2x - y - 3 = 2x_1 - y_1 + 2\xi - \eta - 3 \end{cases} \Rightarrow \begin{cases} \xi + 2\eta + 1 = 0 \\ 2\xi - \eta - 3 = 0 \end{cases} \Rightarrow \begin{cases} \xi = 1 \\ \eta = -1 \end{cases}$

作变换: $\begin{cases} x = x_1 + 1 \\ y = y_1 - 1 \end{cases}$ 则有: $\frac{dy_1}{dx_1} = -\frac{x_1 + 2y_1}{2x_1 - y_1} = -\frac{1 + 2\frac{y_1}{x_1}}{2 - \frac{y_1}{x_1}}$.

令 $y_1 = ux_1$. 则有: $u + x_1\frac{du}{dx_1} = -\frac{1+2u}{2-u}$.

$x_1\frac{du}{dx_1} = -\frac{1+2u}{2-u} - u = \frac{u^2 - 4u - 1}{2-u} \Rightarrow -\frac{(u-2)du}{u^2-4u-1} = \frac{dx_1}{x_1}$

$\ln(u^2 - 4u - 1) = -2\ln x_1 + C_1 \Rightarrow x_1^2(u^2 - 4u - 1) = C$.

$\Rightarrow y_1^2 - 4x_1y_1 - x_1^2 = C$.

$\therefore (y+1)^2 - 4(x-1)(y+1) - (x-1)^2 = C$.

即: $y = 2x - 3 \pm \sqrt{5x^2 - 10x + C}$.

(2) $(3x-2y+1)dx + (x-4y-3)dy = 0$.

解: $\frac{dy}{dx} = -\frac{3x-2y+1}{x-4y-3}$. 令 $\begin{cases} x = x_1 + \xi \\ y = y_1 + \eta \end{cases}$

$\begin{cases} 3x - 2y + 1 = 3x_1 - 2y_1 + 3\xi - 2\eta + 1 \\ x - 4y - 3 = x_1 - 4y_1 + \xi - 4\eta - 3 \end{cases} \Rightarrow \begin{cases} 3\xi - 2\eta + 1 = 0 \\ \xi - 4\eta - 3 = 0 \end{cases} \Rightarrow \begin{cases} \xi = -1 \\ \eta = -1 \end{cases}$

作变换: $\begin{cases} x = x_1 - 1 \\ y = y_1 - 1 \end{cases}$ 则有: $\frac{dy_1}{dx_1} = -\frac{3x_1 - 2y_1}{x_1 - 4y_1} = -\frac{3 - 2\frac{y_1}{x_1}}{1 - 4\frac{y_1}{x_1}}$.

令 $y_1 = ux_1$. 则有: $u + x_1\frac{du}{dx_1} = -\frac{3-2u}{1-4u}$.

$x_1\frac{du}{dx_1} = -\frac{3-2u}{1-4u} - u = \frac{4u^2 + u - 3}{1-4u} \Rightarrow -\frac{(4u-1)du}{4u^2+u-3} = \frac{dx_1}{x_1}$

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$$-\frac{d(4u^2+u-3)}{4u^2+u-3}+\frac{12}{7}\frac{du}{4u-3}-\frac{3}{7}\frac{du}{u+1}=\frac{2dx_1}{x_1}$$

$$-\ln(4u^2+u-3)+\frac{3}{7}\ln(4u-3)-\frac{3}{7}\ln(u+1)-2\ln x_1=C$$

$$x_1^2(4u^2+u-3)+\left(\frac{u+1}{4u-3}\right)^{\frac{3}{7}}=C \Rightarrow \left(4y_1^2+x_1y_1-3x_1^2\right)+\left(\frac{x_1+y_1}{4y_1-3x_1}\right)^{\frac{3}{7}}=C$$

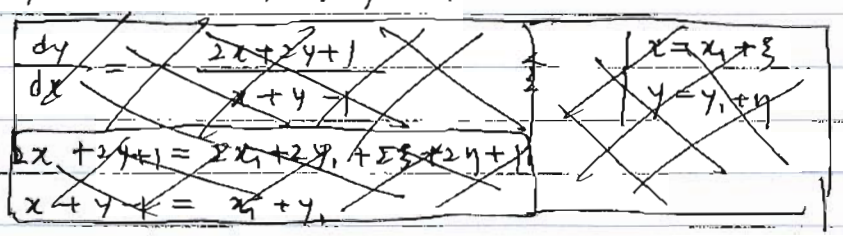
$$\Rightarrow 4(y+1)^2+(x+1)(y+1)-3(x+1)^2+\left(\frac{x+y+2}{4y-3x+1}\right)^{\frac{3}{7}}=C$$

即: $(4y-3x+1)(-x+y+2)+\left(\frac{x+y+2}{4y-3x+1}\right)^{\frac{3}{7}}=C$

$$\therefore (4y-3x+1)^{\frac{10}{7}}(x+y+2)+(x+y+2)^{\frac{3}{7}}=C(4y-3x+1)^{\frac{3}{7}}$$

(3) $(2x+2y+1)dx+(x+y-1)dy=0$

解:




$$\frac{dy}{dx}=-\frac{2x+2y+1}{x+y-1}=-2\frac{3}{x+y-1}$$ 令 $x+y=u$, 则 $\frac{dy}{dx}=\frac{du}{dx}-1$

即: $\frac{du}{dx}-1=-2-\frac{3}{u-1} \Rightarrow \frac{du}{dx}=-1-\frac{3}{u-1}=-\frac{u+2}{u-1}$

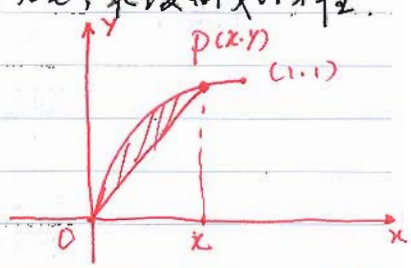
$$\Rightarrow -\frac{u-1}{u+2}du=dx \quad -\left(-1+\frac{3}{u+2}\right)du=dx$$

$$\Rightarrow -u+3\ln(u+2)=x+C \Rightarrow -(x+y)+3\ln(x+y+2)=x+C$$

$$\therefore 3\ln(x+y+2)-2x-y=C$$

14. 上凹曲线 $y=f(x)$ 经过点 $(0,0)$ 和 $(1,1)$, 且对于曲线上任一点 $P(x,y)$ $(0<x<1)$, 曲线上连接 $(0,0)$ 和 $P$ 的弧与连接 $(0,0)$ 和 $P$ 的线段所围面积为 $x^2$, 求该曲线的方程.

解: $S=\int_0^x f(t)dt-\frac{1}{2}xy=x^2$ 两边对 $x$ 求导, 得:




$$f(x)-\frac{1}{2}y-\frac{1}{2}x\frac{dy}{dx}=2x \Rightarrow y-\frac{1}{2}y-\frac{1}{2}x\frac{dy}{dx}=2x$$

$$\Rightarrow y-4x-x\frac{dy}{dx}=0 \Rightarrow \frac{y}{x}-4-\frac{dy}{dx}=0$$

$u-4-u-x\frac{du}{dx}=0. \Rightarrow 4\frac{dx}{x}+du=0 \Rightarrow 4\ln x+u=C.$

由 $y(1)=1$ 即 $u(1)=1 \Rightarrow C=1. \Rightarrow 4\ln x+\frac{y}{x}=1.$

∴ 曲线方程: $y=x-4x\ln x.$

15. 判断下列方程是否为全微分方程，若是全微分方程则求出其通解.

(1) $(5x^4+3xy^2-y^3)dx+(3x^2y-3xy^2+y^2)dy=0.$

解: $f(x,y)=5x^4+3xy^2-y^3. \quad g(x,y)=3x^2y-3xy^2+y^2.$

$\frac{\partial f}{\partial y}=6xy-3y^2=\frac{\partial g}{\partial x}$ ∴是全微分方程.

$u(x,y)=\int_0^x f(x,y)dx+\int_0^y g(0,y)dy$

$=\int_0^x (5x^4+3xy^2-y^3)dx+\int_0^y y^2dy$

$=x^5+\frac{3}{2}x^2y^2-xy^3+\frac{1}{3}y^3.$

∴ 通解: $x^5+\frac{3}{2}x^2y^2-xy^3+\frac{1}{3}y^3=C.$

(2) $(4x^2+2xy+y^2)dx+(x+y)^2dy=0.$

解: $f(x,y)=4x^2+2xy+y^2. \quad g(x,y)=(x+y)^2.$

$\frac{\partial f}{\partial y}=2x+2y.=\frac{\partial g}{\partial x}.$ ∴是全微分方程.

$u(x,y)=\int_0^x f(x,y)dx+\int_0^y g(0,y)dy$

$=\int_0^x (4x^2+2xy+y^2)dx+\int_0^y y^2dy$

$=\frac{4}{3}x^3+x^2y+xy^2+\frac{1}{3}y^3.$

∴ 通解: $\frac{4}{3}x^3+x^2y+xy^2+\frac{1}{3}y^3=C.$

(3) $e^ydx+(xe^y-2y)dy=0.$

解: $f(x,y)=e^y. \quad g(x,y)=xe^y-2y.$

$\frac{\partial f}{\partial y}=e^y \quad \frac{\partial g}{\partial x}=e^y=\frac{\partial f}{\partial y}.$ ∴是全微分方程.

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$u(x,y)=\int_0^x f(x,y)\,dx+\int_0^y g(0,y)\,dy$

$=\int_0^x e^y\,dx+\int_0^y(-2y)\,dy=xe^y-y^2$.

∴ 通解: $xe^y-y^2=C$.

(4) $(x\cos y+\cos x)\frac{dy}{dx}+(\sin y-y\sin x)=0$.

解: $(x\cos y+\cos x)dy+(\sin y-y\sin x)dx=0$.

$f(x,y)=\sin y-y\sin x$. $g(x,y)=x\cos y+\cos x$.

$\frac{\partial f}{\partial y}=\cos y-\sin x$. $\frac{\partial g}{\partial x}=\cos y-\sin x=\frac{\partial f}{\partial y}$ ∴是全微分方程.

$u(x,y)=\int_0^x f(x,y)\,dx+\int_0^y g(0,y)\,dy$

$=\int_0^x(\sin y-y\sin x)\,dx+\int_0^y 1\cdot dy$

$=x\sin y+y\cos x\Big|_0^y+y=x\sin y+y\cos x$

∴ 通解: $x\sin y+y\cos x=C$.

(5) $(3x^2+6xy^2)dx+(6x^2y+4y^2)dy=0$.

解: $f(x,y)=3x^2+6xy^2$. $g(x,y)=6x^2y+4y^2$.

$\frac{\partial f}{\partial y}=12xy$. $\frac{\partial g}{\partial x}=12xy=\frac{\partial f}{\partial y}$ ∴是全微分方程.

$u(x,y)=\int_0^x f(x,y)\,dx+\int_0^y g(0,y)\,dy$

$=\int_0^x(3x^2+6xy^2)\,dx+\int_0^y(4y^2)\,dy$

$=x^3+3x^2y^2+\frac{4}{3}y^3$.

∴ 通解: $x^3+3x^2y^2+\frac{4}{3}y^3=C$.

(6) $y(x-2y)dx-x^2dy=0$

解: $f(x,y)=y(x-2y)$. $g(x,y)=-x^2$.

$\frac{\partial f}{\partial y}=x-4y$. $\frac{\partial g}{\partial x}=-2x\neq\frac{\partial f}{\partial y}$ ∴不是全微分方程.

1b. 用观察法判断下列方程的积分因子，再求出通解：

(1) $ydx - xdy = 0$.

解：$d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2} = 0$. $\therefore \mu(x,y) = \frac{1}{y^2}$. 通解：$\frac{x}{y} = C$.

(2) $y^2(x-3y)dx + (1-3xy^2)dy = 0$.

解：$(x-3y)dx + \left(\frac{1}{y^2} - 3x\right)dy = 0$. $f(x,y) = x-3y$. $g(x,y) = \frac{1}{y^2} - 3x$.

$\frac{\partial f}{\partial y} = -3$. $\frac{\partial g}{\partial x} = -3 = \frac{\partial f}{\partial y}$.

$$u(x,y) = \int_0^x f(x,y)dx + \int_0^y g(0,y)dy$$

$$= \int_0^x (x-3y)dx + \int_0^y \left(\frac{1}{y^2}\right)dy = \frac{1}{2}x^2 - 3xy - \frac{1}{y}.$$

即 $d\left(\frac{1}{2}x^2 - 3xy - \frac{1}{y}\right) = \frac{1}{y^2}\left[y^2(x-3y)dx + (1-3xy^2)dy\right] = 0$.

$\therefore \mu(x,y) = \frac{1}{y^2}$. 通解：$\frac{1}{2}x^2 - 3xy - \frac{1}{y} = C$.

(3) $xdx + ydy = (x^2+y^2)dx$.

解：$2 \cdot \frac{xdx + ydy}{x^2+y^2} = d\left[\ln(x^2+y^2)\right]$. $\therefore \mu(x,y) = \frac{1}{x^2+y^2}$.

则有：$2dx = d\left[\ln(x^2+y^2)\right]$ $\therefore$ 通解：$\ln(x^2+y^2) = 2x + C$.

(4) $(x-y^2)dx + 2xydy = 0$.

解：~~$d\left(\frac{1}{2}x^2 - xy^2\right) = \frac{1}{x^2y^2}\left[xy(x-y^2)dx\right]$~~

$$xdx = y^2dx - 2xydy = x^2 \cdot \frac{y^2dx - 2xydy}{x^2} = x^2 d\left(\frac{y^2}{x}\right).$$

$\Rightarrow$ 积分因子 $\mu(x,y) = \frac{1}{x^2}$. $\frac{xdx}{x^2} + d\left(\frac{y^2}{x}\right) = 0$.

$d\left(\ln x + \frac{y^2}{x}\right) = 0$. $\therefore$ 通解：$\ln x + \frac{y^2}{x} = C$.

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(5) $(2y-3x^2y)dx - x\,dy = 0$.

解: $\dfrac{(2y-3x^2y)dx - x\,dy}{xy} = \left(\dfrac{2}{x}-3x\right)dx - \dfrac{1}{y}dy = d(2\ln x - \dfrac{3}{2}x^2 - \ln y)$

$\therefore \mu(x,y) = \dfrac{1}{xy}$. 通解: $2\ln x - \ln y - \dfrac{3}{2}x^2 = C$.

(6) $y(1+xy)dx + x(1-xy)dy = 0$.

解: $\dfrac{y(1+xy)dx + x(1-xy)dy}{x^2y^2} = \dfrac{1}{x^2y}dx + \dfrac{1}{x}dx + \dfrac{1}{xy^2}dy - \dfrac{1}{y}dy$

$= -\dfrac{1}{y}d\left(\dfrac{1}{x}\right) - \dfrac{1}{x}d\left(\dfrac{1}{y}\right) + d(\ln x) - d(\ln y) = d\left(-\dfrac{1}{xy}\right) + d(\ln x) - d(\ln y)$

$= d\left(\ln x - \ln y - \dfrac{1}{xy}\right)$ $\therefore \mu(x,y) = \dfrac{1}{x^2y^2}$.

通解: $\ln x - \ln y - \dfrac{1}{xy} = C$.

17. 求下列线性方程的通解:

(1) $\dfrac{dy}{dx} + 4y = -4x$.

解: $y(x) = e^{-4x}\left(C + \int(-4x)e^{4x}dx\right) = Ce^{-4x} + e^{-4x}\int(-x)e^{4x}d4x$

$= Ce^{-4x} + e^{-4x}\left[-xe^{4x} + \int e^{4x}dx\right]$

$= Ce^{-4x} - x + \dfrac{1}{4}\cdot e^{-4x}\cdot e^{4x} = Ce^{-4x} - x + \dfrac{1}{4}$.

(2) $x\dfrac{dy}{dx} + 2y = x^2 - 3x + 2$.

解: $\dfrac{dy}{dx} + \dfrac{2}{x}y = x - 3 + \dfrac{2}{x}$. $f(x) = \dfrac{2}{x}$. $g(x) = x - 3 + \dfrac{2}{x}$.

$y(x) = e^{-\int f(x)dx}\left[C + \int g(x)\cdot e^{\int f(x)dx}dx\right]$. $\int f(x)dx = 2\ln x$.

$= e^{-2\ln x}\left[C + \int\left(x - 3 + \dfrac{2}{x}\right)\cdot e^{2\ln x}dx\right] = \dfrac{1}{x^2}\left[C + \int\left(x - 3 + \dfrac{2}{x}\right)\cdot x^2dx\right]$

$= \dfrac{1}{x^2}\left[C + \int(x^3 - 3x^2 + 2x)dx\right] = \dfrac{1}{x^2}\left[C + \dfrac{1}{4}x^4 - x^3 + x^2\right]$

$= \dfrac{x^2}{4} - x + 1 + \dfrac{C}{x^2}$.

(3) $\frac{dy}{dx} + y\tan x = \cos x$.

解: $f(x) = \tan x$. $\int f(x)\,dx = \int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\ln\cos x$.

$g(x) = \cos x$.

$$y(x) = e^{-\int f(x)dx}\left[C + \int g(x)\cdot e^{\int f(x)dx}dx\right]$$

$$= e^{\ln\cos x}\left[C + \int \cos x\cdot e^{-\ln\cos x}\,dx\right]$$

$$= \cos x\left[C + \int \cos x\cdot\frac{1}{\cos x}\,dx\right] = C\cos x + x\cos x$$

$$= \cos x\,(C+x).$$

(4) $\frac{dy}{dx} + y = e^{-2x}$.

解: $f(x) = 1$. $\int f(x)\,dx = x$. $g(x) = e^{-2x}$.

$$y(x) = e^{-\int f(x)dx}\left[C + \int g(x)\cdot e^{\int f(x)dx}\,dx\right]$$

$$= e^{-x}\left[C + \int e^{-2x}\cdot e^{x}\,dx\right] = e^{-x}\left[C + \int e^{-x}\,dx\right]$$

$$= e^{-x}\left[C - e^{-x}\right] = -e^{-2x} + Ce^{-x}.$$

(5) $\frac{dy}{dx} + 2xy = 2x$.

解: $f(x) = 2x$. $\int f(x)\,dx = \int 2x\,dx = x^2$. $g(x) = 2x$.

$$y(x) = e^{-\int f(x)dx}\left[C + \int g(x)\cdot e^{\int f(x)dx}dx\right]$$

$$= e^{-x^2}\left[C + \int 2x\cdot e^{x^2}\,dx\right] = e^{-x^2}\left[C + \int e^{x^2}\,dx^2\right]$$

$$= e^{-x^2}\left[C + e^{x^2}\right] = 1 + Ce^{-x^2}.$$

(6) $(x-2)\frac{dy}{dx} + y = 2(x-2)^2$.

解: $\frac{dy}{dx} + \frac{1}{x-2}\cdot y = 2(x-2)$. $f(x) = \frac{1}{x-2}$. $\int f(x)\,dx = \int \frac{1}{x-2}\,dx = \ln(x-2)$.

$g(x) = 2(x-2)$.

$$y(x) = e^{-\ln(x-2)}\left[C + \int 2(x-2)\, e^{\ln(x-2)}\, dx\right]$$

$$= \frac{1}{x-2}\left[C + 2\int (x-2)^2 dx\right] = \frac{1}{x-2}\left[C + \frac{2}{3}(x-2)^3\right]$$

$$= \frac{2}{3}(x-2)^2 + \frac{C}{x-2}.$$

(7) $(y^2-6x)\frac{dy}{dx} + 2y = 0$.

解: $\frac{dy}{dx} + \frac{2y}{y^2-6x} = 0 \iff \frac{dx}{dy} + \frac{y^2-6x}{2y} = 0$.

即: $\frac{dx}{dy} - \frac{3}{y}x = -\frac{1}{2}y$. $f(y) = -\frac{3}{y}$. $\int f(y)dy = -3\ln y$.

$g(y) = -\frac{1}{2}y$.

$$x(y) = e^{3\ln y}\left[C + \int \left(-\frac{1}{2}y\right)\cdot e^{-3\ln y}\, dy\right]$$

$$= y^3\left[C - \frac{1}{2}\int y\cdot\frac{1}{y^3}dy\right] = y^3\left[C - \frac{1}{2}\int \frac{1}{y^2}dy\right]$$

$$= y^3\left[C + \frac{1}{2y}\right] = \frac{1}{2}y^2 + Cy^3.$$

(8) $y\ln y\, dx + (x - \ln y)\, dy = 0$.

解: $\frac{dx}{dy} + \frac{1}{y\ln y}x = \frac{1}{y}$. $f(y) = \frac{1}{y\ln y}$. $\int f(y)dy = \int \frac{1}{y\ln y}dy = \ln\ln y$.

$g(x) = \frac{1}{y}$.

$$x(y) = e^{-\ln\ln y}\left[C + \int \frac{1}{y}\cdot e^{\ln\ln y}\, dy\right]$$

$$= \frac{1}{\ln y}\left[C + \int \frac{1}{y}\ln y\, dy\right] = \frac{1}{\ln y}\left[C + \frac{1}{2}\ln^2 y\right]$$

$$= \frac{1}{2}\ln y + \frac{C}{\ln y}.$$

18. 求下列线性方程的解:

(1) $\frac{dy}{dx} - y\tan x = \sec x$, $y(0) = 0$

解: $f(x) = -\tan x$. $\int f(x)dx = \int(-\tan x)dx = \ln\cos x$. $g(x) = \sec x$.

$$y(x) = e^{-\ln\cos x}\left[ C + \int \sec x \cdot e^{\ln\cos x}\, dx \right]$$

$$= \frac{1}{\cos x}\left[ C + \int \sec x \cdot \cos x\, dx \right] = \frac{1}{\cos x}\left[ C + x \right].$$

$$y(0) = 0 \quad \Rightarrow \quad 0 = 1 \cdot [C + 0], \quad \Rightarrow \quad C = 0.$$

$$\therefore \quad y(x) = \frac{x}{\cos x} = x \cdot \sec x.$$

(2) $\dfrac{dy}{dx} + \dfrac{y}{x} = \dfrac{\sin x}{x}$ $\qquad y(\pi) = 1.$

Sol: $f(x) = \dfrac{1}{x}$. $\qquad \int f(x)\, dx = \int \frac{1}{x}\, dx = \ln x$. $\qquad g(x) = \dfrac{\sin x}{x}$.

$$y(x) = e^{-\ln x}\left[ C + \int \frac{\sin x}{x} \cdot e^{\ln x}\, dx \right]$$

$$= \frac{1}{x}\left[ C + \int \frac{\sin x}{x} \cdot x\, dx \right] = \frac{1}{x}\left[ C + \int \sin x\, dx \right] = \frac{C - \cos x}{x}.$$

$$y(\pi) = 1. \quad \Rightarrow \quad 1 = \frac{C - \cos\pi}{\pi} = \frac{C+1}{\pi} \quad \Rightarrow \quad C = \pi - 1.$$

$$\therefore \quad xy = -\cos x + \pi - 1.$$

(3) $\dfrac{dy}{dx} + y\cot x = 5e^{\cos x}$ $\qquad y\left(\frac{\pi}{2}\right) = -4.$

Sol: $f(x) = \cot x$. $\qquad \int f(x)\, dx = \int \frac{\cos x}{\sin x}\, dx = \ln\sin x$. $\qquad g(x) = 5e^{\cos x}$.

$$y(x) = e^{-\ln\sin x}\left[ C + \int 5e^{\cos x} \cdot e^{\ln\sin x}\, dx \right]$$

$$= \frac{1}{\sin x}\left[ C + 5\int e^{\cos x} \cdot \sin x\, dx \right] = \frac{1}{\sin x}\left[ C - 5\int e^{\cos x}\, d\cos x \right]$$

$$= \frac{1}{\sin x}\left[ C - 5e^{\cos x} \right].$$

$$y\left(\frac{\pi}{2}\right) = -4. \quad \Rightarrow \quad -4 = \frac{1}{1}\left[ C - 5e^{0} \right] = (C - 5) \quad \Rightarrow \quad C = 1.$$

$$\therefore \quad y(x) = \frac{1}{\sin x}\left(1 - 5e^{\cos x}\right).$$

(4) $\dfrac{dy}{dx} + \dfrac{2 - 3x^2}{x^3}\, y = 1.$ $\qquad y(1) = 0$

解: $f(x)=\frac{2-3x^2}{x^3}$ $\int f(x)dx=\int\frac{2-3x^2}{x^3}dx=-\frac{1}{x^2}-\ln x^3$ $g(x)=1$.

$$y(x)=e^{+\left(\frac{1}{x^2}+\ln x^3\right)}\cdot\left[C+\int 1\cdot e^{-\frac{1}{x^2}-\ln x^3}dx\right]$$

$$=e^{\frac{3x^2\ln x+1}{x^2}}\cdot\left[C+\int e^{-\frac{1}{x^2}}\cdot e^{-\ln x^3}dx\right]$$

$$=e^{\frac{3x^2\ln x+1}{x^2}}\cdot\left[C+\int\frac{1}{x^3}e^{-\frac{1}{x^2}}dx\right]=e^{\frac{3x^2\ln x+1}{x^2}}\left[C-\frac{1}{2}\int e^{-\frac{1}{x^2}}d\frac{1}{x^2}\right]$$

$$=e^{\frac{3x^2\ln x+1}{x^2}}\left[C+\frac{1}{2}e^{-\frac{1}{x^2}}\right].$$

$y(1)=0\Rightarrow 0=e^{+1}\left[C+\frac{1}{2}e^{-1}\right]\Rightarrow C=\frac{1}{2}e^{-1}=-\frac{1}{2e}$

$$\therefore y(x)=e^{\frac{3x^2\ln x+1}{x^2}}\left[-\frac{1}{2}e^{-1}+\frac{1}{2}e^{-\frac{1}{x^2}}\right]$$

$$=\frac{1}{2}e^{\frac{3x^2\ln x+1}{x^2}-\frac{1}{x^2}}-\frac{1}{2}e^{\frac{3x^2\ln x+1}{x^2}-1}$$

$$=\frac{1}{2x^3}-\frac{1}{2}e^{\frac{1+3x^2\ln x-x^2}{x^2}}$$

19. 求微分方程 $(1+x^2)\sin(2y)\frac{dy}{dx}+x\cos^2y+2x\sqrt{1+x^2}=0$ 满足 $y(0)=0$ 的通解.

解: 设 $u=\cos^2y$, 则 $\frac{du}{dx}=2\cos y\cdot(-\sin y)\cdot\frac{dy}{dx}=-\sin 2y\cdot\frac{dy}{dx}$.

∴ 原方程变为, $-(1+x^2)\frac{du}{dx}+ux+2x\sqrt{1+x^2}=0$.

即: $\frac{du}{dx}-\frac{x}{1+x^2}u=\frac{2x}{\sqrt{1+x^2}}$

$f(x)=-\frac{x}{1+x^2}$ $\int f(x)dx=-\int\frac{x}{1+x^2}dx=-\frac{1}{2}\ln(1+x^2)=-\ln\sqrt{1+x^2}$

$g(x)=\frac{2x}{\sqrt{1+x^2}}$

$$u(x)=e^{\ln\sqrt{1+x^2}}\left[C+\int\frac{2x}{\sqrt{1+x^2}}\cdot e^{-\ln\sqrt{1+x^2}}dx\right]$$

$$=\sqrt{1+x^2}\cdot\left[C+\int\frac{2x}{\sqrt{1+x^2}}\cdot\frac{1}{\sqrt{1+x^2}}dx\right]=\sqrt{1+x^2}\left[C+\ln(1+x^2)\right]$$

∴ 通解为: $\cos^2y=\sqrt{1+x^2}\left[C+\ln(1+x^2)\right]$.

20. 曲线 $y=f(x)$ 过原点，且在任一点处的切线斜率为 $2x+y$. 求该曲线的方程.

解: $\frac{dy}{dx}=2x+y$. $y(0)=0$. $f(x)=-1$. $\int f(x)dx=-x$. $g(x)=2x$.

$$y(x)=e^{x}\cdot\left[C+\int 2x\cdot e^{-x}dx\right]=e^{x}\cdot\left[C-2(x+1)e^{-x}\right]$$

$$=Ce^{x}-2(x+1).$$

$y(0)=0 \Rightarrow 0=C-2 \Rightarrow C=2$.

∴ 曲线方程: $y(x)=2e^{x}-2(x+1)$.

21. 曲线积分 $\int_{L} yf(x)dx+[2xf(x)-x^{2}]dy$ 在右半平面内与路径无关，$f(x)$ 在 $[0,+\infty)$ 上具有连续导数，且 $f(1)=1$. 求 $f(x)$ 的方程.

解: $\frac{\partial[yf(x)]}{\partial y}=\frac{\partial[2xf(x)-x^{2}]}{\partial x}$. $x\in[0,+\infty)$. $\left(\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}\right)$.

即: $f(x)=2f(x)+2x\frac{df(x)}{dx}-2x$. 令 $y=f(x)$. 则有:

$$\frac{dy}{dx}+\frac{1}{2x}\cdot y=1. \quad 且\ y(1)=1.$$

$$y(x)=e^{-\int\frac{1}{2x}dx}\left[C+\int 1\cdot e^{\int\frac{1}{2x}dx}dx\right]$$

$$=e^{-\frac{1}{2}\ln x}\cdot\left[C+\int e^{\frac{1}{2}\ln x}dx\right]=\frac{1}{\sqrt{x}}\left[C+\int\sqrt{x}\,dx\right]$$

$$=\frac{C}{\sqrt{x}}+\frac{2}{3}x. \quad 由\ y(1)=1 \Rightarrow 1=C+\frac{2}{3}. \Rightarrow C=\frac{1}{3}.$$

$$\therefore y=f(x)=\frac{1}{3\sqrt{x}}+\frac{2}{3}x.$$

22. 设函数 $f(t)$ 在 $[0,+\infty)$ 上具有连续导数，且满足

$$f(t)=e^{4\pi t^{2}}+\iint_{x^{2}+y^{2}\le 4t^{2}} f\left(\frac{1}{2}\sqrt{x^{2}+y^{2}}\right)dxdy. \quad 求\ f(t).$$

解: 令 $\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$ 则有: $\iint_{x^{2}+y^{2}\le 4t^{2}} f\left(\frac{1}{2}\sqrt{x^{2}+y^{2}}\right)dxdy=\iint_{\substack{0\le r\le 2t\\ 0\le\theta\le 2\pi}} f(r)\cdot 4r\,dr\,d\theta$

$$=8\pi\int_{0}^{t} r f(r)\,dr.$$

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则原表达式为: $f(t) = e^{4\pi t^2} + 8\pi \int_0^t \tau f(\tau)\, d\tau$.

上式两边对 $t$ 求导: $f'(t) = 8\pi t \cdot e^{4\pi t^2} + 8\pi t f(t)$.

即: $f'(t) - 8\pi t \cdot f(t) = 8\pi t \cdot e^{4\pi t^2}$

$$f(t) = e^{-\int(-8\pi t)dt} \cdot \left[ C + \int 8\pi t e^{4\pi t^2} \cdot e^{\int(-8\pi t)dt} dt \right]$$

$$= e^{4\pi t^2} \cdot \left[ C + 8\pi \int t \cdot e^{4\pi t^2} \cdot e^{-4\pi t^2} dt \right]$$

$$= e^{4\pi t^2} \cdot \left[ C + 4\pi t^2 \right].$$

且题有: $f(0) = 1 \Rightarrow 1 = C$.

$\therefore f(t) = (4\pi t^2 + 1) e^{4\pi t^2}$.

23. 求下列 Bernoulli 方程的通解:

(1) $\frac{dy}{dx} - \frac{2y}{x} = y^2$.

解: $f(x) = -\frac{2}{x}$, $g(x) = 1$, $n = 2$. 令 $u = \frac{1}{y}$. 则 $\frac{du}{dx} = -\frac{1}{y^2}\frac{dy}{dx}$.

代入原方程: $-y^2 \frac{du}{dx} - y^2 \frac{2u}{x} = y^2$. 即: $\frac{du}{dx} + \frac{2}{x} u = -1$.

$$u(x) = e^{-\int \frac{2}{x} dx} \cdot \left[ C + \int (-1) \cdot e^{\int \frac{2}{x} dx} dx \right]$$

$$= e^{-2\ln x} \cdot \left[ C - \int e^{2\ln x} dx \right] = \frac{1}{x^2} \left[ C - \int x^2 dx \right]$$

$$= \frac{1}{x^2} \left[ C - \frac{1}{3} x^3 \right] = \frac{C}{x^2} - \frac{1}{3} x = \frac{1}{y}.$$

$\therefore y(x) = \frac{3x^2}{3C - x^3}$.

(2) $\frac{dy}{dx} - 3xy = xy^2$.

解: $f(x) = -3x$. $g(x) = x$. $n = 2$. 令 $u = \frac{1}{y}$. 则 $\frac{du}{dx} = -\frac{1}{y^2}\frac{dy}{dx}$.

代入原方程: $-y^2 \frac{du}{dx} - 3xu \cdot y^2 = xy^2$. 即: $\frac{du}{dx} + 3xu = -x$.

$$u(x) = e^{-\int 3x\,dx}\cdot\left[C+\int(-x)\cdot e^{\int 3x\,dx}\,dx\right]$$

$$= e^{-\frac{3}{2}x^2}\cdot\left[C-\int x\cdot e^{\frac{3}{2}x^2}\,dx\right] = e^{-\frac{3}{2}x^2}\cdot\left[C-\frac{1}{3}\int e^{\frac{3}{2}x^2}\,d\left(\frac{3}{2}x^2\right)\right]$$

$$= e^{-\frac{3}{2}x^2}\cdot\left[C-\frac{1}{3}e^{\frac{3}{2}x^2}\right] = Ce^{-\frac{3}{2}x^2}-\frac{1}{3} = \frac{1}{y}.$$

$$\therefore\ y(x) = \frac{3}{3Ce^{-\frac{3}{2}x^2}-1} = \frac{3e^{\frac{3}{2}x^2}}{3C-e^{\frac{3}{2}x^2}}.$$

(3) $\frac{dy}{dx}+y = y^3(\cos x-\sin x)$

解: $f(x)=1$, $g(x)=\cos x-\sin x$, $n=3$. 令 $u=\frac{1}{y^2}$. $\frac{du}{dx} = -\frac{2}{y^3}\frac{dy}{dx}$.

代入原方程: $-\frac{y^3}{2}\frac{du}{dx}+y^3\cdot u = y^3(\cos x-\sin x)$.

即: $\frac{du}{dx}-2u = 2(\sin x-\cos x)$

$$u(x) = e^{-\int(-2)dx}\cdot\left[C+\int 2(\sin x-\cos x)\cdot e^{\int(-2)dx}\,dx\right]$$

$$= e^{2x}\cdot\left[C+2\int(\sin x-\cos x)\cdot e^{-2x}\,dx\right] = e^{2x}\cdot[C+I].$$

设 $I = 2\int(\sin x-\cos x)e^{-2x}dx = \int(\cos x-\sin x)\,de^{-2x}$

$$= (\cos x-\sin x)e^{-2x}-\int e^{-2x}(-\sin x-\cos x)\,dx$$

$$= (\cos x-\sin x)e^{-2x}-\frac{1}{2}\int(\sin x+\cos x)\,de^{-2x}$$

$$= (\cos x-\sin x-\frac{1}{2}\sin x-\frac{1}{2}\cos x)e^{-2x}+\frac{1}{2}\int(\cos x-\sin x)e^{-2x}dx$$

$$= (\frac{1}{2}\cos x-\frac{3}{2}\sin x)\cdot e^{-2x}-\frac{1}{4}I.$$

$$\therefore\ I = \frac{4}{5}(\frac{1}{2}\cos x-\frac{3}{2}\sin x)\cdot e^{-2x} = (\frac{2}{5}\cos x-\frac{6}{5}\sin x)\cdot e^{-2x}.$$

$$u(x) = e^{2x}\cdot[C+I] = Ce^{2x}+(\frac{2}{5}\cos x-\frac{6}{5}\sin x) = \frac{1}{y^2}.$$

$$\therefore\ \text{通解:}\ \frac{1}{y^2}-\frac{2}{5}\cos x+\frac{6}{5}\sin x)-Ce^{2x} = 0.$$

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(4) $3\frac{dy}{dx}+y=(1-2x)y^4$.

解: $\frac{dy}{dx}+\frac{1}{3}y=\frac{1-2x}{3}y^4$. $f(x)=\frac{1}{3}$. $g(x)=\frac{1-2x}{3}$. $n=4$. 令 $u=\frac{1}{y^3}$.

$\frac{du}{dx}=-\frac{3}{y^4}\frac{dy}{dx}$. 代入原方程,得: $-\frac{1}{3}y^4\cdot\frac{du}{dx}+y^4\cdot\frac{1}{3}u=\frac{1-2x}{3}y^4$.

即: $\frac{du}{dx}-u=2x-1$.

$u(x)=e^{-\int(-1)dx}\cdot\left[C+\int(2x-1)e^{\int(-1)dx}dx\right]$

$=e^{x}\cdot\left[C+\int(2x-1)e^{-x}dx\right]$

$=e^{x}\cdot\left[C-(2x-1)e^{-x}-2e^{-x}\right]=Ce^{x}+2x+1=\frac{1}{y^3}$.

∴ 原方程: $\frac{1}{y^3}=2x+1+Ce^{x}$.

(5) $\frac{dy}{dx}+\frac{y}{x}=y^2\ln x$.

解: $f(x)=\frac{1}{x}$. $g(x)=\ln x$. $n=2$. 令 $u=\frac{1}{y}$. 则: $\frac{du}{dx}=-\frac{1}{y^2}\frac{dy}{dx}$.

代入原方程,得: $-y^2\frac{dy}{dx}+y^2\cdot\frac{u}{x}=y^2\ln x$. 即: $\frac{dy}{dx}-\frac{1}{x}\cdot u=-\ln x$.

$u(x)=e^{-\int(-\frac{1}{x})dx}\cdot\left[C+\int(-\ln x)\cdot e^{\int(-\frac{1}{x})dx}dx\right]$

$=e^{\ln x}\cdot\left[C-\int\ln x\cdot e^{-\ln x}dx\right]=x\cdot\left[C-\int\frac{\ln x}{x}dx\right]$

$=x\cdot\left[C-\int\ln x\,d\ln x\right]=x\cdot\left[C-\frac{1}{2}\ln^2 x\right]=\frac{1}{y}$.

∴ 原方程: $y(x)=\frac{2}{2Cx-x\ln^2 x}$.

(6) $[y+xy^3(1+\ln x)]dx-xdy=0$.

解: $\frac{dy}{dx}-\frac{y}{x}=(1+\ln x)\cdot y^3$. $f(x)=-\frac{1}{x}$. $g(x)=1+\ln x$. $n=3$. 令 $u=\frac{1}{y^2}$.

则: $\frac{du}{dx}=-\frac{2}{y^3}\cdot\frac{dy}{dx}$. 代入原方程,得: $-\frac{1}{2}y^3\frac{du}{dx}-y^3\cdot\frac{u}{x}=(1+\ln x)\cdot y^3$.

即: $\frac{du}{dx}+\frac{2}{x}u=-2(1+\ln x)$.

$$u(x)=e^{-\int\frac{2}{x}dx}\cdot\left[C+\int[-2(1+\ln x)]\cdot e^{\int\frac{2}{x}dx}dx\right]$$

$$=e^{-2\ln x}\cdot\left[C-2\int(1+\ln x)\cdot e^{2\ln x}dx\right]$$

$$=\frac{1}{x^2}\cdot\left[C-2\int(1+\ln x)\cdot x^2dx\right]$$

$$=\frac{1}{x^2}\cdot\left[C-\frac{2}{3}x^3-\frac{2}{3}x^3\ln x+\frac{2}{9}x^3\right]=\frac{C}{x^2}-\frac{4}{9}x-\frac{2x}{3}\ln x=\frac{1}{y^2}.$$

∴ 通解: $y^2(-4x^3+6x^3\ln x-9C)+9x^2=0$.

24. 求微分方程 $y'\cos y-\cos x\cdot\sin^2y=\sin y$ 的通解.

解: $\frac{dy}{dx}\cdot\cos y-\cos x\cdot\sin^2y=\sin y$. 设 $u=\sin y$. 则有: $\frac{du}{dx}=\frac{dy}{dx}\cdot\cos y$.

$\frac{du}{dx}-\cos x\cdot u^2=u$. 即: $\frac{du}{dx}-u=\cos x\cdot u^2$.

$f(x)=-1$. $g(x)=\cos x$. $n=2$. 令 $v=\frac{1}{u}$. 则: $\frac{dv}{dx}=-\frac{1}{u^2}\frac{du}{dx}$.

代入得: $-u^2\frac{dv}{dx}-u^2\cdot v=\cos x\cdot u^2$. 即: $\frac{dv}{dx}+v=-\cos x$.

$$v(x)=e^{-\int1\cdot dx}\cdot\left[C+\int(-\cos x)\cdot e^{\int1\cdot dx}dx\right]$$

$$=e^{-x}\left[C-\int\cos x\cdot e^xdx\right]=e^{-x}\left[C-I\right].$$

$$I=\int\cos x\cdot e^xdx=\int\cos x\,de^x=\cos x\cdot e^x+\int\sin x\cdot e^xdx$$

$$=\cos x\cdot e^x+\int\sin x\,de^x=(\cos x+\sin x)\cdot e^x-\int\cos x\cdot e^xdx$$

$$=(\sin x+\cos x)\cdot e^x-I.\quad\therefore\quad I=\frac{1}{2}(\sin x+\cos x)\cdot e^x$$

即: $v(x)=e^{-x}\left[C-\frac{1}{2}(\sin x+\cos x)e^x\right]=Ce^{-x}-\frac{1}{2}(\sin x+\cos x)=\frac{1}{u}$.

∴ 通解: $\sin y=\dfrac{1}{Ce^{-x}-\frac{1}{2}(\sin x+\cos x)}=\dfrac{2e^x}{2C-e^x(\sin x+\cos x)}$.

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25. 设函数 $y=y(x)$ 是微分方程 $\frac{dy}{dx}+ay=be^{-cx}$ $(a>0,\ c>0)$ 的解，证明：

$$\lim_{x\to+\infty} y(x)=0.$$

证明：$\frac{dy}{dx}+ay=be^{-cx}$ 是线性方程，$f(x)=a$，$g(x)=be^{-cx}$.

$$y(x)=e^{-\int a\,dx}\left[C_1+\int be^{-cx}\cdot e^{\int a\,dx}\,dx\right]$$

$$=e^{-ax}\left[C_1+b\int e^{-cx}e^{ax}\,dx\right]=e^{-ax}\left[C_1+\frac{b}{a-c}e^{(a-c)x}\right]$$

$$=C_1e^{-ax}+\frac{b}{a-c}e^{-cx}.$$

$\because a>0,\ c>0$，$\therefore \lim_{x\to+\infty} y(x)=\lim_{x\to+\infty}\left(C_1e^{-ax}+\frac{b}{a-c}e^{-cx}\right)=0.$

26. 设函数 $f(t)$ 在 $[1,+\infty)$ 上具有连续导数，若曲线 $y=f(x)$，直线 $x=1$，$x=t$ $(t>1)$ 与 $x$ 轴围成的平面图形绕 $x$ 轴旋转一周所成的旋转体体积为：

$$V=\frac{\pi}{3}\left[t^2f(t)-f(1)\right].$$

试求 $f(x)$ 满足的微分方程，并求该方程的满足条件 $y\big|_{x=2}=\frac{2}{9}$ 的解.

解：$y=f(x)$，$x=1$，$x=t$，$(t>1)$ 绕 $x$ 轴旋转一周所围旋转体体积：

$$V=\int_1^t \pi f^2(x)\,dx=\frac{\pi}{3}\left[t^2f(t)-f(1)\right]$$

两边对 $t$ 求导，得：$\pi f^2(t)=\frac{\pi}{3}\left[2tf(t)+t^2f'(t)\right]$ 记 $y=f(t)$.

即：$3y^2=2ty+t^2\frac{dy}{dt}\Rightarrow \frac{dy}{dt}+\frac{2}{t}y=\frac{3}{t^2}y^2.$

记 $u=\frac{1}{y}$，则 $\frac{du}{dt}=-\frac{1}{y^2}\frac{dy}{dt}\Rightarrow -y^2\frac{du}{dt}+y^2\frac{2}{t}u=\frac{3}{t^2}y^2$

即：$\frac{du}{dt}-\frac{2}{t}u=-\frac{3}{t^2}.$

$$u(t)=e^{+\int\frac{2}{t}dt}\left[C+\int\left(-\frac{3}{t^2}\right)e^{\int-\frac{2}{t}dt}\,dt\right]=e^{2\ln t}\left[C-\int\frac{3}{t^2}e^{-2\ln t}\,dt\right]$$

$$=t^2\left[C-\int\frac{3}{t^2}\cdot t^2\,dt\right]=Ct^2+\frac{1}{t}=\frac{1+Ct^3}{t}=\frac{1}{y}.$$

$\therefore y=f(t)=\frac{t}{1+Ct^3}$ 由 $y\big|_{t=2}=\frac{2}{9}\Rightarrow\frac{2}{9}=\frac{2}{1+8C}\Rightarrow C=1.$

$\therefore y=f(x)=\frac{x}{1+x^3}.$

## §3. 二阶线性微分方程

习题 P. 203.

1. 求下列二阶常系数齐次线性微分方程的通解:

(1) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$.

解: $\lambda^2 + 5\lambda + 6 = 0 \Rightarrow \lambda_1 = -2, \lambda_2 = -3$.

$\therefore y(x) = C_1 e^{-2x} + C_2 e^{-3x}$.

(2) $\frac{d^2y}{dx^2} - 9y = 0$.

解: $\lambda^2 - 9 = 0 \Rightarrow \lambda_1 = 3, \lambda_2 = -3$.

$\therefore y(x) = C_1 e^{-3x} + C_2 e^{3x}$.

(3) $\frac{d^2y}{dx^2} + 4y = 0$.

解: $\lambda^2 + 4 = 0 \Rightarrow \lambda_{1,2} = \pm 2i$.

$\therefore y(x) = C_1 \cos 2x + C_2 \sin 2x$.

(4) $4\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 25y = 0$.

解: $4\lambda^2 - 20\lambda + 25 = 0 \Rightarrow \lambda_1 = \lambda_2 = \frac{5}{2}$.

$\therefore y(x) = (C_1 + C_2 x) e^{\frac{5}{2}x}$.

(5) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$.

解: $\lambda^2 - 6\lambda + 9 = 0 \Rightarrow \lambda_1 = \lambda_2 = 3$.

$\therefore y(x) = (C_1 + C_2 x) e^{3x}$.

(6) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 0$.

解: $\lambda^2 + 4\lambda = 0 \Rightarrow \lambda_1 = 0, \lambda_2 = -4$.

$\therefore y(x) = C_1 + C_2 e^{-4x}$.

(7) $\frac{d^3y}{dx^3} - 5\frac{d^2y}{dx^2} + 6\frac{dy}{dx} = 0$

解: $\lambda^3-5\lambda^2+6\lambda=0 \Rightarrow \lambda_1=0,\ \lambda_2=2,\ \lambda_3=3.$

$\therefore y(x)=C_1+C_2e^{2x}+C_3e^{3x}.$

(8) $\frac{d^4y}{dx^4}-ay=0.$

解: $\lambda^4-a=0 \Rightarrow \lambda_{1,2}=\pm\sqrt[4]{a},\ \lambda_{3,4}=\pm\sqrt[4]{a}\,i.\quad (a>0)$

当 $a\le 0$ 时: $\lambda^2=\pm\sqrt{-a}\cdot i \Rightarrow \lambda_{1,2}=\sqrt[4]{-a}\cdot\sqrt{i},\quad \lambda_{3,4}=\sqrt[4]{-a}\cdot i\cdot\sqrt{i}$

$i=\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}=e^{i\cdot\frac{\pi}{2}}.\quad \sqrt{i}=e^{i\frac{\pi}{4}}=\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}=\frac{1}{2}(\sqrt{2}+i\sqrt{2})$

即当 $a\le 0$ 时: $\lambda_{1,2}=\frac{\sqrt{2}}{2}\sqrt[4]{-a}\cdot(1+i),\quad \lambda_{3,4}=\frac{\sqrt{2}}{2}\sqrt[4]{-a}\cdot(-1+i)$

$\therefore$ 当 $a>0$ 时, 通解: $y(x)=C_1e^{\sqrt[4]{a}x}+C_2e^{-\sqrt[4]{a}x}+C_3\sin\sqrt[4]{a}x+C_4\cos\sqrt[4]{a}x.$

当 $a\le 0$ 时, 通解:

$$y(x)=e^{\frac{\sqrt{2}}{2}\sqrt[4]{-a}}\left(C_1\cos\frac{\sqrt{2}}{2}\sqrt[4]{-a}x+C_2\sin\frac{\sqrt{2}}{2}\sqrt[4]{-a}x\right)+e^{-\frac{\sqrt{2}}{2}\sqrt[4]{-a}}\left(C_3\cos\frac{\sqrt{2}}{2}\sqrt[4]{-a}x+C_4\sin\frac{\sqrt{2}}{2}\sqrt[4]{-a}x\right)$$

(9) $\frac{d^4y}{dx^4}+2\frac{d^2y}{dx^2}+y=0.$

解: $\lambda^4+2\lambda^2+1=0 \Rightarrow \lambda_{1,2}=i,\quad \lambda_{3,4}=-i.$

$\therefore y(x)=(C_1+C_2x)\cos x+(C_3+C_4x)\sin x.$

(10) $\frac{d^4y}{dx^4}-2\frac{d^3y}{dx^3}+\frac{d^2y}{dx^2}=0.$

解: $\lambda^4-2\lambda^3+\lambda^2=0 \Rightarrow \lambda_{1,2}=0,\quad \lambda_{3,4}=1.$

$\therefore y(x)=C_1+C_2x+(C_3+C_4x)e^x$

2. 求下列二阶常系数齐次线性微分方程的特解:

(1) $\frac{d^2y}{dx^2}+4y=0,\quad y(0)=0,\quad y'(0)=1.$

解: $\lambda^2+4=0 \Rightarrow \lambda_1=2i,\quad \lambda_2=-2i.$

$y(x)=C_1\cos 2x+C_2\sin 2x,\quad y'(x)=-2C_1\sin 2x+2C_2\cos 2x$

$$\begin{cases} y(0)=0 \\ y'(0)=1 \end{cases} \Rightarrow \begin{cases} C_1=0 \\ 2C_2=1 \end{cases} \Rightarrow \begin{cases} C_1=0 \\ C_2=\frac{1}{2} \end{cases} \quad \therefore y(x)=\frac{1}{2}\sin 2x.$$

(2) $\frac{d^2y}{dx^2} - 9y = 0$. $y(0) = 2$, $y'(0) = 1$.

解: $\lambda^2 - 9 = 0$. $\Rightarrow$ $\lambda_1 = 3$. $\lambda_2 = -3$.

$y(x) = C_1 e^{3x} + C_2 e^{-3x}$. $y'(x) = 3C_1 e^{3x} - 3C_2 e^{-3x}$.

$\begin{cases} y(0) = 2 \\ y'(0) = 1 \end{cases} \Rightarrow \begin{cases} C_1 + C_2 = 2 \\ 3C_1 - 3C_2 = 1 \end{cases} \Rightarrow \begin{cases} C_1 = \frac{7}{6} \\ C_2 = \frac{5}{6} \end{cases}$.

$\therefore y(x) = \frac{7}{6} e^{3x} + \frac{5}{6} e^{-3x}$.

(3) $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$. $y(0) = -1$. $y'(0) = 0$.

解: $\lambda^2 + \lambda + 1 = 0$. $\Rightarrow$ $\lambda_{1,2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$.

$y(x) = e^{-\frac{1}{2}x}\left(C_1 \cos\frac{\sqrt{3}}{2}x + C_2 \sin\frac{\sqrt{3}}{2}x\right)$. $y'(x) = e^{-\frac{1}{2}x}\left(\frac{-C_1+\sqrt{3}C_2}{2}\cos\frac{\sqrt{3}}{2}x - \frac{C_2+\sqrt{3}C_1}{2}\sin\frac{\sqrt{3}}{2}x\right)$

$\begin{cases} y(0) = -1 \\ y'(0) = 0 \end{cases} \Rightarrow \begin{cases} C_1 = -1 \\ \frac{-C_1+\sqrt{3}C_2}{2} = 0 \end{cases} \Rightarrow \begin{cases} C_1 = -1 \\ C_2 = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3} \end{cases}$.

$\therefore y(x) = -e^{-\frac{x}{2}}\left(\cos\frac{\sqrt{3}}{2}x + \frac{\sqrt{3}}{3}\sin\frac{\sqrt{3}}{2}x\right)$

(4) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$. $y(0) = 2$. $y'(0) = 0$.

解: $\lambda^2 - 4\lambda + 4 = 0$. $\Rightarrow$ $\lambda_1 = \lambda_2 = 2$.

$y(x) = (C_1 + C_2 x)e^{2x}$. $y'(x) = (2C_1 + C_2 + 2C_2 x)e^{2x}$.

$\begin{cases} y(0) = 2 \\ y'(0) = 0 \end{cases} \Rightarrow \begin{cases} C_1 = 2 \\ 2C_1 + C_2 = 0 \end{cases} \Rightarrow \begin{cases} C_1 = 2 \\ C_2 = -4 \end{cases}$.

$\therefore y(x) = (2 - 4x)e^{2x}$.

(5) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} - 6y = 0$. $y(0) = 5$. $y'(0) = -1$.

解: $\lambda^2 + 5\lambda - 6 = 0$. $\Rightarrow$ $\lambda_1 = -6$. $\lambda_2 = 1$.

$y(x) = C_1 e^{-6x} + C_2 e^{x}$. $\Rightarrow$ $y'(x) = -6C_1 e^{-6x} + C_2 e^{x}$.

$\begin{cases} y(0) = 5 \\ y'(0) = -1 \end{cases} \Rightarrow \begin{cases} C_1 + C_2 = 5 \\ -6C_1 + C_2 = -1 \end{cases} \Rightarrow \begin{cases} C_1 = \frac{6}{7} \\ C_2 = \frac{29}{7} \end{cases}$.

$\therefore y(x) = \frac{6}{7} e^{-6x} + \frac{29}{7} e^{x}$.

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(6) $\frac{d^3y}{dx^3} - 5\frac{d^2y}{dx^2} + 6\frac{dy}{dx} = 0$ $y(0)=0$ $y'(0)=1$ $y''(0)=2$.

解: $\lambda^3 - 5\lambda^2 + 6\lambda = 0 \Rightarrow \lambda_1 = 0 \quad \lambda_2 = 2 \quad \lambda_3 = 3$.

$y(x) = C_1 + C_2 e^{2x} + C_3 e^{3x}$. $y'(x) = 2C_2 e^{2x} + 3C_3 e^{3x}$

$y''(x) = 4C_2 e^{2x} + 9C_3 e^{3x}$

$$\begin{cases} y(0)=0 \\ y'(0)=1 \\ y''(0)=2 \end{cases} \Rightarrow \begin{cases} C_1 + C_2 + C_3 = 0 \\ 2C_2 + 3C_3 = 1 \\ 4C_2 + 9C_3 = 2 \end{cases} \Rightarrow C_1 = -\frac{1}{2} \quad C_2 = \frac{1}{2} \quad C_3 = 0$$

$\therefore y(x) = -\frac{1}{2} + \frac{1}{2}e^{2x}$.

3. 已知齐次线性方程 $(x-1)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$ 的一个解是 $y = x$, 求非齐次线性方程 $(x-1)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = (x-1)^2$ 的通解.

解: 已知 $y = x$ 是齐次方程 $(x-1)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$ 的一个解.

通过观察可以发现 $y = e^x$ 也是上述齐次方程的解.

显然, $x$ 与 $e^x$ 线性无关. $\therefore$ 齐次方程的通解为: $C_1 x + C_2 e^x$.

由于非齐次方程的系数均为一次多项式, 非齐次项 $(x-1)^2$ 为二次多项式.

$\therefore$ 可设非齐次方程有特解形为: $P_2(x) = a_0 + a_1 x + a_2 x^2$.

代入非齐次方程, 得: $(x-1)2a_2 - x(a_1 + 2a_2 x) + a_0 + a_1 x + a_2 x^2 = (x-1)^2$

对比同次幂系数可得: $a_0 = -1$. $a_1$ 可取任何数 $a_2 = -1$.

$\therefore$ 取非齐次方程特解, $y^*(x) = -1 - x^2$.

则非齐次方程通解为: $y(x) = C_1 x + C_2 e^x - 1 - x^2$.

4. 已知 $y = e^x$ 是齐次线性方程 $(2x-1)\frac{d^2y}{dx^2} - (2x+1)\frac{dy}{dx} + 2y = 0$ 的一个解, 求它的通解.

解: 齐次方程的系数均为 $P_1(x)$. $\therefore$ 可设另一特解形为 $Q_1(x) = a_0 + a_1 x$.

代入方程, 得: $-(2x+1)\cdot a_1 + 2(a_0 + a_1 x) = 0 \Rightarrow a_1 = 2a_0$.

取 $a_0 = 1$. $a_1 = 2$. $\therefore Q_1(x) = 1 + 2x$.

显然 $e^x$ 与 $2x+1$ 线性无关.

$\therefore$ 齐次方程的通解为: $y(x) = C_1 e^x + C_2(1+2x)$.

5. 求非齐次线性方程 $\frac{d^2y}{dx^2}+y=\sec x$ 的通解.

解: 齐次方程 $\frac{d^2y}{dx^2}+y=0$ 的特征方程 $\lambda^2+1=0 \Rightarrow \lambda_{1,2}=\pm i$.

∴ 齐次方程通解为: $C_1\cos x + C_2\sin x$.

设非齐次方程具有形如 $y^*(x)=u(x)\cdot\cos x$ 的特解, 代入原方程:

$y^{*\prime}(x)=u'(x)\cdot\cos x - u(x)\sin x$.

$y^{*\prime\prime}(x)=u''(x)\cdot\cos x - 2u'(x)\sin x - u(x)\cos x$.

$y^{*\prime\prime}(x)+y^*(x)=u''(x)\cdot\cos x - 2u'(x)\cdot\sin x = \sec x$. 设 $v(x)=u'(x)$.

则有: $\frac{dv}{dx}\cos x - 2\sin x\cdot v = \sec x$. 即: $\frac{dv}{dx}-\frac{2\sin x}{\cos x}v=\frac{1}{\cos^2 x}$.

$$\therefore v(x)=e^{\int\frac{2\sin x}{\cos x}dx}\left[C+\int\frac{1}{\cos^2 x}\cdot e^{-\int\frac{2\sin x}{\cos x}dx}dx\right]$$

$$=e^{-2\ln\cos x}\left[C+\int\frac{1}{\cos^2 x}\cdot e^{2\ln\cos x}dx\right]$$

$$=\frac{1}{\cos^2 x}\left[C+\int\frac{1}{\cos^2 x}\cdot\cos^2 x\,dx\right]=\frac{1}{\cos^2 x}(C+x)=\sec^2 x(C+x)$$

$$u(x)=\int v(x)\,dx=\int\sec^2 x(C+x)dx=C\cdot\tan x+\int x\,d\tan x$$

$$=C\cdot\tan x+x\cdot\tan x-\int\frac{\sin x}{\cos x}dx=(C+x)\tan x+\ln\cos x$$

$$\therefore y^*(x)=u(x)\cdot\cos x=(C+x)\sin x+\cos x\cdot\ln\cos x.$$

∴ 原方程的通解为: $y(x)=C_1\cos x+C_2\sin x+x\sin x+\cos x\cdot\ln\cos x$.

6. 求下列二阶常系数非齐次线性方程的通解:

(1) $\frac{d^3y}{dx^3}-5\frac{d^2y}{dx^2}+6\frac{dy}{dx}=x^3-2x+1$.

解: $\lambda^3-5\lambda^2+6\lambda=0 \Rightarrow \lambda_1=0,\ \lambda_2=2,\ \lambda_3=3$.

齐次方程通解: $y=C_1+C_2e^{2x}+C_3e^{3x}$.

设非齐次方程特解: $y^*=a_3x^3+a_2x^2+a_1x+a_0+a_4x^4$, 代入原方程:

$6a_3-5(6a_4x+2a_2)+6(3a_3x^2+2a_2x+a_1)=x^3-2x+1$

即: $6a_3-10a_2+6a_1=1$

$(24a_4x+6a_3)-5(6a_3x+2a_2+12a_4x^2)+6(3a_3x^2+2a_2x+a_1+4a_4x^3)=x^3-2x+1.$

$$\Rightarrow \begin{cases} 24a_4=1 \\ -60a_4+18a_3=0 \\ 24a_4-30a_3+12a_2=-2 \\ 6a_3-10a_2+6a_1=1 \end{cases} \Rightarrow a_1=\frac{41}{216},\quad a_2=\frac{7}{72},\quad a_3=\frac{5}{36},\quad a_4=\frac{1}{24}.$$

∴ 通解: $y(x)=C_1+C_2e^{2x}+C_3e^{3x}+\frac{41}{216}x+\frac{7}{72}x^2+\frac{5}{36}x^3+\frac{1}{24}x^4.$

(2) $\frac{d^2y}{dx^2}-3\frac{dy}{dx}=x.$

解: $\lambda^2-3\lambda=0 \Rightarrow \lambda_1=0,\ \lambda_2=3.$ 齐次方程通解: $y=C_1+C_2e^{3x}.$

设非齐次方程特解: $y^*(x)=a_1x+a_2x^2.$ 代入原方程,得:

$2a_2-3(a_1+2a_2x)=x \Rightarrow \begin{cases} 2a_2-3a_1=0 \\ -6a_2=1 \end{cases} \Rightarrow \begin{cases} a_1=-\frac{1}{9} \\ a_2=-\frac{1}{6} \end{cases}$

∴ 通解: $y(x)=C_1+C_2e^{3x}-\frac{1}{9}x-\frac{1}{6}x^2.$

(3) $\frac{d^2y}{dx^2}-9y=e^{3x}(x^2+1).$

解: $\lambda^2-9=0 \Rightarrow \lambda_1=3,\ \lambda_2=-3.$ 齐次方程通解: $y=C_1e^{3x}+C_2e^{-3x}.$

设非齐次方程特解: $y^*(x)=(a_0+a_1x+a_2x^2)xe^{3x}=(a_0x+a_1x^2+a_2x^3)e^{3x}.$

代入原方程,得: $\begin{cases} 2a_1+6a_0=1 \\ 6a_2+12a_1=0 \\ 18a_2=1 \end{cases} \Rightarrow a_2=\frac{1}{18},\ a_1=-\frac{1}{36},\ a_0=\frac{19}{108}.$

∴ 通解: $y(x)=C_1e^{3x}+C_2e^{-3x}+\left(\frac{19}{108}-\frac{1}{36}x+\frac{1}{18}\right)xe^{3x}.$

(4) $\frac{d^2y}{dx^2}+4y=x\cos 2x.$

解: $\lambda^2+4=0 \Rightarrow \lambda_{1,2}=\pm 2i.$ 齐次方程通解: $y=C_1\cos 2x+C_2\sin 2x.$

设非齐次方程特解: $y^*(x)=(a_0+a_1x)x\cos 2x+(b_0+b_1x)x\sin 2x$

$=(a_0x+a_1x^2)\cos 2x+(b_0x+b_1x^2)\sin 2x.$

$y^{*\prime}(x)=[a_0+(2a_1+2b_0)x+2b_1x^2]\cos 2x+[b_0+(2b_1-2a_0)x-2a_1x^2]\sin 2x$

$y^{*\prime\prime}(x)=[(2a_1+4b_0)+(8b_1-4a_0)x-4a_1x^2]\cos 2x+[(2b_1-4a_0)-(8a_1+4b_0)x-4b_1x^2]\sin 2x.$

代入原方程方程得：$\begin{cases}(2b_1-4a_0)-(8a_1+4b_0)x-4b_1x^2+4(b_0x+b_1x^2)=0\\(2a_1+4b_0)+(8b_1-4a_0)x-4a_1x^2+4(a_0x+a_1x^2)=x\end{cases}$

即：$\begin{cases}2b_1-4a_0=0\\-8a_1=0\\2a_1+4b_0=0\\8b_1=1\end{cases}\Rightarrow\begin{cases}a_0=-\frac{1}{16}\\a_1=0\\b_0=0\\b_1=\frac{1}{8}\end{cases}$

∴通解：$y(x)=\frac{1}{16}x\cos 2x+\frac{1}{8}x^2\sin 2x+C_1\cos 2x+C_2\sin 2x$

(5) $\frac{d^2y}{dx^2}-6\frac{dy}{dx}+9y=xe^x\sin x$

解：$\lambda^2-6\lambda+9=0\Rightarrow\lambda_1=\lambda_2=3$. 齐次方程通解：$y(x)=(C_1+C_2x)e^{3x}$

设非齐次方程特解：$y^*(x)=e^x[(a_0+a_1x)\cos x+(b_0+b_1x)\sin x]$

$y^{*\prime}(x)=e^x\{[(a_0+a_1+b_0)+(a_1+b_1)x]\cos x+[(b_0+b_1-a_0)+(b_1-a_1)x]\sin x\}$

$y^{*\prime\prime}(x)=e^x\{[(2b_0+2a_1+2b_1)+2b_1x]\cos x+[(-2a_0-2a_1+2b_1)-2a_1x]\sin x\}$

代入原方程得：$\begin{cases}3a_0-4b_0-4a_1+2b_1=0\\3a_1-4b_1=0\\4a_0+3b_0-2a_1-4b_1=0\\4a_1+3b_1=1\end{cases}\Rightarrow\begin{cases}a_0=\frac{22}{125}\\a_1=\frac{4}{25}\\b_0=\frac{4}{125}\\b_1=\frac{3}{25}\end{cases}$

∴通解：$y(x)=(C_1+C_2x)e^{3x}+\left[\left(\frac{22}{125}+\frac{4}{25}x\right)\cos x+\left(\frac{4}{125}+\frac{3}{25}x\right)\sin x\right]e^x$

(6) $\frac{d^2y}{dx^2}-4\frac{dy}{dx}-5y=x\sin x$

解：$\lambda^2-4\lambda-5=0\Rightarrow\lambda_1=5,\ \lambda_2=-1$. 齐次方程通解：$y=C_1e^{5x}+C_2e^{-x}$

设非齐次方程特解：$y^*(x)=(a_0+a_1x)\cos x+(b_0+b_1x)\sin x$.

$y^{*\prime}(x)=[(a_1+b_0)+b_1x]\cos x+[(-a_0+b_1)-a_1x]\sin x$.

$y^{*\prime\prime}(x)=[(-a_0+2b_1)-a_1x]\cos x+[(-2a_1-b_0)-b_1x]\sin x$.

代入原方程得：$\begin{cases}-6a_0-4a_1-4b_0+2b_1=0\\-6a_1-4b_1=0\\4a_0-2a_1-6b_0-4b_1=0\\4a_1-6b_1=1\end{cases}\Rightarrow\begin{cases}a_0=-\frac{29}{338}\\a_1=\frac{1}{13}\\b_0=-\frac{1}{169}\\b_1=-\frac{3}{26}\end{cases}$

∴通解：$y(x)=C_1e^{5x}+C_2e^{-x}+\left[\left(-\frac{29}{338}+\frac{1}{13}x\right)\cos x+\left(-\frac{1}{169}-\frac{3}{26}x\right)\sin x\right]$.

(7) $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+6y=(x^2-2)\cos x$.

解: $\lambda^2+5\lambda+6=0 \Rightarrow \lambda_1=-2,\ \lambda_2=-3$. 齐次方程通解: $y=C_1e^{-2x}+C_2e^{-3x}$

设非齐次方程特解: $y^*(x)=(a_0+a_1x+a_2x^2)\cos x+(b_0+b_1x+b_2x^2)\sin x$.

则: $y^{*\prime}(x)=[(a_1+b_0)+(2a_2+b_1)x+b_2x^2]\cos x+[(-a_0+b_1)+(-a_1+2b_2)x-a_2x^2]\sin x$

$y^{*\prime\prime}(x)=[(-a_0+2a_2+2b_1)+(-a_1+4b_2)x-a_2x^2]\cos x+[(-2a_1-b_0+2b_2)+(-4a_2-b_1)x-b_2x^2]\sin x$.

代入原方程,得:

$$\begin{cases} 5a_2+5b_2=1 \\ 5a_1+10a_2+5b_1+4b_2=0 \\ 5a_0+5a_1+2a_2+5b_0+2b_1=-2 \\ -5a_2+5b_2=0 \\ -5a_1-4a_2+5b_1+10b_2=0 \\ -5a_0-2a_1+5b_0+5b_1+2b_2=0 \end{cases} \Rightarrow \begin{cases} a_0=-\frac{57}{250} \\ a_1=-\frac{2}{25} \\ a_2=\frac{1}{10} \\ b_0=-\frac{19}{250} \\ b_1=-\frac{1}{5} \\ b_2=\frac{1}{10} \end{cases}$$

∴通解: $y(x)=C_1e^{-2x}+C_2e^{-3x}+\left(-\frac{57}{250}-\frac{2}{25}x+\frac{1}{10}x^2\right)\cos x+\left(-\frac{19}{250}-\frac{1}{5}x+\frac{1}{10}x^2\right)\sin x$.

(8) $\frac{d^2y}{dx^2}+y=\cos 2x+e^x$.

解: $\lambda^2+1=0 \Rightarrow \lambda_{1,2}=\pm i$. 齐次方程通解: $y=C_1\cos x+C_2\sin x$.

对于 $f_1(x)=\cos 2x$, 设特解: $y_1(x)=A\cos 2x+B\sin 2x$. $y_1''(x)=-4A\cos 2x-4B\sin 2x$.

代入原方程得: $\begin{cases} -4A+A=1 \\ -4B+B=0 \end{cases} \Rightarrow \begin{cases} A=-\frac{1}{3} \\ B=0 \end{cases}$

即: $y_1(x)=-\frac{1}{3}\cos 2x$.

对于 $f_2(x)=e^x$, 设特解: $y_2(x)=ae^x$. $y_2''(x)=ae^x$.

代入原方程,得: $a+a=1 \Rightarrow a=\frac{1}{2}$.

即: $y_2(x)=\frac{1}{2}e^x$.

∴通解为: $y(x)=C_1\cos x+C_2\sin x+\frac{1}{2}e^x-\frac{1}{3}\cos 2x$.

(9) $\frac{d^2y}{dx^2}-y=x^2\sin x\cos x$.

解: $\lambda^2-1=0 \Rightarrow \lambda_1=1,\ \lambda_2=-1$. 齐次方程通解: $y=C_1e^x+C_2e^{-x}$.

$f(x)=x^2\sin x\cos x=\frac{1}{2}x^2\sin 2x$.

设非齐次方程特解：$y^*(x)=(a_0+a_1x+a_2x^2)\cos 2x+(b_0+b_1x+b_2x^2)\sin 2x$

则：$y^{*\prime}(x)=[(a_1+2b_0)+(2a_2+2b_1)x+2b_2x^2]\cos 2x+[(-2a_0+b_1)+(-2a_1+2b_2)x-2a_2x^2]\sin 2x$

$y^{*\prime\prime}(x)=[(-4a_0+2a_2+4b_1)+(-4a_1+8b_2)x-4a_2x^2]\cos 2x+[(-4a_1-4b_0+2b_2)+(-8a_2-4b_1)x-4b_2x^2]\sin 2x$

代入原方程，得：

$$\begin{cases}-4a_0+2a_2+4b_1-a_0=0\\-4a_1+8b_2-a_1=0\\-4a_2-a_2=0\\-4a_1-4b_0+2b_2-b_0=0\\-8a_2-4b_1-b_1=0\\-4b_2-b_2=\frac{1}{2}\end{cases}\Rightarrow\begin{cases}a_0=0\\a_1=-\frac{4}{25}\\a_2=0\\b_0=\frac{11}{125}\\b_1=0\\b_2=-\frac{1}{10}\end{cases}$$

∴ 通解为：$y(x)=C_1e^x+C_2e^{-x}-\frac{4}{25}x\cos 2x+\left(\frac{11}{125}-\frac{1}{10}x^2\right)\sin 2x$.

(10) $4\frac{d^2y}{dx^2}-20\frac{dy}{dx}+25y=\sin^2x$

解：$4\lambda^2-20\lambda+25=0\Rightarrow\lambda_1=\lambda_2=\frac{5}{2}$，齐次方程通解：$y=(C_1+C_2x)e^{\frac{5}{2}x}$

$f(x)=\sin^2x=\frac{1}{2}-\frac{1}{2}\cos 2x=f_1(x)+f_2(x)$.

对于 $f_1(x)=\frac{1}{2}$，有特解：$y_1(x)=a$，代入得：$25a=\frac{1}{2}$，$a=\frac{1}{50}$，∴ $y_1(x)=\frac{1}{50}$.

对于 $f_2(x)=-\frac{1}{2}\cos 2x$，设特解：$y_2(x)=A\cos 2x+B\sin 2x$.

$y_2'(x)=-2A\sin 2x+2B\cos 2x$，$y_2''(x)=-4A\cos 2x-4B\sin 2x$.

代入方程，得：$\begin{cases}4\cdot(-4A)+20\cdot(2B)+25A=-\frac{1}{2}\\4\cdot(-4B)-20\cdot(-2A)+25B=0\end{cases}\Rightarrow\begin{cases}A=-\frac{9}{3362}\\B=\frac{20}{1681}\end{cases}$

∴ 通解为：$y(x)=(C_1+C_2x)e^{\frac{5}{2}x}-\frac{9}{3362}\cos 2x+\frac{20}{1681}\sin 2x+\frac{1}{50}$.

7. 求下列二阶常系数非齐次线性方程的特解：

(1) $\frac{d^2y}{dx^2}-3\frac{dy}{dx}=x^2$，$y(0)=0$，$y'(0)=1$.

解：$\lambda^2-3\lambda=0\Rightarrow\lambda_1=0$，$\lambda_2=3$，齐次通解：$y=C_1+C_2e^{3x}$

设非齐次特解：$y^*(x)=a_0+a_1x+a_2x^2+a_3x^3$，$y^{*\prime}(x)=a_1+2a_2x+3a_3x^2$，$y^{*\prime\prime}(x)=2a_2+6a_3x$.

代入方程，得：$\begin{cases}2a_2-3a_1=0\\6a_3-6a_2=0\\-9a_3=1\end{cases}\Rightarrow\begin{cases}a_1=-\frac{2}{27}\\a_2=-\frac{1}{9}\\a_3=-\frac{1}{9}\end{cases}$

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∴ 通解为: $y(x)=C_1+C_2e^{3x}-\frac{2}{27}x-\frac{1}{9}x^2-\frac{1}{9}x^3$.

$\begin{cases}y(0)=0\\y'(0)=1\end{cases}\Rightarrow\begin{cases}C_1+C_2=0\\3C_2-\frac{2}{27}=1\end{cases}\Rightarrow\begin{cases}C_1=-\frac{29}{81}\\C_2=\frac{29}{81}\end{cases}$

∴ $y(x)=-\frac{29}{81}+\frac{29}{81}e^{3x}-\frac{2}{27}x-\frac{1}{9}x^2-\frac{1}{9}x^3$.

(2) $\frac{d^2y}{dx^2}-9y=e^{3x}(-x^2+1)$. $y(0)=-1$. $y'(0)=0$.

解: $\lambda^2-9=0\Rightarrow\lambda_1=3.\ \lambda_2=-3$. ∴ 齐次通解: $y=C_1e^{3x}+C_2e^{-3x}$.

设非齐次特解: $y^*(x)=(a_0+a_1x+a_2x^2)xe^{3x}=(a_0x+a_1x^2+a_2x^3)e^{3x}$

则: $y^{*\prime}(x)=\left[a_0+(2a_1+3a_0)x+(3a_2+3a_1)x^2+3a_2x^3\right]e^{3x}$

$y^{*\prime\prime}(x)=\left[(2a_1+6a_0)+(9a_0+12a_1+6a_2)x+(9a_1+18a_2)x^2+9a_2x^3\right]e^{3x}$.

代入方程为, 得: $\begin{cases}2a_1+6a_0=1\\9a_0+12a_1+6a_2-9a_0=0\\9a_1+18a_2-9a_1=1\end{cases}\Rightarrow a_0=\frac{19}{108}.\ a_1=-\frac{1}{36}.\ a_2=\frac{1}{18}$.

非齐次通解为: $y(x)=C_1e^{3x}+C_2e^{-3x}+\left(\frac{19}{108}-\frac{1}{36}x+\frac{1}{18}x^2\right)xe^{3x}$

$\begin{cases}y(0)=-1\\y'(0)=0\end{cases}\Rightarrow\begin{cases}C_1+C_2=-1\\3(C_1-C_2)+\frac{19}{108}=0\end{cases}\Rightarrow C_1=-\frac{343}{648}.\ C_2=-\frac{305}{648}$

∴ $y(x)=-\frac{343}{648}e^{3x}-\frac{305}{648}e^{-3x}+\left(\frac{19}{108}-\frac{1}{36}x+\frac{1}{18}x^2\right)xe^{3x}$

(3) $\frac{d^2y}{dx^2}+\frac{dy}{dx}+y=x\sin x$. $y(0)=2$, $y'(0)=1$.

解: $\lambda^2+\lambda+1=0\Rightarrow\lambda_{1,2}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$. 齐次通解: $y=e^{-\frac{1}{2}x}\left(C_1\cos\frac{\sqrt{3}}{2}x+C_2\sin\frac{\sqrt{3}}{2}x\right)$.

设非齐次特解: $y^*(x)=(a_0+a_1x)\cos x+(b_0+b_1x)\sin x$.

则: $y^{*\prime}(x)=\left[(a_1+b_0)+b_1x\right]\cos x+\left[(-a_0+b_1)-a_1x\right]\sin x$

$y^{*\prime\prime}(x)=\left[(-a_0+2b_1)-a_1x\right]\cos x+\left[(-2a_1-b_0)-b_1x\right]\sin x$.

代入原方程为, 得: $\begin{cases}-a_0+2b_1+a_1+b_0+a_0=0\\-a_1+b_1+a_1=0\\-2a_1-b_0-a_0+b_1+b_0=0\\-b_1-a_1+b_1=1\end{cases}\Rightarrow\begin{cases}a_0=2\\a_1=-1\\b_0=1\\b_1=0\end{cases}$

非齐次通解为：$y(x)=e^{-\frac{1}{2}x}\left(C_1\cos\frac{\sqrt{3}}{2}x+C_2\sin\frac{\sqrt{3}}{2}x\right)+(2-x)\cos x+\sin x$

$$\begin{cases} y(0)=2 \\ y'(0)=1 \end{cases} \Rightarrow \begin{cases} C_1=0 \\ \frac{\sqrt{3}}{2}C_2=1 \end{cases} \Rightarrow \begin{array}{l} C_1=0 \\ C_2=\frac{2}{\sqrt{3}} \end{array}$$

$$\therefore\ y(x)=e^{-\frac{1}{2}x}\cdot\frac{2}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}x+(2-x)\cos x+\sin x.$$

(4) $\frac{d^2y}{dx^2}-6\frac{dy}{dx}+9y=xe^x\sin x$，$y(0)=2$，$y'(0)=0$.

解：由6(5)可知其通解为：

$$y(x)=(C_1+C_2x)e^{3x}+\left[\left(\frac{22}{125}+\frac{4}{25}x\right)\cos x+\left(\frac{4}{125}+\frac{3}{25}x\right)\sin x\right]e^x$$

$$\begin{cases} y(0)=2 \\ y'(0)=0 \end{cases} \Rightarrow \begin{cases} C_1+\frac{22}{125}=2 \\ C_2+3C_1+\frac{46}{125}=0 \end{cases} \Rightarrow C_1=\frac{228}{125},\quad C_2=-\frac{146}{25}$$

$$\therefore\ y(x)=\left(\frac{228}{125}-\frac{146}{25}x\right)e^{3x}+\left[\left(\frac{22}{125}+\frac{4}{25}x\right)\cos x+\left(\frac{4}{125}+\frac{3}{25}x\right)\sin x\right]e^x.$$

(5) $\frac{d^2y}{dx^2}+\frac{dy}{dx}+y=\cos^2x$，$y(0)=2$，$y'(0)=1$.

解：$\lambda^2+\lambda+1=0 \Rightarrow \lambda_{1,2}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$. 齐次通解：$y=e^{-\frac{1}{2}x}\left(C_1\cos\frac{\sqrt{3}}{2}x+C_2\sin\frac{\sqrt{3}}{2}x\right)$.

$f(x)=\cos^2x=\frac{1}{2}+\frac{1}{2}\cos2x=f_1(x)+f_2(x)$.

对于$f_1(x)=\frac{1}{2}$，有特解$y_1(x)=a$，代入可得：$a=\frac{1}{2}$，$\therefore y_1(x)=\frac{1}{2}$.

对于$f_2(x)=\frac{1}{2}\cos2x$，设特解$y_2(x)=A\cos2x+B\sin2x$.

$y_2'(x)=2B\cos2x-2A\sin2x$，$y_2''(x)=-4A\cos2x-4B\sin2x$

代入原方程可得：$\begin{cases} A+2B-4A=\frac{1}{2} \\ B-2A-4B=0 \end{cases} \Rightarrow A=-\frac{3}{26},\quad B=\frac{1}{13}$.

$$y_2(x)=-\frac{3}{26}\cos2x+\frac{1}{13}\sin2x.$$

非齐次通解：$y(x)=e^{-\frac{1}{2}x}\left(C_1\cos\frac{\sqrt{3}}{2}x+C_2\sin\frac{\sqrt{3}}{2}x\right)-\frac{3}{26}\cos2x+\frac{1}{13}\sin2x+\frac{1}{2}$.

$$\begin{cases} y(0)=2 \\ y'(0)=1 \end{cases} \Rightarrow \begin{cases} C_1-\frac{3}{26}+\frac{1}{2}=2 \\ -\frac{1}{2}C_1+\frac{\sqrt{3}}{2}C_2+\frac{2}{13}=1 \end{cases} \Rightarrow C_1=\frac{21}{13},\quad C_2=\frac{43}{13\sqrt{3}}$$

$$\therefore\ y(x)=e^{-\frac{1}{2}x}\left[\frac{21}{13}\cos\frac{\sqrt{3}}{2}x+\frac{43}{13\sqrt{3}}\sin\frac{\sqrt{3}}{2}x\right]-\frac{3}{26}\cos2x+\frac{1}{13}\sin2x+\frac{1}{2}.$$

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8. 如图所示，一质量为m的小球，用长为l的细线挂在O点，在地球引力下作往复摆动，如忽略线的质量，而且θ比较小，以至于可假设 $\sin\theta \approx \theta$，求此摆球的运动方程。




解：由动量矩定理可知：质点动量矩的改变量：$J\dot{\omega} = J\ddot{\theta}$（垂直纸面向里）等于所受外力所产生的力矩：$-mgl\sin\theta$（上为负）。

$J = ml^2$，则有：$ml^2\ddot{\theta} = -mgl\sin\theta$。

当θ较小时，$\sin\theta \approx \theta$，则有运动微分方程：$\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta = 0$。

（单摆微振动方程，$\omega = \sqrt{\frac{g}{l}}$）。

9. 当λ为何值时，边值问题 $\begin{cases} y'' + \lambda y = 0 \\ y(0) = 0,\ y'(\pi) = 0 \end{cases}$ 有非零解，并求此边值问题

解：特征方程 $\mu^2 + \lambda = 0$。

当 $\lambda > 0$ 时：$\mu_{1,2} = \pm\sqrt{\lambda}i$，通解：$y(x) = C_1\cos\sqrt{\lambda}x + C_2\sin\sqrt{\lambda}x$。

$$\begin{cases} y(0) = 0 \\ y'(\pi) = 0 \end{cases} \Rightarrow \begin{cases} C_1 = 0 \\ C_2\cdot\sqrt{\lambda}\cos\sqrt{\lambda}\pi = 0 \end{cases} \Rightarrow \begin{cases} C_1 = 0 \\ \text{若 } C_2 \neq 0 \text{ 则 } \cos\sqrt{\lambda}\pi = 0 \end{cases}$$

$\Rightarrow -\sqrt{\lambda}\pi = \left(k + \frac{1}{2}\right)\pi$，$k = 0, -1, -2, \cdots$，即 $\lambda = \left(k + \frac{1}{2}\right)^2$，$k = 0, -1, -2, \cdots$

当 $\lambda \leq 0$ 时，$\mu_{1,2} = \pm\sqrt{-\lambda}$，通解：$y(x) = C_1 e^{\sqrt{-\lambda}x} + C_2 e^{-\sqrt{-\lambda}x}$。

$$\begin{cases} y(0) = 0 \\ y'(\pi) = 0 \end{cases} \Rightarrow \begin{cases} C_1 + C_2 = 0 \\ C_1\cdot\sqrt{-\lambda}\, e^{\sqrt{-\lambda}\pi} + C_2\cdot(-\sqrt{-\lambda})\cdot e^{-\sqrt{-\lambda}\pi} = 0 \end{cases} \Rightarrow \begin{cases} C_1 = 0 \\ C_2 = 0 \end{cases}$$

即无非零解。

∴ 仅当 $\lambda > 0$ 时有非零解，且 $\lambda = \left(k + \frac{1}{2}\right)^2$，$k = 0, 1, 2, \cdots$，

此时：$y(x) = C_2\sin\left(k + \frac{1}{2}\right)x$，$k = 0, 1, 2, \cdots$。

10. 设 $f(x)$ 是连续函数，且满足 $f(x) = e^x + \int_0^x (x - t)f(t)\,dt$，求 $f(x)$。

解：$f(x) = e^x + \int_0^x (x - t)f(t)\,dt = e^x + x\int_0^x f(t)\,dt - \int_0^x t f(t)\,dt$。

两边对 $x$ 求导：$f'(x) = e^x + \int_0^x f(t)\,dt + x\cdot f(x) - xf(x) = e^x + \int_0^x f(t)\,dt$。

两边再对 $x$ 求导：$f''(x) = e^x + f(x)$。令 $y = f(x)$，则得：$\frac{d^2y}{dx^2} - y = e^x$

$\lambda^2 - 1 = 0 \Rightarrow \lambda_{1,2} = \pm 1$，齐次方程通解：$y = C_1 e^x + C_2 e^{-x}$。

设非齐次特解：$y^*(x) = axe^x$，则：$y^{*\prime}(x) = a(1 + x)e^x$，$y^{*\prime\prime}(x) = a(2 + x)e^x$。

代入后得：$a(2 + x) - ax = 1 \Rightarrow a = \frac{1}{2}$，∴ $y^*(x) = \frac{1}{2}xe^x$。

非齐次方程通解：$y(x) = C_1 e^x + C_2 e^{-x} + \frac{1}{2}xe^x$。

且由有：$\begin{cases} f(0)=e^0+0=1 \\ f'(0)=e^0+0=1 \end{cases} \Rightarrow \begin{cases} C_1+C_2=1 \\ C_1-C_2+\frac{1}{2}=1 \end{cases} \Rightarrow \begin{cases} C_1=\frac{3}{4} \\ C_2=\frac{1}{4} \end{cases}$

$\therefore f(x)=\frac{3}{4}e^x+\frac{1}{4}e^{-x}+\frac{1}{2}xe^x$.

11. 设 $f(x)$ 在 $(-\infty,+\infty)$ 上具有连续的二阶导数，且 $f(0)=0$，$f'(0)=1$，试确定 $f(x)$，使得在 $R^2$ 上成立：$du(x,y)=y[f''(x)+3x]dx+[2f'(x)+f(x)]dy$，并求 $u(x,y)$.

解：成立全微分，则有：$\frac{\partial P}{\partial y}=\frac{\partial Q}{\partial x}$. 其中 $\begin{cases} P=y[f''(x)+3x] \\ Q=2f'(x)+f(x) \end{cases}$

$\frac{\partial P}{\partial y}=f''(x)+3x$，$\frac{\partial Q}{\partial x}=2f''(x)+f'(x)$. 即成立：$f''(x)+f'(x)-3x=0$.

$\frac{d^2f}{dx^2}+\frac{df}{dx}=3x$. 令 $u=f'(x)$，则：$\frac{du}{dx}+u=3x$.

$u(x)=e^{-\int 1\cdot dx}\left[C_1+\int 3x\cdot e^{\int 1\cdot dx}dx\right]=e^{-x}\left[C_1+3\int xe^x dx\right]$

$=e^{-x}\left[C_1+3xe^x-3e^x\right]=C_1e^{-x}+3(x-1)=\frac{df}{dx}$.

$f(x)=\int\left[C_1e^{-x}+3(x-1)\right]dx=\int C_1e^{-x}dx+\int 3(x-1)dx$

$=-C_1e^{-x}+\frac{3}{2}(x-1)^2+C_2=-C_1e^{-x}+\frac{3}{2}x^2-3x+C_2$.

$\begin{cases} f(0)=0 \\ f'(0)=1 \end{cases} \Rightarrow \begin{cases} -C_1+C_2=0 \\ C_1-3=1 \end{cases} \Rightarrow \begin{cases} C_1=4 \\ C_2=4 \end{cases}$

$\therefore f(x)=-4e^{-x}+\frac{3}{2}x^2-3x+4$.

$f'(x)=4e^{-x}+3x-3$，$f''(x)=-4e^{-x}+3$. 则有：

$du(x,y)=y\left[-4e^{-x}+3+3x\right]dx+\left[8e^{-x}+6x-6-4e^{-x}+\frac{3}{2}x^2-3x+4\right]dy$

$=y\left[-4e^{-x}+3x+3\right]dx+\left[4e^{-x}+\frac{3}{2}x^2+3x-2\right]dy$

$=y\,d\left[4e^{-x}+\frac{3}{2}x^2+3x\right]+\left[4e^{-x}+\frac{3}{2}x^2+3x-2\right]dy$.

$=y\,d\left[(4e^{-x}+\frac{3}{2}x^2+3x)\right]+dy\left[4e^{-x}+\frac{3}{2}x^2+3x\right]-2dy$

$=d\left[y(8e^{-x}+3x^2+6x-2)\right]$ $d\left[y(4e^{-x}+\frac{3}{2}x^2+3x)\right]-d(2y)$

$\therefore u(x,y)=\frac{1}{2}y(8e^{-x}+3x^2+6x)-2y+C$.

$=y(4e^{-x}+\frac{3}{2}x^2+3x-4)+C$

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12. 设有方程 $\varphi(x)y''+y'-2y=e^x$，其中 $\varphi(x)=\begin{cases}0 & x<0\\ 1 & x>0\end{cases}$.

求在 $(-\infty,+\infty)$ 上的连续函数 $y=y(x)$，使之在 $(-\infty,0)$ 和 $(0,+\infty)$ 上都满足方程，且满足条件 $y(1)=-\frac{2}{3}e$，$y(-1)=-\frac{1}{e}$.

解：当 $x<0$ 时，方程为：$y'-2y=e^x$.

$$y(x)=e^{2x}\left[C+\int e^x e^{-2x}dx\right]=Ce^{2x}-e^x.$$

由 $y(-1)=-\frac{1}{e}$ $\Rightarrow$ $-\frac{1}{e}=Ce^{-2}-e^{-1}=\frac{C}{e^2}-\frac{1}{e}$. $\Rightarrow$ $C=0$.

即：$y_1(x)=-e^x$ $(x<0)$

当 $x>0$ 时，方程为：$y''+y'-2y=e^x$

$\lambda^2+\lambda-2=0$. $\Rightarrow$ $\lambda_1=-2$，$\lambda_2=1$. 齐次通解：$y=C_1e^{-2x}+C_2e^x$.

设特解：$y^*(x)=axe^x$. $y^{*\prime}(x)=a(x+1)e^x$. $y^{*\prime\prime}(x)=a(x+2)e^x$

代入方程得：$a(x+2)+a(x+1)-2ax=1$. $\Rightarrow$ $a=\frac{1}{3}$.

即通解：$y(x)=C_1e^{-2x}+C_2e^x+\frac{1}{3}xe^x$

∵ $f(x)$ 在 $(-\infty,+\infty)$ 上连续，而当 $x<0$ 时，$y(x)=e^x$.

∴ $\lim\limits_{x\to0}y(x)=y(0)=-1$.

由 $\begin{cases}y(0)=-1\\ y(1)=-\frac{2}{3}e\end{cases}$ $\Rightarrow$ $\begin{cases}C_1+C_2=-1\\ \frac{C_1}{e^2}+C_2e+\frac{1}{3}e=-\frac{2}{3}e\end{cases}$ $\Rightarrow$ $\begin{cases}C_1=0\\ C_2=-1\end{cases}$

即有通解：$y_2(x)=-e^x+\frac{1}{3}xe^x$ $(x>0)$

∴ $y(x)=\begin{cases}-e^x & x\le0\\ -e^x+\frac{1}{3}xe^x & x>0\end{cases}$.

13. 求下列 Euler 方程的解：

(1) $x^2\frac{d^2y}{dx^2}-x\frac{dy}{dx}+2y=0$.

解：令 $x=e^t$，即 $t=\ln x$，则 $\frac{dy}{dx}=\frac{dy}{dt}\cdot\frac{1}{x}$. $\frac{d^2y}{dx^2}=\frac{1}{x^2}\left(\frac{d^2y}{dt^2}-\frac{dy}{dt}\right)$.

代入得：$\frac{d^2y}{dt^2}-\frac{dy}{dt}-\frac{dy}{dt}+2y=0$. $\Rightarrow$ $\frac{d^2y}{dt^2}-2\frac{dy}{dt}+2y=0$.

$\lambda^2-2\lambda+2=0$. $\Rightarrow$ $\lambda_{1,2}=1\pm i$. $\Rightarrow$ $\bar{y}(t)=e^t(C_1\cos t+C_2\sin t)$.

∴ $y(x)=x(C_1\cos\ln x+C_2\sin\ln x)$.

(2) $x^3\frac{d^3y}{dx^3} - x^2\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} - 2y = 0$.

解: 令 $x = e^t$. 则 $t = \ln x$. $\frac{dy}{dx} = \frac{1}{x}\frac{dy}{dt}$. $\frac{d^2y}{dx^2} = \frac{1}{x^2}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)$.

$\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = -\frac{2}{x^3}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right) + \frac{1}{x^2}\left(\frac{d^3y}{dt^3} - \frac{d^2y}{dt^2}\right)\frac{1}{x} = \frac{1}{x^3}\left(\frac{d^3y}{dt^3} - \frac{3d^2y}{dt^2} + \frac{2dy}{dt}\right)$

代入方程为得: $\frac{d^3y}{dt^3} - 3\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - \frac{d^2y}{dt^2} + \frac{dy}{dt} + 2\frac{dy}{dt} - 2y = 0$.

即: $\frac{d^3y}{dt^3} - 4\frac{d^2y}{dt^2} + 5\frac{dy}{dt} - 2y = 0$.

$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$. $\Rightarrow$ $\lambda_1 = 2$, $\lambda_2 = \lambda_3 = 1$.

$\Rightarrow$ $y(t) = C_1e^{2t} + (C_2 + C_3t)e^t$.

$\therefore$ $y(x) = C_1x^2 + C_2x + C_3x\ln x$.

(3) $\frac{d^2y}{dx^2} + \frac{2}{x}\frac{dy}{dx} - \frac{n(n+1)}{x^2}y = 0$.

解: 令 $x = e^t$. 则 $t = \ln x$. $\frac{dy}{dx} = \frac{1}{x}\frac{dy}{dt}$. $\frac{d^2y}{dx^2} = \frac{1}{x^2}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)$.

代入方程为得: $\frac{d^2y}{dt^2} - \frac{dy}{dt} + 2\frac{dy}{dt} - n(n+1)y = 0$ 即: $\frac{d^2y}{dt^2} + \frac{dy}{dt} - n(n+1)y = 0$.

$\lambda^2 + \lambda - n(n+1) = 0$. $\Rightarrow$ $\lambda_1 = n$. $\lambda_2 = -(n+1)$.

则: $y(t) = C_1e^{nt} + C_2e^{-(n+1)t}$

$\therefore$ $y(x) = C_1x^n + C_2x^{-(n+1)}$.

(4) $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 0$.

解: 令 $1+x = e^t$. 则 $t = \ln(1+x)$. $\frac{dy}{dx} = \frac{1}{1+x}\frac{dy}{dt}$. $\frac{d^2y}{dx^2} = \frac{1}{(1+x)^2}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)$.

代入方程为得: $\frac{d^2y}{dt^2} - \frac{dy}{dt} + \frac{dy}{dt} + y = 0$. 即: $\frac{d^2y}{dt^2} + y = 0$.

$\lambda^2 + 1 = 0$. $\Rightarrow$ $\lambda_{1,2} = \pm i$. $\Rightarrow$ $y(t) = C_1\cos t + C_2\sin t$.

$\therefore$ $y(x) = C_1\cos\ln(1+x) + C_2\sin\ln(1+x)$. $(1+x>0)$.

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(5) $x^3\frac{d^3y}{dx^3}+2x\frac{dy}{dx}-2y=x^2\ln x+3x$.

解：令 $x=e^t$，则 $t=\ln x$. $\frac{dy}{dx}=\frac{1}{x}\frac{dy}{dt}$. $\frac{d^2y}{dx^2}=\frac{1}{x^2}\left(\frac{d^2y}{dt^2}-\frac{dy}{dt}\right)$.

$\frac{d^3y}{dx^3}=\frac{1}{x^3}\left(\frac{d^3y}{dt^3}-3\frac{d^2y}{dt^2}+2\frac{dy}{dt}\right)$. 代入方程有：

$\frac{d^3y}{dt^3}-3\frac{d^2y}{dt^2}+2\frac{dy}{dt}+2\frac{dy}{dt}-2y=te^{2t}+3e^t$.

即：$\frac{d^3y}{dt^3}-3\frac{d^2y}{dt^2}+4\frac{dy}{dt}-2y=te^{2t}+3e^t$.

$\lambda^3-3\lambda^2+4\lambda-2=0 \Rightarrow \lambda_1=1,\ \lambda_{2,3}=1\pm i$.

齐次方程通解：$y=c_1e^t+(c_2\cos t+c_3\sin t)e^t$.

对于 $f_1(t)=te^{2t}$，设特解：$y_1^*(t)=(a_0+a_1t)e^{2t}$.

$y_1^{*\prime}(t)=[(2a_0+a_1)+2a_1t]e^{2t}$. $y_1^{*\prime\prime}(t)=[(4a_0+4a_1)+4a_1t]e^{2t}$

$y_1^{*\prime\prime\prime}(t)=[(8a_0+12a_1)+8a_1t]e^{2t}$.

代入方程有：$\begin{cases}8a_0+12a_1-12a_0-12a_1+8a_0+4a_1-2a_0=0\\8a_1-12a_1+8a_1-2a_1=1\end{cases}\Rightarrow\begin{cases}a_0=-1\\a_1=\frac{1}{2}\end{cases}$

即有：$y_1^*(t)=\left(-1+\frac{1}{2}t\right)e^{2t}$.

对于 $f_2(t)=3e^t$，设特解：$y_2^*(t)=ate^t$. $y_2^{*\prime}(t)=a(1+t)e^t$.

$y_2^{*\prime\prime}(t)=a(2+t)e^t$. $y_2^{*\prime\prime\prime}(t)=a(3+t)e^t$.

代入方程有，$a(3+t)-3a(2+t)+4a(1+t)-2at=3 \Rightarrow a=3$.

即有：$y_2^*(t)=3te^t$.

通解：$y(t)=c_1e^t+e^t(c_2\cos t+c_3\sin t)+3te^t+\left(-1+\frac{1}{2}t\right)e^{2t}$.

$\therefore y(x)=c_1x+x(c_2\cos\ln x+c_3\sin\ln x)+3x\ln x+x^2\left(\frac{1}{2}\ln x-1\right)$.

(6) $x^2\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+2y=\ln^2x-2\ln x$.

解：令 $x=e^t$，则 $t=\ln x$. $\frac{dy}{dx}=\frac{1}{x}\frac{dy}{dt}$. $\frac{d^2y}{dx^2}=\frac{1}{x^2}\left(\frac{d^2y}{dt^2}-\frac{dy}{dt}\right)$.

代入方程有：$\frac{d^2y}{dt^2}-\frac{dy}{dt}-2\frac{dy}{dt}+2y=t^2-2t$ 即：$\frac{d^2y}{dt^2}-3\frac{dy}{dt}+2y=t^2-2t$.

$\lambda^2-3\lambda+2=0 \Rightarrow \lambda_1=1,\ \lambda_2=2$. 齐次通解：$y=c_1e^t+c_2e^{2t}$

设非齐次特解：$y^*(t)=a_0+a_1t+a_2t^2$. $y^{*\prime}(t)=a_1+2a_2t$. $y^{*\prime\prime}(t)=2a_2$.

代入方程后得：$2a_2-3a_1-6a_2t+2a_0+2a_1t+2a_2t^2\equiv t^2-2t$.

$$\begin{cases}2a_2-3a_1+2a_0=0\\-6a_2+2a_1=-2\\2a_2=1\end{cases}\Rightarrow a_0=\frac{1}{4},\quad a_1=\frac{1}{2},\quad a_2=\frac{1}{2}.$$

即：$y^*(t)=\frac{1}{4}+\frac{1}{2}t+\frac{1}{2}t^2$.

通解：$y(t)=C_1e^t+C_2e^{2t}+\frac{1}{4}+\frac{1}{2}t+\frac{1}{2}t^2$.

$\therefore y(x)=C_1x+C_2x^2+\frac{1}{4}+\frac{1}{2}\ln x+\frac{1}{2}\ln^2x$.

## §4. 可降阶的高阶微分方程

习题. P.273

1. 求下列微分方程的通解：

(1) $\frac{d^2y}{dx^2}=\ln x-\cos^2x$.

解：设 $\frac{dy}{dx}=u(x)$. 则 $\frac{du}{dx}=\ln x-\cos^2x$.

$$u(x)=\int(\ln x-\cos^2x)dx=\int\ln x\,dx-\frac{1}{2}\int(1+\cos2x)dx$$

$$=x\ln x-x-\frac{1}{2}x-\frac{1}{4}\sin2x+C_1=x\ln x-\frac{3}{2}x-\frac{1}{4}\sin2x+C_1.$$

$$y(x)=\int u(x)dx=\int x\ln x\,dx-\int\frac{3}{2}x\,dx-\frac{1}{4}\int\sin2x\,dx+C_1x$$

$$=\frac{1}{2}x^2\ln x-\frac{1}{4}x^2-\frac{3}{4}x^2-\frac{1}{4}\sin^2x+C_1x+C_2$$

$\therefore y(x)=\frac{1}{2}x^2(\ln x-2)-\frac{1}{4}\sin^2x+C_1x+C_2$.

(2) $\frac{d^3y}{dx^3}-xe^x=0$.

解：$\frac{d^2y}{dx^2}=\int xe^xdx=xe^x-e^x+C_1$

$\frac{dy}{dx}=\int xe^xdx-\int e^xdx+C_1x=xe^x-e^x-e^x+C_1x+C_2=xe^x-2e^x+C_1x+C_2$

$y(x)=\int xe^xdx-2\int e^xdx+\frac{1}{2}C_1x^2+C_2x+C_3=xe^x-3e^x+C_1x^2+C_2x+C_3$.

$\therefore y(x)=(x-3)e^x+C_1x^2+C_2x+C_3$.

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(3) $x\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$.

解：设 $u(x)=\frac{dy}{dx}$. 则：$x\frac{du}{dx}+u=0$. $\Rightarrow \frac{du}{u}+\frac{dx}{x}=0$.

$\Rightarrow d\ln(xu)=0$. $\Rightarrow xu=C_1$ $\Rightarrow u(x)=\frac{C_1}{x}$.

$\Rightarrow y(x)=\int u(x)dx = C_1\ln x + C_2$.

(4) $y\frac{d^2y}{dx^2}=\left(\frac{dy}{dx}\right)^2-1$.

解：设 $u(x)=\frac{dy}{dx}$. 则：$\frac{du}{dy}=\frac{du}{dx}\cdot\frac{dx}{dy}=\frac{d^2y}{dx^2}\cdot\frac{1}{u}$. 即：$u\frac{du}{dy}=\frac{d^2y}{dx^2}$.

代入方程得：$yu\frac{du}{dy}=u^2-1$. $\Rightarrow \frac{u\,du}{u^2-1}=\frac{dy}{y}$ $\Rightarrow \frac{1}{2}\ln(u^2-1)=\ln y+C_1$.

$\Rightarrow u^2-1=(C_1y)^2$ $\Rightarrow \left(\frac{dy}{dx}\right)^2=C_1^2y^2+1$. $\Rightarrow \frac{dy}{dx}=\pm\sqrt{1+C_1^2y^2}$.

$\frac{dy}{\sqrt{1+C_1^2y^2}}=\pm dx$. $\Rightarrow \frac{d(C_1y)}{\sqrt{1+C_1^2y^2}}=\pm d(C_1x)$.

$\Rightarrow \ln\left(C_1y+\sqrt{1+C_1^2y^2}\right)=\pm C_1x+C_2$.

即：$C_1y+\sqrt{1+C_1^2y^2}=C_2e^{\pm C_1x}$.

(5) $\frac{d^2y}{dx^2}=\frac{1}{\sqrt{y}}$

解：设 $\frac{dy}{dx}=p$. 则：$\frac{d^2y}{dx^2}=p\frac{dp}{dy}$. 即：$p\frac{dp}{dy}=\frac{1}{\sqrt{y}}$ $\Rightarrow p\,dp=y^{-\frac{1}{2}}dy$.

$\frac{1}{2}p^2=2\sqrt{y}+C$. $\Rightarrow p=\sqrt{4\sqrt{y}+C}=\frac{dy}{dx}$.

$\frac{dy}{\sqrt{4\sqrt{y}+C}}=dx$ $\Rightarrow$ $\frac{4\sqrt{y}\,d(4\sqrt{y}+C)}{\sqrt{4\sqrt{y}+C}}=8\,dx$

$\left(\sqrt{4\sqrt{y}+C}-\frac{C}{\sqrt{4\sqrt{y}+C}}\right)d(4\sqrt{y}+C)=8\,dx$.

$\frac{2}{3}(4\sqrt{y}+C)\sqrt{4\sqrt{y}+C}-2C\sqrt{4\sqrt{y}+C}=8dx+C_1$

即：$(2\sqrt{y}-C)\sqrt{4\sqrt{y}+C}=6x+C_1$.

(6) $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^3 + \frac{dy}{dx}$.

解: 设 $\frac{dy}{dx} = u$. 则: $\frac{du}{dx} = u^3+u$. $\Rightarrow \frac{du}{u^3+u} = dx$. $\Rightarrow \left(\frac{1}{u} - \frac{u}{u^2+1}\right)du = dx$.

$$\ln u - \frac{1}{2}\ln(u^2+1) = x + C \Rightarrow \frac{u}{\sqrt{1+u^2}} = Ce^x \Rightarrow \frac{u^2}{1+u^2} = C^2e^{2x}$$

$$\frac{1}{1+u^2} = 1 - C^2e^{2x} \Rightarrow u^2 = \frac{C^2e^{2x}}{1-C^2e^{2x}} \Rightarrow u = \frac{Ce^x}{\sqrt{1-C^2e^{2x}}}$$

即 $\frac{dy}{dx} = \frac{Ce^x}{\sqrt{1-C^2e^{2x}}}$. $y(x) = \int \frac{Ce^x}{\sqrt{1-C^2e^{2x}}}dx = \int \frac{1}{\sqrt{1-C^2e^{2x}}}d\,Ce^x = \arcsin Ce^x + C_1$.

$\therefore y(x) = \arcsin Ce^x + C_1$.

(7) $(y'')^4 + 5y'' = x$.

解: 设 $y'' = u$. 即 $\frac{d^2y}{dx^2} = u$. $\Rightarrow u^4+5u = x$. $\Rightarrow \frac{dx}{du} = 4u^3+5$

$$u = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\frac{d}{du}\left(\frac{dy}{dx}\right)}{\frac{dx}{du}} = \frac{\frac{d}{du}\left(\frac{dy}{dx}\right)}{4u^3+5} \Rightarrow \frac{d}{du}\left(\frac{dy}{dx}\right) = 4u^4+5u$$

$$\frac{dy}{dx} = \int (4u^4+5u)\,du = \frac{4}{5}u^5 + \frac{5}{2}u^2 + C_1.$$

$$\frac{dy}{dx} = \frac{\frac{dy}{du}}{\frac{dx}{du}} = \frac{\frac{dy}{du}}{4u^3+5} = \frac{4}{5}u^5 + \frac{5}{2}u^2 + C_1 \Rightarrow \frac{dy}{du} = \frac{16}{5}u^8 + 14u^5 + 4C_1u^3 + \frac{15}{2}u^2 + 5C_1$$

$$y(u) = \int\left(\frac{16}{5}u^8 + 14u^5 + 4C_1u^3 + \frac{15}{2}u^2 + 5C_1\right)du$$

$$= \frac{16}{45}u^9 + \frac{7}{3}u^6 + C_1u^4 + \frac{15}{6}u^3 + 5C_1u + C_2.$$

$\therefore$ 参数形式的通解为: $\begin{cases} x(t) = 4t^3+5 \\ y(t) = \frac{16}{45}t^9 + \frac{7}{3}t^6 + C_1t^4 + \frac{15}{6}t^3 + 5C_1t + C_2 \end{cases}$.

(8) $xy''' + y'' = x^2$.

解: 设 $y'' = u$. 则: $y''' = \frac{du}{dx}$. 代入方程得: $x\frac{du}{dx} + u = x^2 \Rightarrow \frac{du}{dx} + \frac{u}{x} = x$.

$$u(x) = e^{\int -\frac{1}{x}dx}\cdot\left[C_1 + \int x\cdot e^{\int \frac{1}{x}dx}dx\right] = e^{-\ln x}\cdot\left[C_1 + \int x\cdot e^{\ln x}dx\right]$$

$$= \frac{1}{x}\cdot\left[C_1 + \int x\cdot x\,dx\right] = \frac{C_1}{x} + \frac{1}{3}x^2 \Rightarrow \frac{d^2y}{dx^2} = \frac{C_1}{x} + \frac{1}{3}x^2$$

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$$\frac{dy}{dx} = \int \left(\frac{C_1}{x} + \frac{1}{3}x^2\right) dx = C_1 \ln x + \frac{1}{9}x^3 + C_2.$$

$$y(x) = \int \left(C_1 \ln x + \frac{1}{9}x^3 + C_2\right) dx = C_1 x \ln x - C_1 x + \frac{1}{36}x^4 + C_2 x + C_3.$$

$$\therefore y(x) = \frac{1}{36}x^4 + C_1 x(\ln x - 1) + C_2 x + C_3.$$

(9) $y''' = 2(y''-1)\cot x$.

解: 设 $y'' = u$. 则: $\frac{du}{dx} = 2(u-1)\cot x \Rightarrow \frac{du}{u-1} = \frac{2\cos x}{\sin x} dx$

$$\ln(u-1) = 2\ln \sin x + C_1. \Rightarrow u = 1 + C_1 \sin^2 x$$

$$\frac{d^2y}{dx^2} = 1 + C_1 \sin^2 x. \Rightarrow \frac{dy}{dx} = \int (1 + C_1 \sin^2 x) dx = x + \frac{C_1}{2}x - \frac{C_1}{4}\sin 2x + C_2.$$

$$y = \int \left(x + \frac{C_1}{2}x - \frac{C_1}{4}\sin 2x + C_2\right) dx = \frac{1}{2}x^2 + \frac{C_1}{4}x^2 + \frac{C_1}{8}\cos 2x + C_2 x + C_3.$$

$$\therefore y(x) = \frac{x^2}{2} + C_1\left(\frac{x^2}{4} - \frac{1}{8}\cos 2x\right) + C_2 x + C_3.$$

(10) $xyy'' + x(y')^2 - yy' = 0$.

解: 设 $y^2 = u$. 则: $2yy' = u' \Rightarrow yy' = \frac{1}{2}u'$, 再对 $x$ 求导一次:

$$yy'' + (y')^2 = \frac{1}{2}u''. \quad \text{代入原方程, 得: } \frac{1}{2}xu'' - \frac{1}{2}u' = 0.$$

即: $x\frac{du'}{dx} - u' = 0$. $\frac{du'}{u'} = \frac{dx}{x}$. $\Rightarrow u' = C_1 x$.

$$\frac{du}{dx} = C_1 x. \Rightarrow u(x) = \frac{1}{2}C_1 x^2 + C_2$$

$$\therefore y^2(x) = C_1 x^2 + C_2. \quad \text{原解为: } y(x) = \pm\sqrt{C_1 x^2 + C_2}.$$

(11) $yy'' - (y')^2 = y^2 \ln y$.

解: $\frac{yy'' - (y')^2}{y^2} = \frac{d}{dx}\left(\frac{y'}{y}\right)$ 代入原方程得: $\frac{d}{dx}\left(\frac{y'}{y}\right) = \ln y$.

$$\Rightarrow d\left(\frac{1}{y}\frac{dy}{dx}\right) = \ln y \cdot dx. \Rightarrow \frac{d}{dx}\left(\frac{d\ln y}{dx}\right) = \ln y.$$

令 $u = \ln y \Rightarrow \frac{d^2u}{dx^2} = u. \Rightarrow \frac{d^2u}{dx^2} - u = 0 \quad \lambda^2 - 1 = 0. \quad \lambda_{1,2} = \pm 1$

∴ $u(x) = C_1 e^{x} + C_2 e^{-x}$. 即: $\ln y = C_1 e^{x} + C_2 e^{-x}$.

(2) $y''' + 4y'' = 0$.

解: $\frac{dy''}{dx} + 4y'' = 0 \Rightarrow \ln y'' = -4x + C_1 \Rightarrow y'' = C_1 e^{-4x}$.

$y' = \int C_1 e^{-4x} dx = C_1 e^{-4x} + C_2$.

∴ $y(x) = \int (C_1 e^{-4x} + C_2) dx = C_1 e^{-4x} + C_2 x + C_3$.

(3) $y^{(4)} - y''' - 2y'' = x^2$.

解: 设 $y'' = u$. 则: $\frac{d^2u}{dx^2} - \frac{du}{dx} - 2u = x^2$. $\lambda^2 - \lambda - 2 = 0 \Rightarrow \lambda_1 = 2, \lambda_2 = -1$.

齐次方程通解: $u = C_1 e^{2x} + C_2 e^{-x}$. 设非齐次特解: $u^*(x) = a_0 + a_1 x + a_2 x^2$.

代入得: $2a_2 - a_1 - 2a_2 x - 2a_0 - 2a_1 x - 2a_2 x^2 = x^2$.

$\Rightarrow a_2 = -\frac{1}{2}$. $a_1 = \frac{1}{2}$. $a_0 = -\frac{3}{4}$.

即: $u^*(x) = -\frac{3}{4} + \frac{1}{2}x - \frac{1}{2}x^2$.

∴ $u(x) = C_1 e^{2x} + C_2 e^{-x} - \frac{1}{2}x^2 + \frac{1}{2}x - \frac{3}{4}$.

$\frac{dy}{dx} = \int u(x) dx = C_1 e^{2x} + C_2 e^{-x} - \frac{1}{6}x^3 + \frac{1}{4}x^2 - \frac{3}{4}x + C_3$.

$y(x) = \int \frac{dy}{dx} dx = C_1 e^{2x} + C_2 e^{-x} - \frac{1}{24}x^4 + \frac{1}{12}x^3 - \frac{3}{8}x^2 + C_3 x + C_4$.

2. 求下列微分方程的特解:

(1) $\frac{d^2y}{dx^2} - xe^x = 0$ $y(0) = y'(0) = 0$.

解: $\frac{dy}{dx} = \int xe^x dx = xe^x - e^x + C_1 = (x-1)e^x + C_1$.

$\left.\frac{dy}{dx}\right|_{x=0} = 0 \Rightarrow -e^0 + C_1 = 0 \Rightarrow C_1 = +1$.

$y(x) = \int (xe^x - e^x + 1) dx = (x-1)e^x - e^x + x + C_2 = (x-2)e^x + x + C_2$.

$y(0) = 0 \Rightarrow -2e^0 + 0 + C_2 = 0 \Rightarrow C_2 = 2$.

∴ $y(x) = (x-2)e^x + x + 2$.

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(2) $\frac{d^3y}{dx^3} - xe^x = 0$. $y(1) = y'(1) = y''(1) = 0$.

解: $\frac{d^3y}{dx^3} = xe^x$. $\frac{d^2y}{dx^2} = \int xe^x dx = xe^x - e^x + C_1$.

$y''(1) = 0$. $\Rightarrow$ $e - e + C_1 = 0$. $\Rightarrow$ $C_1 = 0$.

$\frac{d^2y}{dx^2} = xe^x - e^x$. $\frac{dy}{dx} = \int (xe^x - e^x) dx = xe^x - 2e^x + C_2$.

$y'(1) = 0$. $\Rightarrow$ $e - 2e + C_2 = 0$. $\Rightarrow$ $C_2 = e$.

$\frac{dy}{dx} = xe^x - 2e^x + e$. $y(x) = \int (xe^x - 2e^x + e) dx = xe^x - 3e^x + ex + C_3$.

$y(1) = 0$. $\Rightarrow$ $e - 3e + e + C_3 = 0$. $\Rightarrow$ $C_3 = e$.

$\therefore y(x) = xe^x - 3e^x + ex + e$.

(3) $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$. $y(0) = 0$. $y'(0) = -1$.

解: 设 $\frac{dy}{dx} = u(x)$. 则: $\frac{du}{dx} = u^2$. $\frac{du}{u^2} = dx$. $\Rightarrow$ $-\frac{1}{u} = x + C_1$.

$u(x) = -\frac{1}{x + C_1}$. $u(0) = y'(0) = -1$ $\Rightarrow$ $-\frac{1}{C_1} = -1$ $\Rightarrow$ $C_1 = 1$.

$\frac{dy}{dx} = u(x) = -\frac{1}{x+1}$. $y(x) = -\int \frac{1}{x+1} dx = -\ln(x+1) + C_2$.

$y(0) = 0$. $\Rightarrow$ $-\ln 1 + C_2 = 0$ $\Rightarrow$ $C_2 = 0$.

$\therefore y(x) = -\ln(x+1)$.

(4) $\frac{d^2y}{dx^2} = 3\sqrt{y}$. $y(0) = 1$. $y'(0) = 2$.

解: 设 $\frac{dy}{dx} = p$. 则 $\frac{d^2y}{dx^2} = \frac{d}{dx}p = \frac{dp}{dy}\frac{dy}{dx} = \frac{dp}{dy}p = 3\sqrt{y}$.

$p\,dp = 3\sqrt{y}\,dy$. $\Rightarrow$ $\frac{1}{2}p^2 = 2y\sqrt{y} + C_1$. $\Rightarrow$ $p^2 = 4y\sqrt{y} + 2C_1$.

$\begin{cases} y(0) = 1 \\ y'(0) = 2 \end{cases}$ 即 当 $x = 0$ 时有 $\begin{matrix} y = 1 \\ y' = p = 2 \end{matrix}$ $\Rightarrow$ $4 = 4 + 2C_1$ $\Rightarrow$ $C_1 = 0$.

$\left(\frac{dy}{dx}\right)^2 = p^2 = 4y\sqrt{y} = 4y^{\frac{3}{2}}$. $\Rightarrow$ $\frac{dy}{dx} = 2 \cdot y^{\frac{3}{4}}$ $\Rightarrow$ $y^{-\frac{3}{4}} dy = 2\,dx$.

$4y^{\frac{1}{4}} = 2x + C_2$. $\quad y = \left(\frac{2x+C_2}{4}\right)^4$.

$y(0)=1$. $\Rightarrow \quad 1 = \left(\frac{C_2}{4}\right)^4 \Rightarrow \quad C_2 = 4$.

$\therefore \quad y(x) = \left(\frac{x}{2}+1\right)^4$.

(5) $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 1$. $\quad y(0) = y'(0) = 0$.

解: 设 $\frac{dy}{dx} = u(x)$. 则: $\frac{du}{dx} + u^2 = 1$. $\quad \frac{du}{1-u^2} = dx$.

$\ln \frac{1+u}{1-u} = 2x + C_1$. $\quad u(0) = y'(0) = 0$. $\Rightarrow \quad C_1 = 0$.

$\Rightarrow \ln \frac{1+u}{1-u} = 2x$. $\Rightarrow \frac{1+u}{1-u} = e^{2x}$ $\Rightarrow \quad u(x) = \frac{e^{2x}-1}{1+e^{2x}}$

即: $\frac{dy}{dx} = \frac{e^{2x}-1}{e^{2x}+1}$. $\quad y(x) = \int \frac{e^{2x}-1}{e^{2x}+1} dx = \int \left(\frac{2e^{2x}}{e^{2x}+1} - 1\right) dx$.

$\Rightarrow y(x) = \ln(1+e^{2x}) - x + C_2$. 由 $y(0)=0$. $\Rightarrow C_2 = -\ln 2$.

$\therefore \quad y(x) = \ln(1+e^{2x}) - x - \ln 2$.

(6) $y^3 \frac{d^2y}{dx^2} + 1 = 0$. $\quad y(1) = 1$. $\quad y'(1) = 0$.

解: 设 $\frac{dy}{dx} = p$. $\quad \frac{d^2y}{dx^2} = \frac{dp}{dy} \cdot p \quad \Rightarrow \quad y^3 \cdot p \frac{dp}{dy} + 1 = 0$.

$p\,dp = -\frac{1}{y^3} dy$. $\Rightarrow \quad p^2 = \frac{1}{y^2} + C_1$. 当 $x=1$ 时. $\begin{cases} y=1 \\ p=0 \end{cases} \Rightarrow C_1 = -1$.

即: $p^2 = \frac{1}{y^2} - 1 = \frac{1-y^2}{y^2}$. $\Rightarrow \quad \frac{dy}{dx} = p = \frac{\sqrt{1-y^2}}{y}$.

$\Rightarrow \quad \frac{y\,dy}{\sqrt{1-y^2}} = dx$. $\Rightarrow \quad 2\sqrt{1-y^2} = -2x + C_2$. $\quad y(1)=1 \Rightarrow C_2 = 2$.

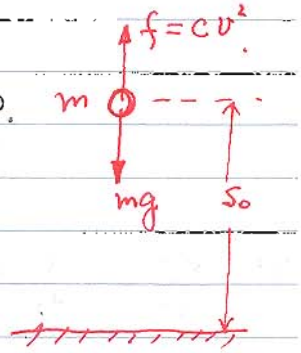
$\Rightarrow \quad 2\sqrt{1-y^2} = -2x + 2$. 即: $\quad 1-y^2 = (1-x)^2$.

$\therefore \quad y(x) = \sqrt{2x - x^2}$.

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3. 设一质量为m的物体，在空气中由静止开始下落，如果空气阻力与物体下落速度v的平方成正比，试求物体下落的距离s与时间t的函数关系.




解：运动方程：$m\frac{dv}{dt}=mg-cv^2$　　$v=\frac{ds}{dt}$　　$s(0)=s_0$　　$s'(0)=v(0)=0$

$$\frac{dv}{dt}=g-\frac{c}{m}v^2 \qquad \frac{dv}{1-\frac{c}{mg}v^2}=g\,dt$$

$$\left(\frac{1}{1-\sqrt{\frac{c}{mg}}v}+\frac{1}{1+\sqrt{\frac{c}{mg}}v}\right)d\sqrt{\frac{c}{mg}}v=2\sqrt{\frac{c}{mg}}g\,dt$$

$$\ln\frac{1+\sqrt{\frac{c}{mg}}v}{1-\sqrt{\frac{c}{mg}}v}=2gt\sqrt{\frac{c}{mg}}+C_1 \qquad 由\ v(0)=0 \Rightarrow C_1=0$$

则有：$$\frac{1+\sqrt{\frac{c}{mg}}v}{1-\sqrt{\frac{c}{mg}}v}=e^{2gt\sqrt{\frac{c}{mg}}} \qquad 记\ k=\sqrt{\frac{c}{mg}}$$

即有：$$\frac{1+kv}{1-kv}=e^{2kgt} \Rightarrow kv=\frac{e^{2kgt}-1}{e^{2kgt}+1}=\frac{2e^{2kgt}}{1+e^{2kgt}}-1$$

$$\Rightarrow k\frac{ds}{dt}=\frac{2e^{2kgt}}{1+e^{2kgt}}-1$$

$$\Rightarrow s=\int\left(\frac{2}{k}\cdot\frac{e^{2kgt}}{1+e^{2kgt}}dt-\frac{t}{k}\right)$$

$$=\frac{2}{k}\cdot\frac{1}{2kg}\ln(1+e^{2kgt})-\frac{t}{k}+C_2$$

$$=\frac{m}{c}\ln(1+e^{2gt\sqrt{\frac{c}{mg}}})-\sqrt{\frac{mg}{c}}t+C_2$$

由 $s(0)=s_0 \Rightarrow C_2=s_0-\frac{m}{c}\ln 2$

$$\therefore s(t)=s_0-\sqrt{\frac{mg}{c}}t+\frac{m}{c}\ln\frac{1}{2}(1+e^{2gt\sqrt{\frac{c}{mg}}})$$

即：$$s(t)=s_0-\sqrt{\frac{mg}{c}}t+\frac{m}{c}\ln\frac{1+e^{2gt\sqrt{\frac{c}{mg}}}}{2}$$

4. 求 $y^2y''+1=0$ 的积分曲线方程，使得该积分曲线通过 $(0,\frac{1}{2})$ 点，且在该点处的切线斜率为2.

解：$y^2\frac{d^2y}{dx^2}+1=0$　　$y(0)=\frac{1}{2}$　　$y'(0)=2$

令 $\frac{dy}{dx}=p$. 则 $\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{dp}{dy}\cdot\frac{dy}{dx}=\frac{dp}{dy}\cdot p$.

代入得: $y^2\cdot p\frac{dp}{dy}+1=0$. $\quad p\,dp=-\frac{1}{y^2}dy$

$\frac{1}{2}p^2=\frac{1}{y}+C_1$ $\quad \begin{cases} y(0)=\frac{1}{2} \\ y'(0)=2 \end{cases}$ 即 $\begin{cases} y=\frac{1}{2} \\ p=2 \end{cases} \Rightarrow C_1=0$.

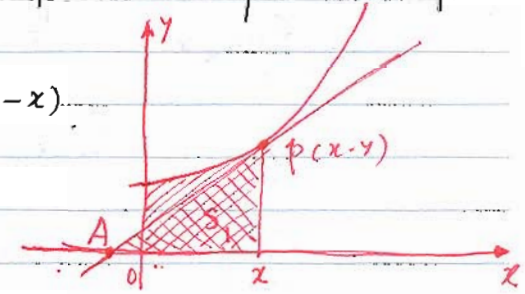
$p^2=\frac{2}{y}$. $\quad p=\pm\sqrt{\frac{2}{y}}$ $\quad \frac{dy}{dx}=\pm\frac{\sqrt{2}}{\sqrt{y}}$ $\quad$ ($\because x=0$ 时 $y=\frac{1}{2}$, $p=2$.)

$\sqrt{y}\,dy=\pm\sqrt{2}\,dx \Rightarrow \frac{2}{3}y\sqrt{y}=\pm\sqrt{2}x+C_2$.

$y(0)=\frac{1}{2} \Rightarrow \frac{1}{3\sqrt{2}}=C_2$. 即: $\frac{2}{3}y\sqrt{y}=\pm\sqrt{2}x+\frac{1}{3\sqrt{2}}$

$\therefore y(x)=\frac{1}{2}\sqrt[3]{(6x+1)^2}$.

5. 设函数 $y(x)$ $(x\geq 0)$ 二阶可导, 且 $y'(x)>0$, $y(0)=1$. 过曲线 $y=y(x)$ 上任一点 $P(x,y)$ 作该曲线的切线及 $x$ 轴的垂线. 将上述两直线与 $x$ 轴所围成的三角形的面积记为 $S_1$, 区间 $[0,x]$ 上以 $y=y(x)$ 为曲边的曲边梯形面积记为 $S_2$. 并设 $2S_2-S_1$ 恒为 1. 求曲线 $y=y(x)$ 的方程.




解: $S_2=\int_0^x f(t)\,dt$. $\quad$ 切线 AP 方程: $\bar{y}-y=y'(x)\cdot(\bar{x}-x)$.

设 A 点横坐标 $\bar{x}_A$. 则令 $\bar{y}=0$ 得: $\bar{x}_A=x-\frac{y}{y'(x)}$.

$S_1=\frac{1}{2}\cdot(x-\bar{x}_A)\cdot y=\frac{1}{2}\cdot\frac{y}{y'(x)}\cdot y$

$2S_1-S_2=\frac{y^2}{y'(x)}-\int_0^x f(t)\,dt=1$. 两边求导得: $\frac{2yy'^2-y^2y''}{y'^2}-y=0$.

即: $yy'^2-y^2y''=0$. $\quad$ 令 $y'=p$. 则有: $y''=p\cdot\frac{dp}{dy}$ 代入得:

$yp^2-y^2\cdot p\frac{dp}{dy}=0 \Rightarrow \frac{dp}{dy}=\frac{p}{y} \Rightarrow p=cy$.

即: $\frac{dy}{dx}=cy$. $\Rightarrow y(x)=Ce^{cx}$. 由 $y(0)=1 \Rightarrow C=1$.

$\therefore$ 曲线方程: $y(x)=e^{cx}$ $(c>0)$. ($\because y'(x)>0$)

## §5. 微分方程的幂级数解法.

习题. P.219 1.

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1. 应用幂级数的性质求解下列微分方程：

(1) $x\frac{dy}{dx}=3y+3$.

解：设 $y=\sum_{n=0}^{\infty}a_n x^n$. 则 $\frac{dy}{dx}=\sum_{n=1}^{\infty}n a_n x^{n-1}$. $x\frac{dy}{dx}=\sum_{n=1}^{\infty}n a_n x^n$.

$x\frac{dy}{dx}=3y+3 \Rightarrow \sum_{n=1}^{\infty}n a_n x^n=\sum_{n=0}^{\infty}3a_n x^n+3=\sum_{n=1}^{\infty}3a_n x^n+3(a_0+1)$.

$\Rightarrow \sum_{n=1}^{\infty}(n-3)a_n x^n=3(a_0+1)$. $\Rightarrow \begin{cases} \text{当 } n\neq 3 \text{ 时. } a_n=0 \\ a_0=-1 \end{cases}$

$\therefore y=a_0+a_3x^3$ 即：$y(x)=-1+Cx^3$.

(2) $(x-3)\frac{dy}{dx}-xy=0$.

解：设 $y=\sum_{n=0}^{\infty}a_n x^n$. 则 $\frac{dy}{dx}=\sum_{n=1}^{\infty}n a_n x^{n-1}$.

$(x-3)\frac{dy}{dx}-xy=(x-3)\cdot\sum_{n=1}^{\infty}n a_n x^{n-1}-x\cdot\sum_{n=0}^{\infty}a_n x^n$

$=\sum_{n=1}^{\infty}n a_n x^n-3\sum_{n=1}^{\infty}n a_n x^{n-1}-\sum_{n=0}^{\infty}a_n x^{n+1}$

$=\sum_{n=1}^{\infty}n a_n x^n-3\sum_{n=0}^{\infty}(n+1)a_{n+1}x^n-\sum_{n=1}^{\infty}a_{n-1}x^n$

$=\sum_{n=1}^{\infty}[n a_n-3(n+1)a_{n+1}-a_{n-1}]x^n-3a_1=0$.

$\Rightarrow \begin{cases} n a_n-3(n+1)a_{n+1}-a_{n-1}=0. \\ a_1=0. \end{cases} \Rightarrow \begin{cases} a_{n+1}=\frac{n a_n-a_{n-1}}{3(n+1)} \\ a_1=0 \end{cases}$ $a_0$ 任意取值. 令 $a_0=C$.

则有：$a_0=C$. $a_1=0$. $a_2=-\frac{1}{6}C$. $a_3=-\frac{1}{27}C$. $a_4=\frac{1}{216}C$. $a_5=\frac{1}{270}C$.

$a_6=\frac{1}{1296}C$. $a_7=\frac{1}{22680}C$. ……

$\therefore y(x)=C\left(1-\frac{1}{6}x^2-\frac{1}{27}x^3+\frac{1}{216}x^4+\frac{1}{270}x^5+\frac{1}{1296}x^6+\frac{1}{22680}x^7+\cdots\right)$

(3) $\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=0$.

解：设 $y=\sum_{n=0}^{\infty}a_n x^n$. 则：$\frac{dy}{dx}=\sum_{n=1}^{\infty}n a_n x^{n-1}=\sum_{n=0}^{\infty}(n+1)a_{n+1}x^n$.

$$\frac{d^2y}{dx^2}=\sum_{n=1}^{\infty} n(n+1)a_{n+1}x^{n}=\sum_{n=0}^{\infty}(n+1)(n+2)a_{n+2}x^{n}.$$

由 $\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=0 \Rightarrow \sum_{n=0}^{\infty}\left[(n+1)(n+2)a_{n+2}-3(n+1)a_{n+1}+2a_n\right]x^n=0.$

$$\Rightarrow (n+1)(n+2)a_{n+2}-3(n+1)a_{n+1}+2a_n=0. \Rightarrow a_{n+2}=\frac{3(n+1)a_{n+1}-2a_n}{(n+1)(n+2)}$$

$$a_2=\frac{3a_1-2a_0}{2!}=\frac{2^2(a_1-a_0)-a_1+2a_0}{2!}$$

$$a_3=\frac{7a_1-6a_0}{3!}=\frac{2^3(a_1-a_0)-a_1+2a_0}{3!}$$

$$a_4=\frac{15a_1-14a_0}{4!}=\frac{2^4(a_1-a_0)-a_1+2a_0}{4!}$$

$$\vdots$$

$$a_n=\frac{2^n(a_1-a_0)-a_1+2a_0}{n!} \quad \text{令} \quad a_1-a_0=C_1,\quad 2a_0-a_1=C_2.$$

则 $a_n=\frac{C_1\cdot 2^n+C_2}{n!}$ $\therefore y(x)=\sum_{n=0}^{\infty}\frac{C_2+C_1\cdot 2^n}{n!}x^n$

即: $y(x)=C_1\sum_{n=0}^{\infty}\frac{(2x)^n}{n!}+C_2\sum_{n=0}^{\infty}\frac{x^n}{n!}$

(4) $x\frac{d^2y}{dx^2}-(x+m)\frac{dy}{dx}+my=0.$ $m$为正整数.

解: 设 $y=\sum_{n=0}^{\infty}a_nx^n$. 则 $\frac{dy}{dx}=\sum_{n=0}^{\infty}(n+1)a_{n+1}x^n$. $\frac{d^2y}{dx^2}=\sum_{n=0}^{\infty}(n+1)(n+2)a_{n+2}x^n$.

由原方程可得: ~~$n(n+1)(n+2)a_{n+2}-(x+m)(n+1)$~~ $my=\sum_{n=1}^{\infty}ma_nx^n+ma_0$

$$x\frac{d^2y}{dx^2}=\sum_{n=0}^{\infty}(n+1)(n+2)a_{n+2}x^{n+1}=\sum_{n=1}^{\infty}n(n+1)a_{n+1}x^n.$$

$$(x+m)\frac{dy}{dx}=x\frac{dy}{dx}+m\frac{dy}{dx}=\sum_{n=0}^{\infty}(n+1)a_{n+1}x^{n+1}+m\sum_{n=0}^{\infty}(n+1)a_{n+1}x^n$$

$$=\sum_{n=1}^{\infty}na_nx^n+\sum_{n=1}^{\infty}m(n+1)a_{n+1}x^n+ma_1.$$

由原方程可得: $\sum_{n=1}^{\infty}\left[n(n+1)a_{n+1}-na_n-m(n+1)a_{n+1}+ma_n\right]x^n-ma_1+ma_0=0.$

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$\Rightarrow a_0 = a_1$. 且 $(n-m)\left[(n+1)a_{n+1} = a_n\right] = 0$.

当 $n = 2, 3, \cdots, m-1$ 时: $a_{n+1} = \dfrac{a_n}{n+1} = \dfrac{a_1}{(n+1)!}$

即: $a_2 = \dfrac{a_1}{2!}$, $a_3 = \dfrac{a_1}{3!}$, $a_4 = \dfrac{a_1}{4!}$, $\cdots$, $a_m = \dfrac{a_1}{m!}$.

当 $n = m$ 时: $a_{m+1}$ 待定.

当 $n = m+1, m+2, \cdots, \infty$ 时: $a_{n+1} = \dfrac{a_n}{n+1}$

即: $a_{m+2} = \dfrac{a_{m+1}}{m+2}$, $a_{m+3} = \dfrac{a_{m+1}}{(m+3)(m+2)}$, $a_{m+4} = \dfrac{a_{m+1}}{(m+4)(m+3)(m+2)}$, $\cdots$

$$\therefore y(x) = \sum_{n=0}^{\infty} a_n x^n = \cancel{a_0 + a_1 x + \sum_{n=2}^{m} \frac{a_1}{(n+1)!} x^{n+1}} + \cancel{a_1 + \sum_{n=1}^{m-1} \frac{a_1}{n!} x^n}$$

$$= \sum_{n=0}^{m} \frac{a_1}{n!} x^n + a_{m+1} x^{m+1} + \frac{a_{m+1}}{m+2} x^{m+2} + \frac{a_{m+1}}{(m+3)(m+2)} x^{m+3} + \cdots$$

$$= \sum_{n=0}^{m} \frac{a_1 x^n}{n!} + a_{m+1} x^{m+1} \left(1 + \frac{x}{m+2} + \frac{x^2}{(m+3)(m+2)} + \frac{x^3}{(m+4)(m+3)(m+2)} + \cdots\right)$$

$$= a_1 \sum_{n=0}^{m} \frac{x^n}{n!} + a_{m+1} x^{m+1} \sum_{k=0}^{\infty} \frac{(m+1)!}{(m+1+k)!} x^k$$

即: $y(x) = C_1 \sum_{n=0}^{m} \dfrac{x^n}{n!} + C_2 \sum_{n=m+1}^{\infty} \dfrac{(m+1)!}{n!} x^n$

2. 用幂级数解法求解下列定解问题:

解: $\begin{cases} \dfrac{dy}{dx} = e^{-x^2} \\ y(0) = 1. \end{cases}$

设 $y = \sum_{n=0}^{\infty} a_n x^n$. 则: $\dfrac{dy}{dx} = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$

而 $e^{-x^2} = \sum_{n=0}^{\infty} \dfrac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} \dfrac{(-1)^n}{n!} x^{2n}$.

由原方程可得: $\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n = \sum_{n=0}^{\infty} \dfrac{(-1)^n}{n!} x^{2n}$.

$$\Rightarrow \begin{cases} a_{2k} = 0 \\ (2k+1) a_{2k+1} = \dfrac{(-1)^k}{k!} \end{cases} \quad k = 1, 2, 3, \cdots.$$

则: $y(x) = a_0 + a_1 x + \sum_{k=1}^{\infty} \dfrac{(-1)^k}{(2k+1) \cdot k!} x^{2k+1}$.

$\begin{cases} y(0) = 1 \Rightarrow a_0 = 1 \\ y'(0) = e^0 = 1 \Rightarrow a_1 = 1 \end{cases}$

$\therefore y(x) = 1 + x + \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k+1)\cdot k!} x^{2k+1}$

即: $y(x) = 1 + \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)\cdot k!} x^{2k+1}$

(2) $\begin{cases} x\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0 \\ y(0) = 0, \quad y'(0) = 1. \end{cases}$

解: 设 $y = \sum_{n=0}^{\infty} a_n x^n$. 则: $\frac{dy}{dx} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$. $x\frac{dy}{dx} = \sum_{n=1}^{\infty} n a_n x^n$.

$\frac{d^2y}{dx^2} = \sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n$. $x\frac{d^2y}{dx^2} = \sum_{n=1}^{\infty} n(n+1) a_{n+1} x^n$.

由原方程可得: $\sum_{n=1}^{\infty} [n(n+1) a_{n+1} - n a_n + a_n] x^n + a_0 = 0$.

$\Rightarrow \begin{cases} a_0 = 0. \\ a_{n+1} = \frac{n-1}{n(n+1)} a_n, \quad n = 1, 2, \cdots \end{cases} \Rightarrow \begin{cases} a_0 = 0. \\ a_2 = 0, a_1 = 0, a_3 = 0, \cdots a_n = 0, \\ n = 4, 5, \cdots \infty. \end{cases}$

$\therefore y(x) = a_1 x$. 由 $y'(0) = 1 \Rightarrow a_1 = 1$.

$\therefore y(x) = x$.

## §6. 常系数线性微分方程组简介.

习题. P.288-3.

1. 求下列齐次线性方程组的通解:

(1) $\begin{cases} \frac{dy_1}{dx} = -5y_1 - y_2 \\ \frac{dy_2}{dx} = 2y_1 - 3y_2 \end{cases}$

解: $\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{bmatrix} -5 & -1 \\ 2 & -3 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. $|A - \lambda I| = \begin{vmatrix} -5-\lambda & -1 \\ 2 & -3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_{1,2} = -4 \pm i$.

$\Lambda = \begin{bmatrix} -4+i & 0 \\ 0 & -4-i \end{bmatrix}$. $T = (\vec{\alpha}_1, \vec{\alpha}_2) = \begin{bmatrix} 1 & 1 \\ -1-i & -1+i \end{bmatrix}$. 则有: $T^{-1}AT = \Lambda$.

$\frac{d}{dx}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \Lambda\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \Rightarrow \begin{cases} z_1 = c_1 e^{\lambda_1 x} \\ z_2 = c_2 e^{\lambda_2 x} \end{cases}$ 又 $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = T\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$.

则满足: $\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = T\frac{d}{dx}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = T\Lambda\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = T\Lambda T^{-1}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$

即原方程的 $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = T\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{bmatrix} 1 & 1 \\ -1-i & -1+i \end{bmatrix}\begin{pmatrix} C_1 e^{\lambda_1 x} \\ C_2 e^{\lambda_2 x} \end{pmatrix}$

$$= \begin{pmatrix} C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} \\ C_1(-1-i)e^{\lambda_1 x} + C_2(-1+i)e^{\lambda_2 x} \end{pmatrix}$$

$$= \begin{pmatrix} C_1 e^{-4x}(\cos x + i\sin x) + C_2 e^{-4x}(\cos x - i\sin x) \\ C_1 e^{-4x}(-1-i)(\cos x + i\sin x) + C_2 e^{-4x}(-1+i)(\cos x - i\sin x) \end{pmatrix}$$

$$= e^{-4x}\begin{pmatrix} (C_1+C_2)\cos x + i(C_1-C_2)\sin x \\ (C_1+C_2)(\sin x - \cos x) - i(C_1-C_2)(\sin x + \cos x) \end{pmatrix}$$

令复系数 $A_1 = C_1 + C_2$, $A_2 = i(C_1 - C_2)$, 则有:

$$\begin{cases} y_1(x) = e^{-4x}(A_1\cos x + A_2\sin x) \\ y_2(x) = e^{-4x}(A_1(\sin x - \cos x) - A_2(\sin x + \cos x)) \end{cases}$$

(2) $$\begin{cases} \frac{dy_1}{dx} = -5y_1 - y_2 \\ \frac{dy_2}{dx} = -3y_1 + y_2 \end{cases}$$

解: $\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{bmatrix} -5 & -1 \\ -3 & 1 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. $|A-\lambda I| = \begin{vmatrix} -5-\lambda & -1 \\ -3 & 1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_{1,2} = -2 \pm 2\sqrt{3}$.

$\Lambda = \begin{bmatrix} -2+2\sqrt{3} & 0 \\ 0 & -2-2\sqrt{3} \end{bmatrix}$ $T = [\vec{u_1}\ \vec{u_2}] = \begin{bmatrix} 1 & 1 \\ -3-2\sqrt{3} & -3+2\sqrt{3} \end{bmatrix}$ 则有: $T^{-1}AT = \Lambda$.

$\frac{d}{dx}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \Lambda\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \Rightarrow \begin{cases} z_1 = C_1 e^{\lambda_1 x} \\ z_2 = C_2 e^{\lambda_2 x} \end{cases}$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = T\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{bmatrix} 1 & 1 \\ -3-2\sqrt{3} & -3+2\sqrt{3} \end{bmatrix}\begin{pmatrix} C_1 e^{\lambda_1 x} \\ C_2 e^{\lambda_2 x} \end{pmatrix} = \begin{pmatrix} C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} \\ C_1(-3-2\sqrt{3})e^{\lambda_1 x} + C_2(-3+2\sqrt{3})e^{\lambda_2 x} \end{pmatrix}$$

$$\therefore \begin{cases} y_1(x) = C_1 e^{(-2+2\sqrt{3})x} + C_2 e^{(-2-2\sqrt{3})x} \\ y_2(x) = (-3-2\sqrt{3})C_1 e^{(-2+2\sqrt{3})x} + (-3+2\sqrt{3})C_2 e^{(-2-2\sqrt{3})x} \end{cases}$$

(3) $\begin{cases} \frac{dy_1}{dx} = 2y_1 + 4y_2 \\ \frac{dy_2}{dx} = -y_1 + 2y_2 \end{cases}$

解: $\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{bmatrix} 2 & 4 \\ -1 & 2 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. $|A-\lambda I| = \begin{vmatrix} 2-\lambda & 4 \\ -1 & 2-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_{1,2} = 2 \pm 2i$

$\Lambda = \begin{bmatrix} 2+2i & 0 \\ 0 & 2-2i \end{bmatrix}$. $T = \begin{bmatrix} 2 & 2 \\ i & -i \end{bmatrix}$. 则有: $T^{-1}AT = \Lambda$.

$\frac{d}{dx}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \Lambda\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \Rightarrow \begin{cases} z_1 = C_1 e^{\lambda_1 x} \\ z_2 = C_2 e^{\lambda_2 x} \end{cases}$

$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = T\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{bmatrix} 2 & 2 \\ i & -i \end{bmatrix}\begin{pmatrix} C_1 e^{\lambda_1 x} \\ C_2 e^{\lambda_2 x} \end{pmatrix} = \begin{pmatrix} 2C_1 e^{\lambda_1 x} + 2C_2 e^{\lambda_2 x} \\ iC_1 e^{\lambda_1 x} - iC_2 e^{\lambda_2 x} \end{pmatrix}$

$\therefore \begin{cases} y_1(x) = 2e^{2x}\left[C_1(\cos 2x + i\sin 2x) + C_2(\cos 2x - i\sin 2x)\right] \\ \quad = 2e^{2x}\left[(C_1+C_2)\cos 2x + i(C_1-C_2)\sin 2x\right] \\ y_2(x) = e^{2x}\left[C_1(i\cos 2x - \sin 2x) + C_2(-i\cos 2x - \sin 2x)\right] \\ \quad = e^{2x}\left[i(C_1-C_2)\cos 2x - (C_1+C_2)\sin 2x\right] \end{cases}$

令复常数 $A_1 = C_1 + C_2$. $A_2 = i(C_1 - C_2)$. 则有:

$\begin{cases} y_1(x) = 2e^{2x}(A_1\cos 2x + A_2 \sin 2x) \\ y_2(x) = e^{2x}(-A_1\sin 2x + A_2\cos 2x) \end{cases}$

(4) $\begin{cases} \frac{dy_1}{dx} = 2y_1 + y_2 + 2y_3 \\ \frac{dy_2}{dx} = -y_1 - 2y_3 \\ \frac{dy_3}{dx} = y_3 \end{cases}$

解: $\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ -1 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$

$|A-\lambda I| = \begin{vmatrix} 2-\lambda & 1 & 2 \\ -1 & -\lambda & -2 \\ 0 & 0 & 1-\lambda \end{vmatrix} = -(1-\lambda)^3 = 0. \Rightarrow \lambda_1 = \lambda_2 = \lambda_3 = 1$

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$(A-\lambda I)\vec{u}=\begin{bmatrix}1&1&2\\-1&-1&-2\\0&0&0\end{bmatrix}\begin{pmatrix}u_1\\u_2\\u_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$ 线性无关的特征向量有两个：

$\vec{\xi}_1=\begin{pmatrix}-1\\1\\0\end{pmatrix}$ $\vec{\xi}_2=\begin{pmatrix}-2\\0\\1\end{pmatrix}$

求 $\vec{\xi}_1$ 的根向量：$(A-\lambda I)\vec{\xi}_{11}=\vec{\xi}_1$.

即：$\begin{pmatrix}-1&-1&2\\-1&-1&-2\\0&0&0\end{pmatrix}\begin{pmatrix}\xi_{11}^{(1)}\\\xi_{11}^{(2)}\\\xi_{11}^{(3)}\end{pmatrix}=\begin{pmatrix}-1\\1\\0\end{pmatrix}$ $\Rightarrow$ $\vec{\xi}_{11}=\begin{pmatrix}-1\\0\\0\end{pmatrix}$ 此时有：

$A\vec{\xi}_{11}=\vec{\xi}_1+\lambda\vec{\xi}_{11}$.

$\Rightarrow \left(A\vec{\xi}_1 \quad A\vec{\xi}_{11} \quad A\vec{\xi}_2\right)=\left(\vec{\xi}_1 \quad \vec{\xi}_{11} \quad \vec{\xi}_2\right)\begin{bmatrix}\lambda&1&0\\0&\lambda&0\\0&0&\lambda\end{bmatrix}=A\left(\vec{\xi}_1 \quad \vec{\xi}_{11} \quad \vec{\xi}_2\right)$.

记 $T=(\vec{\xi}_1 \ \vec{\xi}_{11} \ \vec{\xi}_2)$. 则有：

$AT=T\begin{bmatrix}\lambda&1&0\\0&\lambda&0\\0&0&\lambda\end{bmatrix}$.

且 T 可逆 $\left(|T|=\begin{vmatrix}-1&-1&-2\\1&0&0\\0&0&1\end{vmatrix}\neq 0\right)$.

记 $\begin{pmatrix}y_1\\y_2\\y_3\end{pmatrix}=T\begin{pmatrix}z_1\\z_2\\z_3\end{pmatrix}$. 则：$\frac{d}{dx}\begin{pmatrix}z_1\\z_2\\z_3\end{pmatrix}=T^{-1}\frac{d}{dx}\begin{pmatrix}y_1\\y_2\\y_3\end{pmatrix}=T^{-1}AT\begin{pmatrix}z_1\\z_2\\z_3\end{pmatrix}=\begin{bmatrix}\lambda&1&0\\0&\lambda&0\\0&0&\lambda\end{bmatrix}\begin{pmatrix}z_1\\z_2\\z_3\end{pmatrix}$

$\Rightarrow\begin{cases}\frac{dz_1}{dx}=\lambda z_1+z_2\\ \frac{dz_2}{dx}=\lambda z_2\\ \frac{dz_3}{dx}=\lambda z_3\end{cases}\Rightarrow\begin{cases}z_1=(C_1+C_2x)e^{\lambda x}\\ z_2=C_2e^{\lambda x}\\ z_3=C_3e^{\lambda x}\end{cases}$

则有：

$\begin{pmatrix}y_1\\y_2\\y_3\end{pmatrix}=T\begin{pmatrix}z_1\\z_2\\z_3\end{pmatrix}=\begin{bmatrix}-1&-1&-2\\1&0&0\\0&0&1\end{bmatrix}\begin{pmatrix}(C_1+C_2x)e^{\lambda x}\\C_2e^{\lambda x}\\C_3e^{\lambda x}\end{pmatrix}=\begin{pmatrix}-e^{\lambda x}(C_1+C_2+C_2x+2C_3)\\e^{\lambda x}(C_1+C_2x)\\C_3e^{\lambda x}\end{pmatrix}$

$\therefore\begin{cases}y_1(x)=-(C_1+C_2+2C_3+C_2x)e^{\lambda x}\\ y_2(x)=(C_1+C_2x)e^{\lambda x}\\ y_3(x)=C_3e^{\lambda x}\end{cases}$

(5) $\begin{cases} \frac{dy_1}{dx} = -y_1 + y_2 + y_3 \\ \frac{dy_2}{dx} = y_1 - y_2 + y_3 \\ \frac{dy_3}{dx} = y_1 + y_2 - y_3 \end{cases}$

解: $\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$.

$|A-\lambda I| = \begin{vmatrix} -1-\lambda & 1 & 1 \\ 1 & -1-\lambda & 1 \\ 1 & 1 & -1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 1,\ \lambda_2 = \lambda_3 = -2$.

$\vec{\xi}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},\ \vec{\xi}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix},\ \vec{\xi}_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

令 $T = (\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3) = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. 则有:

$AT = T\Lambda = T\begin{bmatrix} 1 & & \\ & -2 & \\ & & -2 \end{bmatrix} \Rightarrow T^{-1}AT = \begin{bmatrix} 1 & & \\ & -2 & \\ & & -2 \end{bmatrix}$.

令 $\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = T\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$. 则: $\frac{d}{dx}\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = T^{-1}\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = T^{-1}AT\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \Lambda\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$.

$\Rightarrow \frac{d}{dx}\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{bmatrix} 1 & & \\ & -2 & \\ & & -2 \end{bmatrix}\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} \Rightarrow \begin{cases} z_1 = C_1 e^{x} \\ z_2 = C_2 e^{-2x} \\ z_3 = C_3 e^{-2x} \end{cases}$

$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = T\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{pmatrix} C_1 e^{x} \\ C_2 e^{-2x} \\ C_3 e^{-2x} \end{pmatrix} = \begin{pmatrix} C_1 e^{x} - (C_2 + C_3) e^{-2x} \\ C_1 e^{x} + C_2 e^{-2x} \\ C_1 e^{x} + C_3 e^{-2x} \end{pmatrix}$

$\therefore \begin{cases} y_1(x) = C_1 e^{x} - (C_2 + C_3) e^{-2x} \\ y_2(x) = C_1 e^{x} + C_2 e^{-2x} \\ y_3(x) = C_1 e^{x} + C_3 e^{-2x} \end{cases}$

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(6) $\begin{cases} \frac{dy_1}{dx} = 14y_1 + 66y_2 - 42y_3 \\ \frac{dy_2}{dx} = 4y_1 + 2y_2 - 14y_3 \\ \frac{dy_3}{dx} = 10y_1 + 55y_2 - 33y_3. \end{cases}$

解: $\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{bmatrix} 14 & 66 & -42 \\ 4 & 2 & -14 \\ 10 & 55 & -33 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$

$|A-\lambda I| = \begin{vmatrix} 14-\lambda & 66 & -42 \\ 4 & 2-\lambda & -14 \\ 10 & 55 & -33-\lambda \end{vmatrix} = -(\lambda-2)(\lambda^2+19\lambda+464) = 0 \Rightarrow$

$\lambda_1 = 2, \quad \lambda_{2,3} = -\frac{19}{2} \pm \frac{\sqrt{1495}}{2}i.$

对应特征向量: $\vec{\xi}_1 = \begin{pmatrix} 7 \\ 0 \\ 2 \end{pmatrix}$ $\vec{\xi}_{2,3} = \begin{pmatrix} 132 \\ 23 \pm \sqrt{1495}\,i \\ 110 \end{pmatrix}$

$\therefore T = [\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3] = \begin{bmatrix} 7 & 132 & 132 \\ 0 & 23+\sqrt{1495}\,i & 23-\sqrt{1495}\,i \\ 2 & 110 & 110 \end{bmatrix}$.

则有 $T^{-1}AT = \Lambda = \begin{bmatrix} 2 & & \\ & -\frac{19}{2}+\frac{\sqrt{1495}}{2}i & \\ & & -\frac{19}{2}-\frac{\sqrt{1495}}{2}i \end{bmatrix}$

$\therefore \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = T\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{bmatrix} 7 & 132 & 132 \\ 0 & 23+\sqrt{1495}\,i & 23-\sqrt{1495}\,i \\ 2 & 110 & 110 \end{bmatrix}\begin{pmatrix} C_1 e^{2x} \\ C_2 e^{-\frac{19}{2}x}\left(\cos\frac{\sqrt{1495}}{2}x + i\sin\frac{\sqrt{1495}}{2}x\right) \\ C_3 e^{-\frac{19}{2}x}\left(\cos\frac{\sqrt{1495}}{2}x - i\sin\frac{\sqrt{1495}}{2}x\right) \end{pmatrix}$

$= \begin{pmatrix} 7C_1e^{2x} + 132e^{-\frac{19}{2}x}\left[C_2\left(\cos\frac{\sqrt{1495}}{2}x + i\sin\frac{\sqrt{1495}}{2}x\right) + C_3\left(\cos\frac{\sqrt{1495}}{2}x - i\sin\frac{\sqrt{1495}}{2}x\right)\right] \\ e^{-\frac{19}{2}x}\left[C_2(23+\sqrt{1495}\,i)\left(\cos\frac{\sqrt{1495}}{2}x + i\sin\frac{\sqrt{1495}}{2}x\right) + C_3(23-\sqrt{1495}\,i)\left(\cos\frac{\sqrt{1495}}{2}x - i\sin\frac{\sqrt{1495}}{2}x\right)\right] \\ 2C_1e^{2x} + 110\cdot e^{-\frac{19}{2}x}\left[C_2\left(\cos\frac{\sqrt{1495}}{2}x + i\sin\frac{\sqrt{1495}}{2}x\right) + C_3\left(\cos\frac{\sqrt{1495}}{2}x - i\sin\frac{\sqrt{1495}}{2}x\right)\right] \end{pmatrix}$

$= \begin{pmatrix} 7C_1e^{2x} + 132e^{-\frac{19}{2}x}\left[(C_2+C_3)\cos\frac{\sqrt{1495}}{2}x + i(C_2-C_3)\sin\frac{\sqrt{1495}}{2}x\right] \\ e^{-\frac{19}{2}x}\left[(C_2+C_3)(23\cos\frac{\sqrt{1495}}{2}x - \sqrt{1495}\sin\frac{\sqrt{1495}}{2}x) + i(C_2-C_3)(23\sin\frac{\sqrt{1495}}{2}x + \sqrt{1495}\sin\frac{\sqrt{1495}}{2}x)\right] \\ 2C_1e^{2x} + 110\cdot e^{-\frac{19}{2}x}\left[(C_2+C_3)\cos\frac{\sqrt{1495}}{2}x + i(C_2-C_3)\sin\frac{\sqrt{1495}}{2}x\right] \end{pmatrix}$

令复常数 $A_1=C_2+C_3$, $A_2=(C_2-C_3)i$, 则有:

$$\begin{cases} y_1(x)=7C_1e^{2x}+132e^{-\frac{19}{2}x}\left[A_1\cos\frac{\sqrt{1495}}{2}x+A_2\sin\frac{\sqrt{1495}}{2}x\right] \\ y_2(x)=e^{-\frac{19}{2}x}\left[A_1\left(23\cos\frac{\sqrt{1495}}{2}x-\sqrt{1495}\sin\frac{\sqrt{1495}}{2}x\right)\right. \\ \qquad\left.+A_2\left(23\sin\frac{\sqrt{1495}}{2}x+\sqrt{1495}\cos\frac{\sqrt{1495}}{2}x\right)\right] \\ y_3(x)=2C_1e^{2x}+110e^{-\frac{19}{2}x}\left[A_1\cos\frac{\sqrt{1495}}{2}x+A_2\sin\frac{\sqrt{1495}}{2}x\right] \end{cases}$$

2. 求下列非齐次线性方程组的通解:

(1) $\begin{cases} \frac{dy_1}{dx}=-5y_1-y_2+e^x \\ \frac{dy_2}{dx}=y_1-3y_2+e^{2x} \end{cases}$

解: $\frac{d}{dx}\begin{pmatrix}y_1\\y_2\end{pmatrix}=\begin{bmatrix}-5&-1\\1&-3\end{bmatrix}\begin{pmatrix}y_1\\y_2\end{pmatrix}=A\begin{pmatrix}y_1\\y_2\end{pmatrix}$. $|A-\lambda I|=\begin{vmatrix}-5-\lambda&-1\\1&-3-\lambda\end{vmatrix}=0\Rightarrow\lambda_1=\lambda_2=-4$.

$\vec{\xi}_1=\begin{pmatrix}1\\-1\end{pmatrix}$ 满足 $(A-\lambda I)\vec{\xi}_1=0$. 求特征向量 $\vec{\xi}_{11}$: $(A-\lambda I)\vec{\xi}_{11}=\vec{\xi}_1$.

解得: $\vec{\xi}_{11}=\begin{pmatrix}-1\\0\end{pmatrix}$. 则令 $T=[\vec{\xi}_1\ \vec{\xi}_{11}]\Rightarrow AT=[A\vec{\xi}_1\ A\vec{\xi}_{11}]=[\vec{\xi}_1\ \vec{\xi}_{11}]\begin{bmatrix}\lambda&1\\0&\lambda\end{bmatrix}$

即: $AT=TJ\Rightarrow T^{-1}AT=J$.

令 $\begin{pmatrix}y_1\\y_2\end{pmatrix}=T\begin{pmatrix}z_1\\z_2\end{pmatrix}$ 则: $\frac{d}{dx}\begin{pmatrix}z_1\\z_2\end{pmatrix}=T^{-1}\frac{d}{dx}\begin{pmatrix}y_1\\y_2\end{pmatrix}=T^{-1}AT\begin{pmatrix}z_1\\z_2\end{pmatrix}=J\begin{pmatrix}z_1\\z_2\end{pmatrix}$

$$\Rightarrow\begin{cases}\frac{dz_1}{dx}=\lambda z_1+z_2\\ \frac{dz_2}{dx}=\lambda z_2\end{cases}\Rightarrow\begin{cases}z_1=(C_1+C_2x)e^{\lambda x}\\ z_2=C_2e^{\lambda x}\end{cases}$$

则有: $\begin{pmatrix}y_1\\y_2\end{pmatrix}=T\begin{pmatrix}z_1\\z_2\end{pmatrix}=\begin{bmatrix}1&-1\\-1&0\end{bmatrix}\begin{pmatrix}(C_1+C_2x)e^{\lambda x}\\C_2e^{\lambda x}\end{pmatrix}=\begin{pmatrix}(C_1-C_2+C_2x)e^{\lambda x}\\-(C_1+C_2x)e^{\lambda x}\end{pmatrix}$

即齐次方程的通解为: $\begin{cases}y_1=(C_1-C_2+C_2x)e^{-4x}\\ y_2=-(C_1+C_2x)e^{-4x}\end{cases}$

设非齐次特解为: $\begin{cases}y_1^*=A_1e^x+A_2e^{2x}\\ y_2^*=B_1e^x+B_2e^{2x}\end{cases}$ 代入方程得:

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$$\begin{cases} \frac{dy_1^*}{dx} = A_1e^x + 2A_2e^{2x} = -5A_1e^x - 5A_2e^{2x} - B_1e^x - B_2e^{2x} + e^x \\ \frac{dy_2^*}{dx} = B_1e^x + 2B_2e^{2x} = A_1e^x + A_2e^{2x} - 3B_1e^x - 3B_2e^{2x} + e^{2x} \end{cases}$$

$$\Rightarrow \begin{cases} (6A_1 + B_1 - 1)e^x + (7A_2 + B_2)e^{2x} = 0 \\ (A_1 - 4B_1)e^x + (A_2 - 5B_2 + 1)e^{2x} = 0 \end{cases}$$

由 $\begin{cases} 6A_1 + B_1 - 1 = 0 \\ A_1 - 4B_1 = 0 \end{cases} \Rightarrow \begin{cases} A_1 = \frac{4}{25} \\ B_1 = \frac{1}{25} \end{cases}$ 由 $\begin{cases} 7A_2 + B_2 = 0 \\ A_2 - 5B_2 + 1 = 0 \end{cases} \Rightarrow \begin{cases} A_2 = -\frac{1}{36} \\ B_2 = \frac{7}{36} \end{cases}$

∴通解: $\begin{cases} y_1(x) = (C_1 - C_2 + C_2x)e^{-4x} + \frac{4}{25}e^x - \frac{1}{36}e^{2x} \\ y_2(x) = -(C_1 + C_2x)e^{-4x} + \frac{1}{25}e^x + \frac{7}{36}e^{2x} \end{cases}$

(2) $\begin{cases} \frac{dy_1}{dx} = y_2 \\ \frac{dy_2}{dx} = y_1 + e^x + e^{-x} \end{cases}$

解: $\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ $|A - \lambda I| = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 1,\ \lambda_2 = -1$

$\vec{\xi}_1 = \begin{pmatrix} 1 \\ +1 \end{pmatrix}$, $\vec{\xi}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ 记 $T = [\vec{\xi}_1\ \vec{\xi}_2] = \begin{bmatrix} 1 & 1 \\ +1 & -1 \end{bmatrix}$ 则有: $T^{-1}AT = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} = \Lambda$

记 $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = T\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$ 则有: $\frac{d}{dx}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \Lambda\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} C_1e^x \\ C_2e^{-x} \end{pmatrix} \Rightarrow \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{bmatrix} 1 & 1 \\ +1 & -1 \end{bmatrix}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} C_1e^x + C_2e^{-x} \\ +C_1e^x - C_2e^{-x} \end{pmatrix}$$

即齐次方程通解为: $\begin{cases} y_1 = C_1e^x + C_2e^{-x} \\ y_2 = +C_1e^x - C_2e^{-x} \end{cases}$

设非齐次特解: $\begin{cases} y_1^* = A_1e^x + A_2xe^x + A_3e^{-x} + A_4xe^{-x} \\ y_2^* = B_1e^x + B_2xe^x + B_3e^{-x} + B_4xe^{-x} \end{cases}$ 代入原方程组得:

$$\begin{cases} \frac{dy_1^*}{dx} = (A_1 + A_2 + A_2x)e^x + (A_4 - A_3 - A_4x)e^{-x} = (B_1 + B_2x)e^x + (B_3 + B_4x)e^{-x} \\ \frac{dy_2^*}{dx} = (B_1 + B_2 + B_2x)e^x + (B_4 - B_3 - B_4x)e^{-x} = (A_1 + A_2x + 1)e^x + (A_3 + A_4x + 1)e^{-x} \end{cases}$$

$$\Rightarrow \begin{cases} A_1+A_2=B_1 \\ A_2=B_2 \\ A_4-A_3=B_3 \\ -A_4=B_4 \\ B_1+B_2=A_1+1 \\ B_2=A_2 \\ B_4-B_3=A_3+1 \\ -B_4=A_4 \end{cases} \Rightarrow \text{特解：} \begin{cases} A_1=0 \\ A_2=\frac{1}{2} \\ A_3=0 \\ A_4=-\frac{1}{2} \\ B_1=\frac{1}{2} \\ B_2=\frac{1}{2} \\ B_3=-\frac{1}{2} \\ B_4=\frac{1}{2} \end{cases}$$

∴ 通解：$\begin{cases} y_1(x)=C_1e^{x}+C_2e^{-x}+\frac{1}{2}xe^{x}-\frac{1}{2}xe^{-x} \\ y_2(x)=C_1e^{x}-C_2e^{-x}+\frac{1}{2}(1+x)e^{x}-\frac{1}{2}(1-x)e^{-x} \end{cases}$

(3) $\begin{cases} \frac{dy_1}{dx}=-5y_1-y_2+7e^{x}-27 \\ \frac{dy_2}{dx}=2y_1-3y_2-e^{x}+12 \end{cases}$

解：$\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}=\begin{bmatrix} -5 & -1 \\ 2 & -3 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}=A\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ $|A-\lambda I|=\begin{vmatrix} -5-\lambda & -1 \\ 2 & -3-\lambda \end{vmatrix}=0 \Rightarrow \lambda_{1,2}=-4\pm i$

$\vec{z}_{1,2}=\begin{pmatrix} -1 \\ 1\pm i \end{pmatrix}$ 令 $T=[\vec{z}_1\ \vec{z}_2]=\begin{bmatrix} -1 & -1 \\ 1+i & 1-i \end{bmatrix}$ 则有：$T^{-1}AT=\Lambda=\begin{bmatrix} -4+i & \\ & -4-i \end{bmatrix}$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}=T\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}=\begin{bmatrix} -1 & -1 \\ 1+i & 1-i \end{bmatrix}\begin{pmatrix} C_1e^{-4x}(\cos x+i\sin x) \\ C_2e^{-4x}(\cos x-i\sin x) \end{pmatrix}$$

$$=\begin{pmatrix} -e^{-4x}\left[(C_1+C_2)\cos x+i(C_1-C_2)\sin x\right] \\ e^{-4x}\left[(C_1+C_2)(\cos x-\sin x)+i(C_1-C_2)(\cos x+\sin x)\right] \end{pmatrix}$$

令复常数 $A_1=C_1+C_2$，$A_2=i(C_1-C_2)$，则齐次通解为：

$$\begin{cases} y_1=-e^{-4x}\left[A_1\cos x+A_2\sin x\right] \\ y_2=e^{-4x}\left[A_1(\cos x-\sin x)+A_2(\cos x+\sin x)\right] \end{cases}$$

设非齐次特解：$\begin{cases} y_1^{*}=a_1e^{x}+a_2x+a_3 \\ y_2^{*}=b_1e^{x}+b_2x+b_3 \end{cases}$ 代入后得：

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$$\frac{dy_1^*}{dx} = a_1e^x + a_2 = -5(a_1e^x + a_2x + a_3) - (b_1e^x + b_2x + b_3) + 7e^x - 2.$$

$$\frac{dy_2^*}{dx} = b_1e^x + b_2 = 2(a_1e^x + a_2x + a_3) - 3(b_1e^x + b_2x + b_3) - e^x + 12.$$

$$\Rightarrow \begin{cases} 6a_1 + b_1 - 7 = 0 \\ 2a_1 - 4b_1 - 1 = 0 \\ -5a_2 - b_2 = 0 \\ 2a_2 - 3b_2 = 0 \\ a_2 + 5a_3 + b_3 + 2 = 0 \\ -b_2 = 2a_3 + 3b_3 - 12 = 0 \end{cases} \Rightarrow \begin{cases} a_1 = \frac{29}{26} \\ b_1 = \frac{4}{13} \\ a_2 = 0 \\ b_2 = 0 \\ a_3 = -\frac{93}{17} \\ b_3 = \frac{6}{17} \end{cases}$$

$$\therefore \text{通解:} \begin{cases} y_1(x) = -e^{-4x}(A_1\cos x + A_2\sin x) + \frac{29}{26}e^x - \frac{93}{17} \\ y_2(x) = e^{-4x}\left[A_1(\cos x - \sin x) + A_2(\cos x + \sin x)\right] + \frac{4}{13}e^x + \frac{6}{17} \end{cases}$$

(4) $$\begin{cases} \frac{dy_1}{dx} + y_1 + y_2 = x^2 \\ \frac{dy_2}{dx} + y_2 + y_3 = 2x \\ \frac{dy_3}{dx} + y_3 = x \end{cases}$$

解:
$$\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = A\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}. \qquad |A-\lambda I| = \begin{vmatrix} -1-\lambda & -1 & 0 \\ 0 & -1-\lambda & -1 \\ 0 & 0 & -1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_{1,2,3} = -1$$

由 $\frac{dy_3}{dx} = -y_3 \Rightarrow y_3 = C_3e^{-x}$.

$$\frac{dy_2}{dx} + y_2 = -y_3 = -C_3e^{-x} \Rightarrow y_2 = e^{-x}\left[C_2 + \int C_3e^{-x}e^x dx\right] = (C_2 - C_3x)e^{-x}.$$

$$\frac{dy_1}{dx} + y_1 = -y_2 = -(C_2 - C_3x)e^{-x} \Rightarrow y_1 = e^{-x}\left[C_1 - C_2x + \frac{1}{2}C_3x^2\right].$$

即齐次方程的通解:
$$\begin{cases} y_1 = e^{-x}\left[C_1 - C_2x + \frac{1}{2}C_3x^2\right] \\ y_2 = e^{-x}(C_2 - C_3x) \\ y_3 = C_3e^{-x} \end{cases}$$

设非齐次特解: $\begin{cases} y_1^* = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \\ y_2^* = b_0 + b_1 x + b_2 x^2 + b_3 x^3 \\ y_3^* = c_0 + c_1 x + c_2 x^2 + c_3 x^3 \end{cases}$ 代入原方程, 可得:

$$\begin{cases} \dfrac{dy_1^*}{dx} = a_1 + 2a_2 x + 3a_3 x^2 = x^2 - a_0 - a_1 x - a_2 x^2 - a_3 x^3 - b_0 - b_1 x - b_2 x^2 - b_3 x^3 \\ \dfrac{dy_2^*}{dx} = b_1 + 2b_2 x + 3b_3 x^2 = 2x - b_0 - b_1 x - b_2 x^2 - b_3 x^3 - c_0 - c_1 x - c_2 x^2 - c_3 x^3 \\ \dfrac{dy_3^*}{dx} = c_1 + 2c_2 x + 3c_3 x^2 = x - c_0 - c_1 x - c_2 x^2 - c_3 x^3 \end{cases}$$

对比同次项系数有, 可得: $\begin{cases} a_3 = b_3 = c_3 = 0 \\ a_2 = 1, \quad b_2 = c_2 = 0 \\ a_1 = -3, \quad b_1 = c_1 = 1 \\ a_0 = 3, \quad b_0 = 0, \quad c_0 = -1 \end{cases}$

∴ 通解: $\begin{cases} y_1(x) = \left[ C_1 - C_2 x + \frac{1}{2} C_3 x^2 \right] e^{-x} + x^2 - 3x + 3 \\ y_2(x) = \left[ C_2 - C_3 x \right] e^{-x} + x \\ y_3(x) = \left[ C_3 \right] e^{-x} + x - 1 \end{cases}$

3、利用 Euler 方程的求解法求下列方程组的通解:

(1) $\begin{cases} x \dfrac{dy_1}{dx} = y_1 + 3y_2 \\ x \dfrac{dy_2}{dx} = y_1 - y_2 \end{cases}$

解: 令 $x = e^t \Rightarrow t = \ln x$, 则有: $\dfrac{dy_1}{dt} = \dfrac{dy_1}{dx} \cdot \dfrac{dx}{dt} = x \dfrac{dy_1}{dx}$.

$$\frac{dy_2}{dt} = \frac{dy_2}{dx} \cdot \frac{dx}{dt} = x \frac{dy_2}{dx}.$$

原方程组为: $\begin{cases} \dfrac{dy_1}{dt} = y_1 + 3y_2 \\ \dfrac{dy_2}{dt} = y_1 - y_2 \end{cases} \Rightarrow \dfrac{d}{dt} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{bmatrix} 1 & 3 \\ 1 & -1 \end{bmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$.

$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 3 \\ 1 & -1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 2, \ \lambda_2 = -2$. 对应的: $\vec{\xi}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, $\vec{\xi}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

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令 $T=[\vec{\xi}_1\ \vec{\xi}_2]=\begin{bmatrix}3 & 1\\ 1 & -1\end{bmatrix}$. 则有: $T^{-1}AT=\Lambda=\begin{bmatrix}2 & \\ & -2\end{bmatrix}$.

$$\begin{pmatrix}y_1(t)\\ y_2(t)\end{pmatrix}=T\begin{pmatrix}z_1(t)\\ z_2(t)\end{pmatrix}=\begin{bmatrix}3 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}C_1e^{2t}\\ C_2e^{-2t}\end{bmatrix}=\begin{bmatrix}3C_1e^{2t}+C_2e^{-2t}\\ C_1e^{2t}-C_2e^{-2t}\end{bmatrix}$$

$\therefore$ 原方程: $\begin{cases}y_1(x)=3C_1e^{2t}+C_2e^{-2t}=3C_1x^2+C_2x^{-2}\\ y_2(x)=C_1e^{2t}-C_2e^{-2t}=C_1x^2-C_2x^{-2}\end{cases}$.

(2) $\begin{cases}x\frac{dy_1}{dx}+6y_1-y_2-3y_3=0\\ x\frac{dy_2}{dx}+3y_1-6y_2-9y_3=0\\ x\frac{dy_3}{dx}+y_1+y_2-2y_3=0\end{cases}$

解: 令 $x=e^t$. 则有: $x\frac{dy_i}{dx}=\frac{dy_i}{dt}$ $i=1,2,3$.

$$\frac{d}{dt}\begin{pmatrix}y_1\\ y_2\\ y_3\end{pmatrix}=\begin{bmatrix}-6 & 1 & 3\\ -3 & 6 & 9\\ -1 & -1 & 2\end{bmatrix}\begin{pmatrix}y_1\\ y_2\\ y_3\end{pmatrix}=A\begin{pmatrix}y_1\\ y_2\\ y_3\end{pmatrix}$$

$$|A-\lambda I|=\begin{vmatrix}-6-\lambda & 1 & 3\\ -3 & 6-\lambda & 9\\ -1 & -1 & 2-\lambda\end{vmatrix}=0.\Rightarrow\begin{cases}\lambda_1=-1\\ \lambda_2=1\\ \lambda_3=2\end{cases}$$ 对应: $\vec{\xi}_1=\begin{pmatrix}1\\ 2\\ 1\end{pmatrix}$. $\vec{\xi}_2=\begin{pmatrix}1\\ 1\\ 2\end{pmatrix}$.

令 $T=[\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3]=\begin{bmatrix}1 & 1 & 1\\ 2 & 1 & -1\\ 1 & 2 & 3\end{bmatrix}$ $\vec{\xi}_3=\begin{pmatrix}1\\ -1\\ 3\end{pmatrix}$.

则有: $T^{-1}AT=\Lambda=\begin{bmatrix}-1 & & \\ & 1 & \\ & & 2\end{bmatrix}$.

$$\begin{pmatrix}y_1(t)\\ y_2(t)\\ y_3(t)\end{pmatrix}=T\begin{pmatrix}z_1(t)\\ z_2(t)\\ z_3(t)\end{pmatrix}=\begin{bmatrix}1 & 1 & 1\\ 2 & 1 & -1\\ 1 & 2 & 3\end{bmatrix}\begin{bmatrix}C_1e^{-t}\\ C_2e^{t}\\ C_3e^{2t}\end{bmatrix}=\begin{pmatrix}C_1e^{-t}+C_2e^{t}+C_3e^{2t}\\ 2C_1e^{-t}+C_2e^{t}-C_3e^{2t}\\ C_1e^{-t}+2C_2e^{t}+3C_3e^{2t}\end{pmatrix}$$

∴ 通解为 $\begin{cases} y_1(x) = C_1 e^{-t} + C_2 e^{t} + C_3 e^{2t} = \dfrac{C_1}{x} + C_2 x + C_3 x^2 \\ y_2(x) = 2C_1 e^{-t} + C_2 e^{t} - C_3 e^{2t} = \dfrac{2C_1}{x} + C_2 x - C_3 x^2 \\ y_3(x) = C_1 e^{-t} + 2C_2 e^{t} + 3C_3 e^{2t} = \dfrac{C_1}{x} + 2C_2 x + 3C_3 x^2 \end{cases}$